

GROUP 51

EXTERIOR

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SPECIFICATION(S)**SERVICE SPECIFICATIONS**

M1511000301102

Item	Standard value
Stop position of the windshield wiper arm/blade assembly mm (in)	A (Passenger's side): Ceramic end line ± 5 (0.20)
	B (Driver's side): Front deck garnish end 45 ± 5 (1.77 ± 0.20)

LUBRICANT

M1511000400098

Item	Specified lubricant	Quantity
Wiper motor link rod Contact joint between link rod and wiper motor link plate	Multipurpose grease SAE J310, NLGI No.2 or equivalent	As required

ADHESIVE

M1511000501902

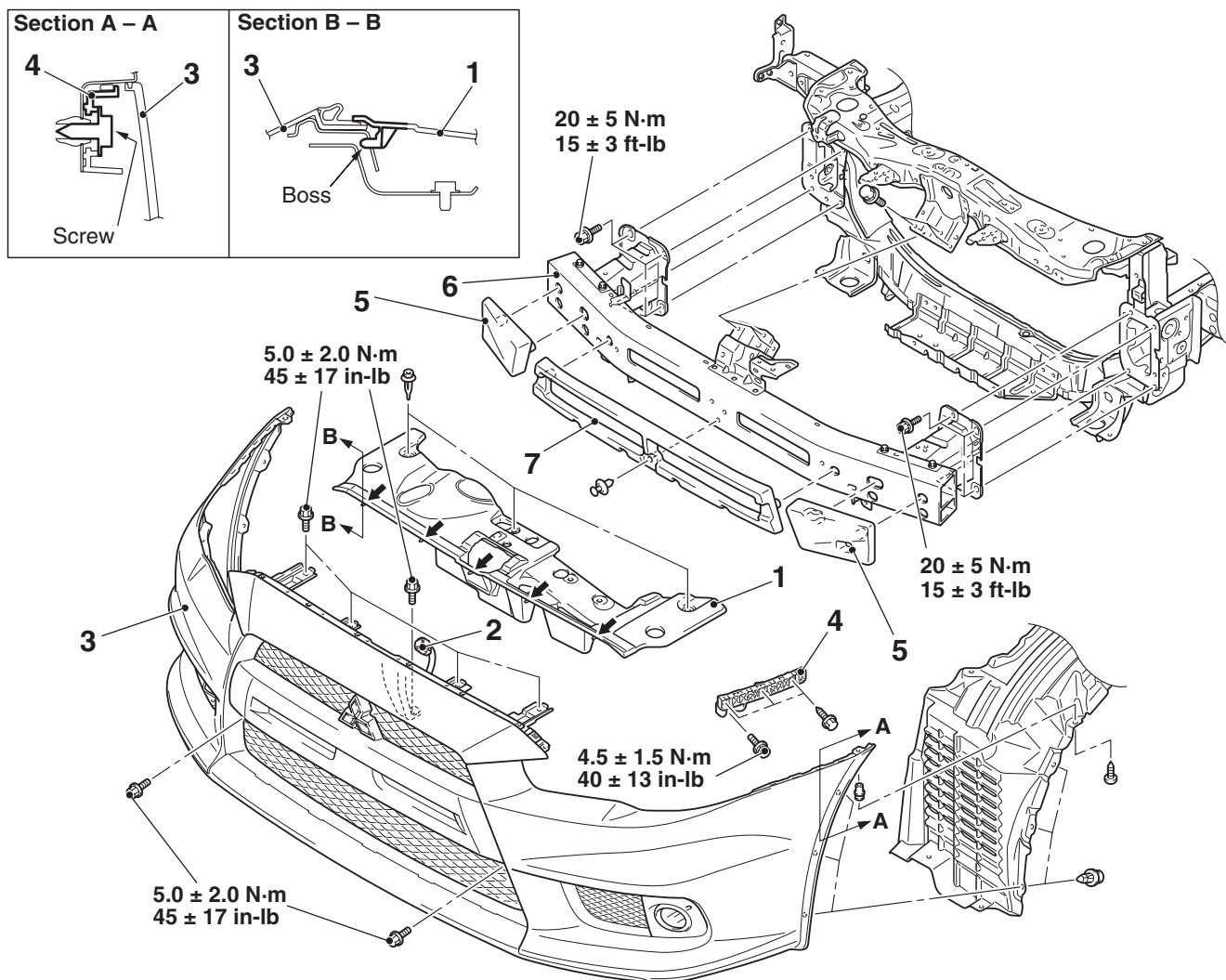
Item	Specification
Front three-diamond mark mm (in)	Adhesive tape: Double-sided tape 0.6 (0.02) thickness
Side air dam mm (in)	Adhesive tape: Double-sided tape 4 (0.16) width and 1.2 (0.05) thickness Adhesive tape remover: 3M™ AAD Part number 8906 or equivalent

FRONT BUMPER ASSEMBLY AND RADIATOR GRILLE REMOVAL AND INSTALLATION

M1511025400330

Pre-removal and post-installation operation

- Engine room under cover front A and engine room under cover center removal and installation (Refer to P.51-16.)
- Air cleaner intake duct removal and installation (Refer to GROUP 15, Air cleaner P.15-10)



◀ : Bosses (triangle marks indicating the boss locations are inscribed on the top surface)

AC708707AB

Removal steps

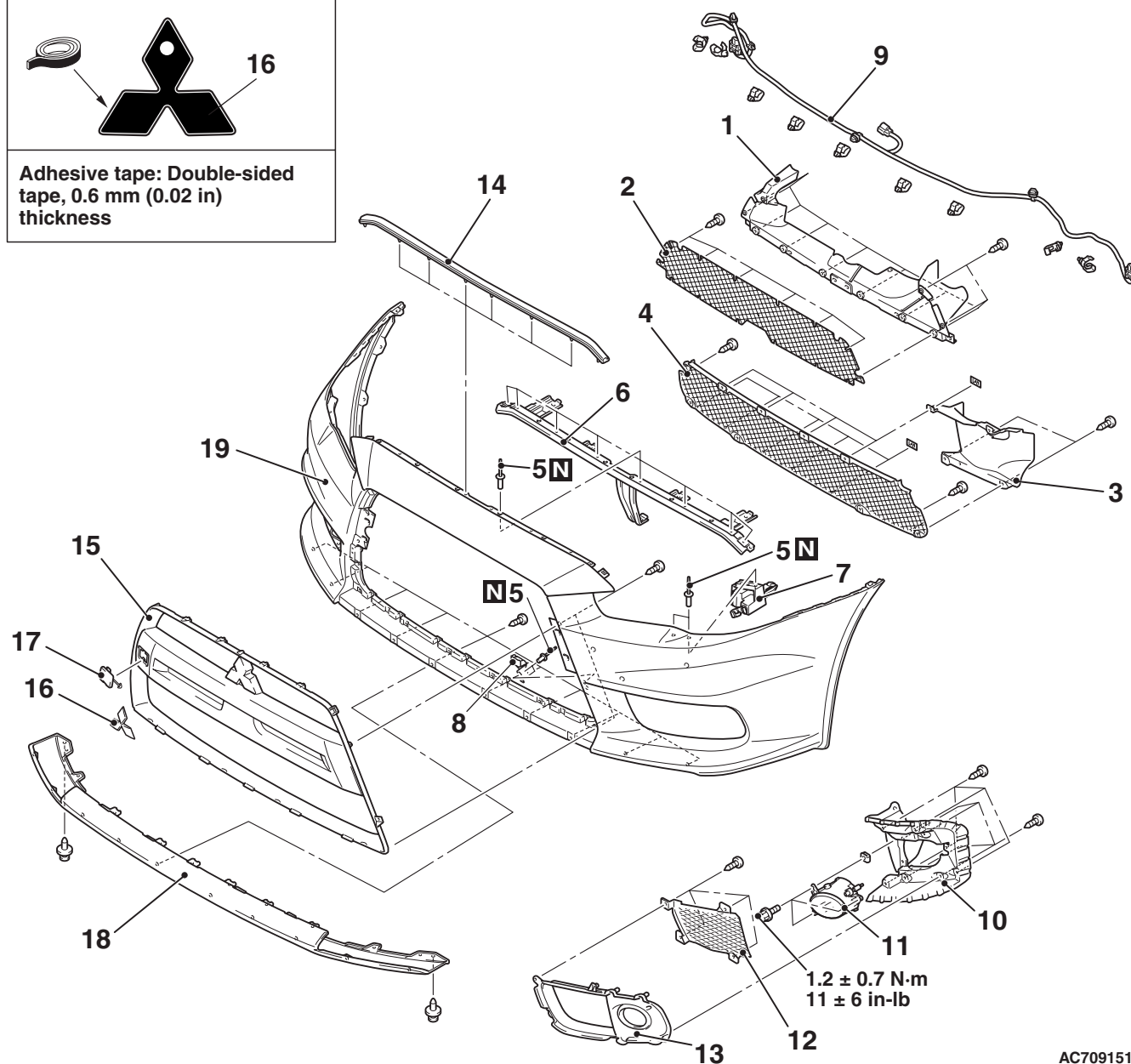
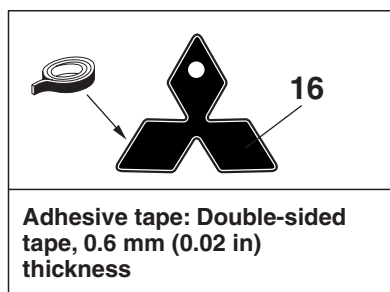
1. Headlight support panel cover
2. Front bumper harness connector connection
3. Front bumper and radiator grille assembly
4. Front bumper side bracket
 - Horn (Refer to GROUP 54A, Horn P.54A-268)

Removal steps (Continued)

- Ambient temperature sensor (Refer to GROUP 55, Ambient temperature sensor P.55-127)
- 5. Front bumper core
- 6. Front bumper reinforcement
- 7. Front bumper air guide center duct

DISASSEMBLY AND ASSEMBLY

M1511025500241



AC709151AB

Disassembly steps

- <<A>> >>A<<
1. Front bumper grille cover A
 2. Front bumper net A
 3. Front bumper grille cover B
 4. Front bumper net B
 5. Rivet
 6. Front bumper reinforcement
 7. Front bumper support
 8. Front bumper bracket
 9. Front fog light harness
 10. Front fog light bracket

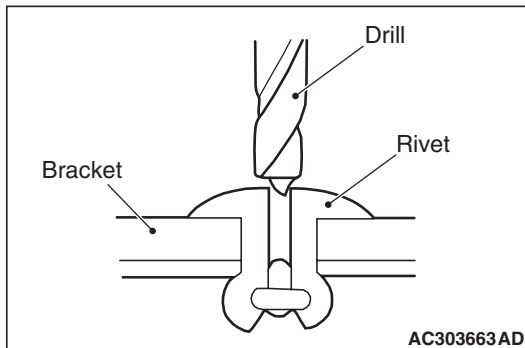
Disassembly steps (Continued)

11. Front fog light assembly
12. Front bumper net C
13. Front bumper garnish
14. Hood weather strip front
15. Radiator grille
16. Front three-diamond mark
17. Front bumper cover
18. Air dam skirt panel
19. Front bumper face

DISASSEMBLY SERVICE POINT

<<A>> RIVETS REMOVAL

Use a drill [ϕ 4.0 mm(0.16 in)] to make a hole in the rivet to break it, and remove the rivet.

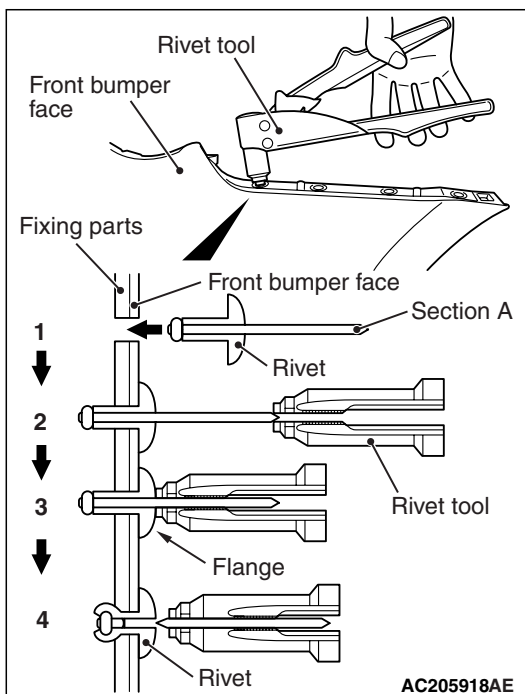


ASSEMBLY SERVICE POINT

>>A<< RIVETS INSTALLATION

Use a rivet tool to connect the parts with rivets by the following procedures.

1. Insert the rivet into a corresponding location.
2. Set the rivet tool at a section A of rivet.
3. While pushing the flange surface of the rivet onto parts to be fixed with the rivet tool, press the handle of the tool.
4. Thin part of section A of the rivet will be cut off and the parts is fixed in position.



REAR BUMPER ASSEMBLY

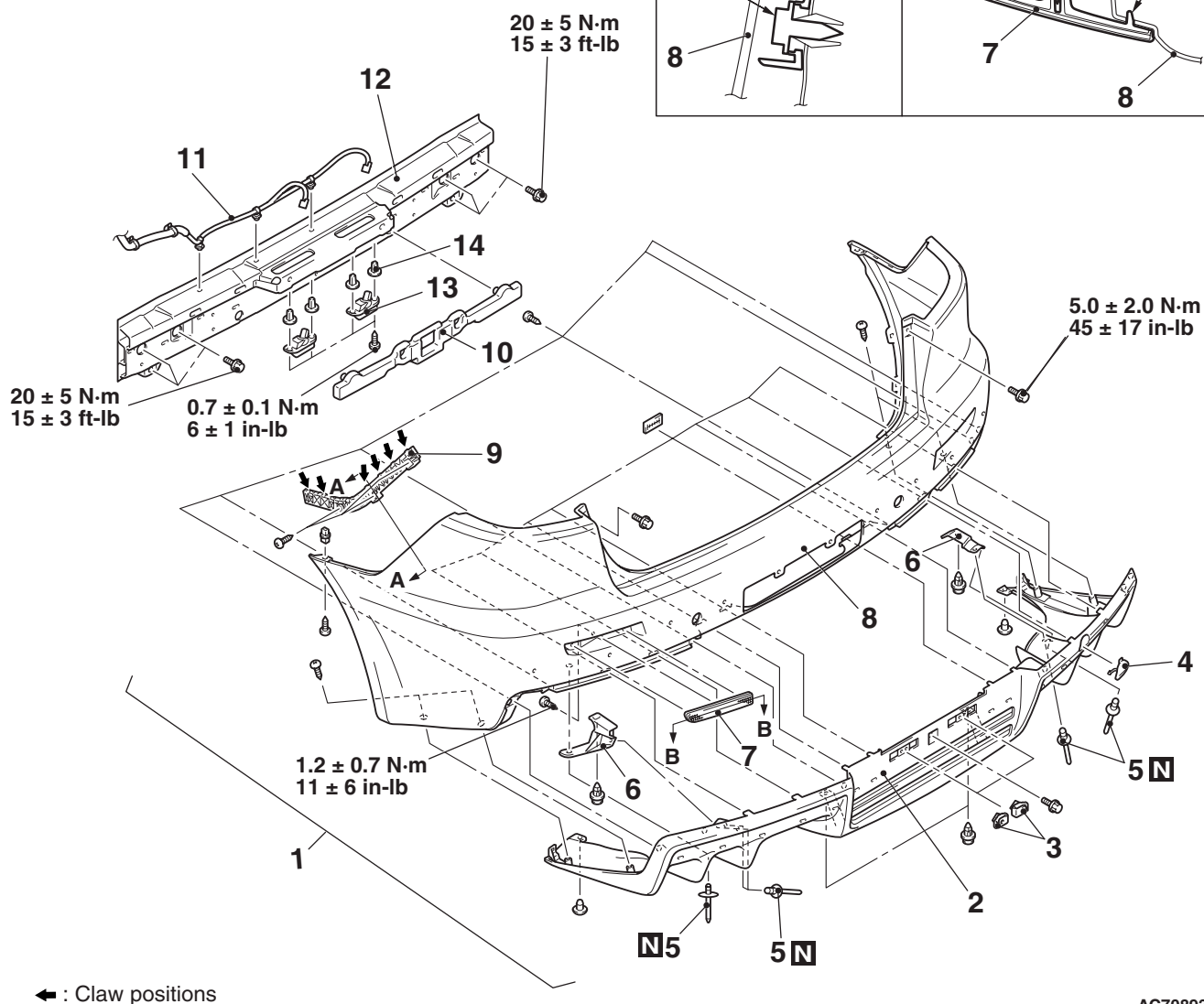
REMOVAL AND INSTALLATION

M1511001901680

Pre-removal and post-installation operation

- Rear splash shield removal and installation (Refer to GROUP 42A-Splash shield [P.42A-13](#))
- Rear combination light removal and installation (Refer to GROUP 54A-Rear combination light [P.54A-216](#))

<Vehicles for USA>



← : Claw positions

AC708933 AC

Removal steps

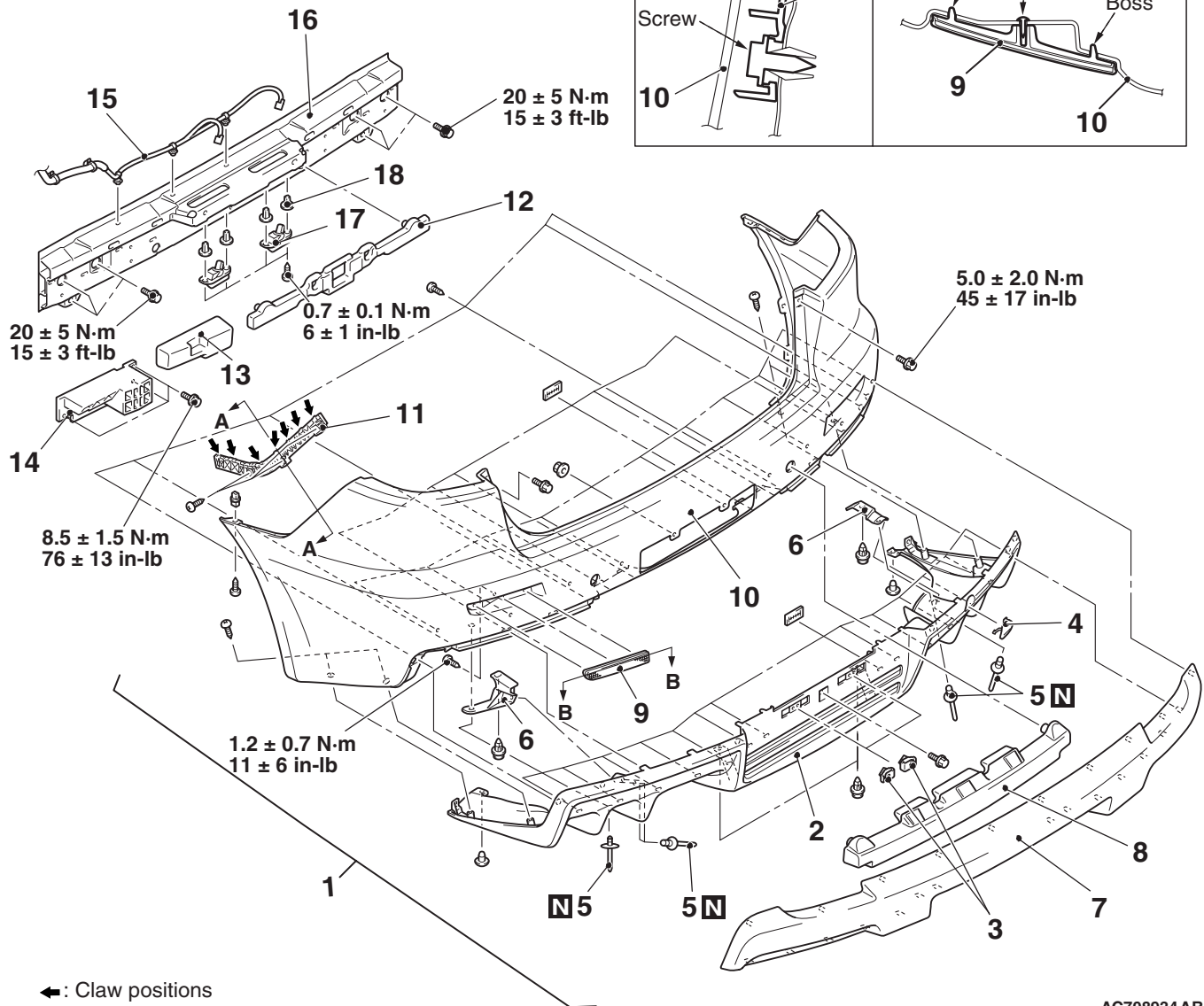
1. Rear bumper assembly
2. Rear bumper extension
3. License plate bracket
4. Rear bumper cover
5. Rivet
6. Rear bumper garnish side bracket
7. Reflector

Removal steps (Continued)

8. Rear bumper face
9. Rear bumper side bracket
10. Rear bumper core lower
11. Rear floor harness
12. Rear bumper reinforcement
13. License plate light assembly
14. Grommet

<<A>> >>A<<

<Vehicles for Canada>



AC708934 AB

Removal steps

1. Rear bumper assembly
2. Rear bumper extension
3. License plate bracket
4. Rear bumper cover
5. Rivet
6. Rear bumper garnish side bracket
7. Rear bumper protector
8. Rear bumper core center
9. Reflector

Removal steps (Continued)

10. Rear bumper face
11. Rear bumper side bracket
12. Rear bumper core lower
13. Rear bumper core upper
14. Rear bumper core side
15. Rear floor harness
16. Rear bumper reinforcement
17. License plate light assembly
18. Grommet

<<A>> >>A<<

REMOVAL SERVICE POINT

<<A>> RIVETS REMOVAL

Refer to FRONT BUMPER DISASSEMBLY SERVICE POINT (P.51-4).

INSTALLATION SERVICE POINT

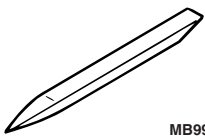
>>A<< RIVETS INSTALLATION

Refer to FRONT BUMPER ASSEMBLY SERVICE POINT
(P.51-4).

GARNISHES AND MOLDINGS

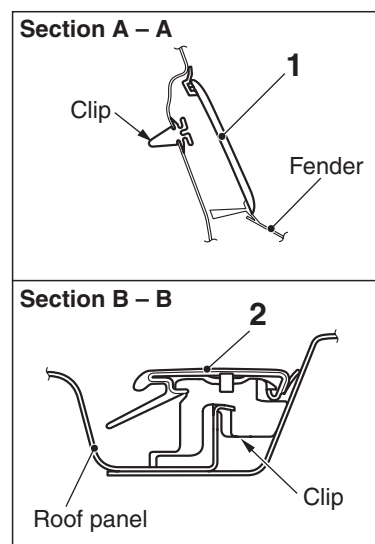
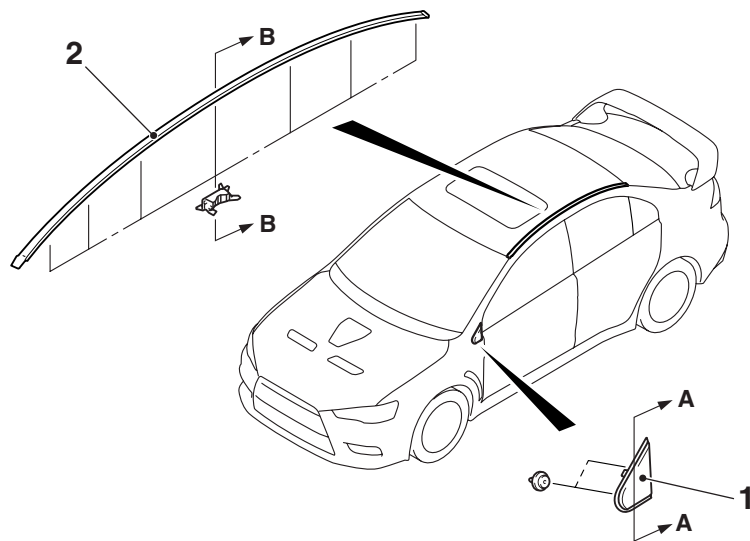
SPECIAL TOOL

M1511000602292

Tool	Tool number and name	Supersession	Application
 MB990784	MB990784 Ornament remover	General service tool	Removal of front delta garnish

REMOVAL AND INSTALLATION

M1511004700950



AC705108AB

<<A>>

Front delta garnish removal

1. Front delta garnish

<>

Roof drip molding removal

2. Roof drip molding

Required Special Tool:

- MB990784: Ornament remover

REMOVAL SERVICE POINTS

<<A>> FRONT DELTA GARNISH REMOVAL

Use the ornament remover (MB990784) to remove the front delta garnish.

<> ROOF DRIP MOLDING REMOVAL

The following tools are recommended for removing the roof drip moldings.

Ornament remover set (parts number JS000004)

1. Remover A (parts number JS000001)

2. Remover B (parts number JS000003)

3. Remover C (parts number JS000002)

INQUIRY:

Service Tool Group Manager

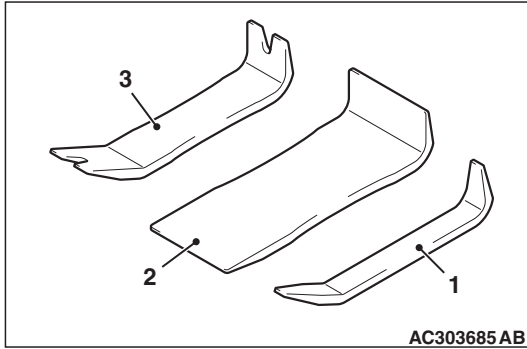
No.2 Sales Promotion Dept.

Marketing Operation Office

MITSUBISHI AUTOMOTIVE ACCESSORIES & PRODUCTS
CO.,LTD.

FAX: +81-3-5445-0846

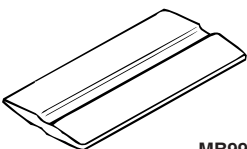
e-mail: sst.map@mitsubishi-motors.com



DOOR SASH TAPE

SPECIAL TOOL

M1511000602300

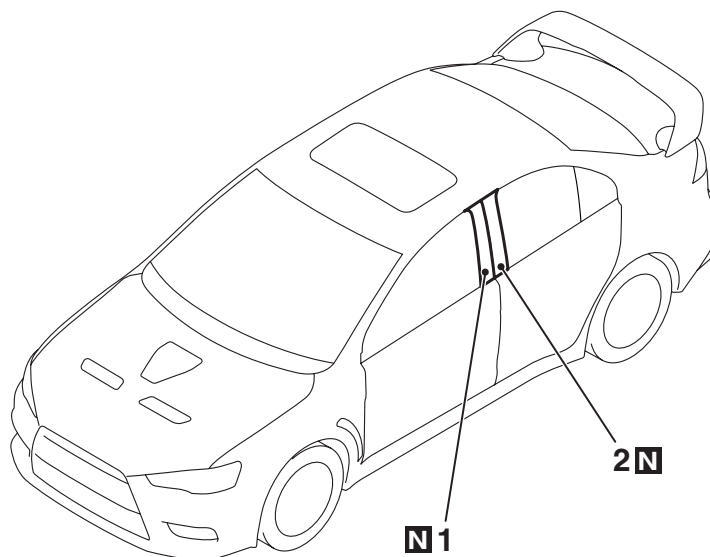
Tool	Tool number and name	Supersession	Application
 MB990528	MB990528 Stripe tape spatula	General service tool	Installation of door sash tape

REMOVAL AND INSTALLATION

M1511024100477

Pre-removal and post-installation operation

- Door Window Glass Runchannel Removal and Installation (Refer to GROUP 42A –Window Glass Runchannel and Door Opening Weatherstrip P.42A-136).
- Door Beltline Molding Removal and Installation (Refer to GROUP 42A –Window Glass Runchannel and Door Opening Weatherstrip P.42A-136).



AC706342AB

- Removal steps**
- <<A>> >>A<< 1. Front door sash tape, rear
<<A>> >>A<< 2. Rear door sash tape, front

Required Special Tool:

- MB990528: Stripe tape spatula

REMOVAL SERVICE POINT**<<A>> DOOR SASH TAPES REMOVAL****⚠ CAUTION**

Pay attention to keep from getting burned by hot door panel or tapes.

1. Use a hair drier to warm the tape.
2. Peel the tip of the tape with your finger, and then peel off the tape in parallel with the application surface.

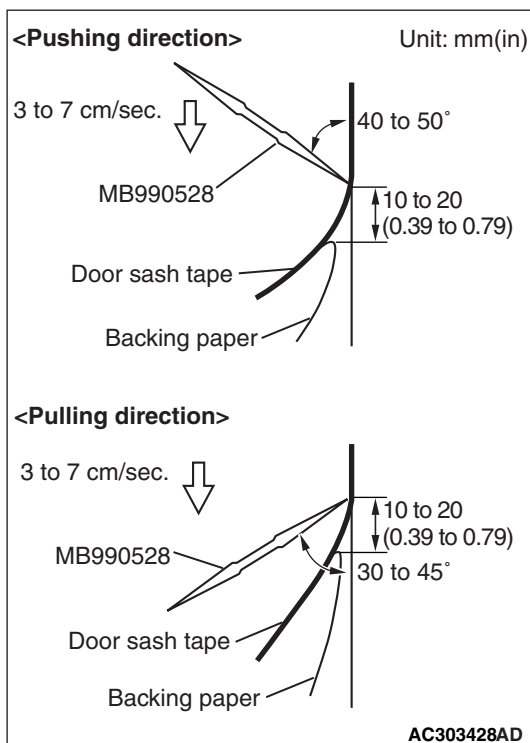
INSTALLATION SERVICE POINT

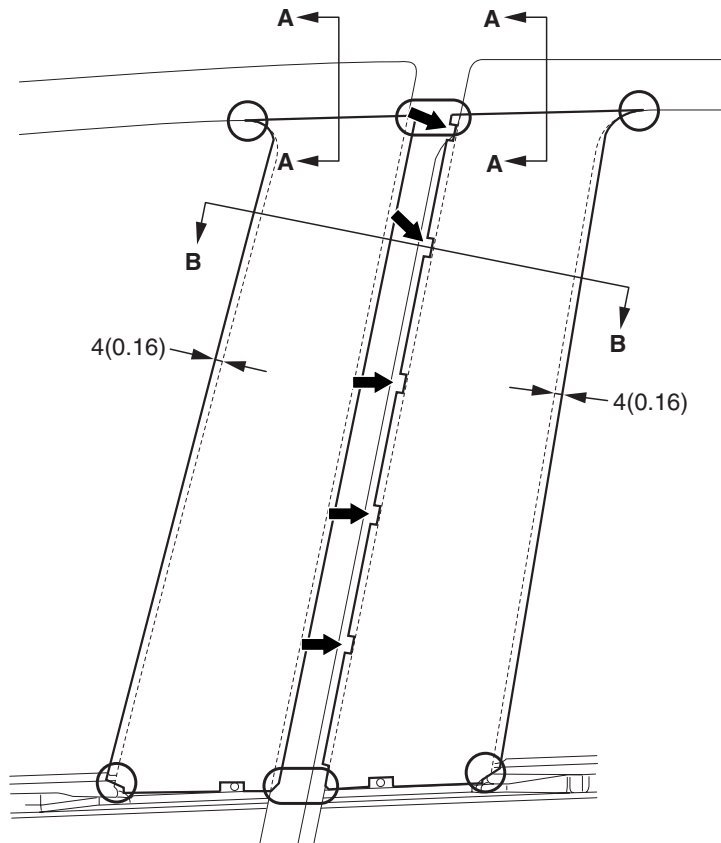
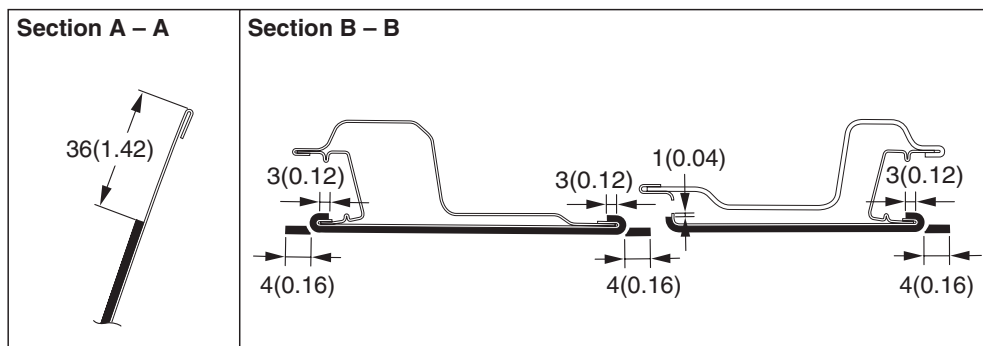
>>A<< DOOR SASH TAPES INSTALLATION

CAUTION

- The ambient temperature should be 20(68° F) to 30° C(86° F). Ensure that the working area is clean. Ideally, the tape application should be done at ambient temperature of 25° C(77° F).
- If ambient temperature is less than 15° C(59° F), heat the tape and application surface to a temperature of 20(68) to 30° C(86° F). If ambient temperature is 35° C(95° F) or higher, cool down them. The adhesive property of the tape is deteriorated at low temperature, so the tape may come loose easily. Meanwhile, it gets softened at hot temperature.
- When beginning to apply the tape, pay particular attention. If the end of the tape cannot be applied to the specified position with an accuracy of less than 1 mm(0.04 in), it may cause the poor appearance or adhesion.
- Use the special tool MB990528 to apply the tape with a steady pace and pressure. If you do not apply the tape with a steady pace or pressure, or abort the application, a shallow groove (lateral groove called as "Shock line") may be present on the tape surface. Meanwhile, if you apply it too quickly, air bubbles may be formed under the tape.

1. Wrap a soft cloth (synthetic fiber) around the tip of the special tool.
2. Use white gasoline to degrease the tape application surface.
3. Wipe away dirt from the tape.

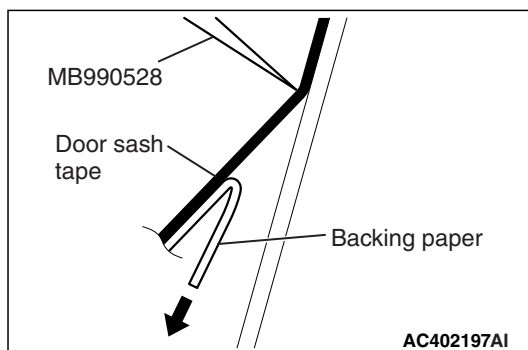




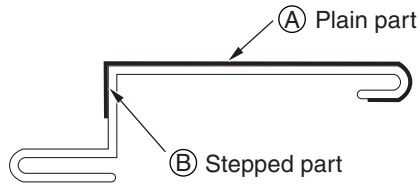
- : Tape locating points
- ➡ : T-stud for door opening weatherstrip attaching locations (5 places in all).

AC607960AB

4. Apply the door sash tape according to the procedure below.
- (1) Position the tape at the upper and lower locating points.
 - (2) Peel of backing strip from the top of the tape and attach it temporarily.
 - (3) Peel off the backing strip to the half length of the tape.
 - (4) Apply the tape using the special tool while peeling off the remaining backing strip.

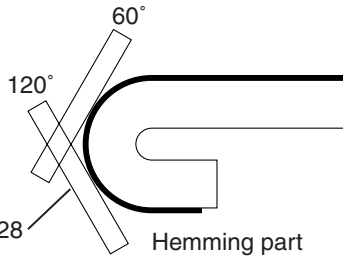


Rear door sash tape front



AC506860AB

- (5) For the rear door sash tape, apply it to plain surface (A). Then apply it to stepped surface (B).



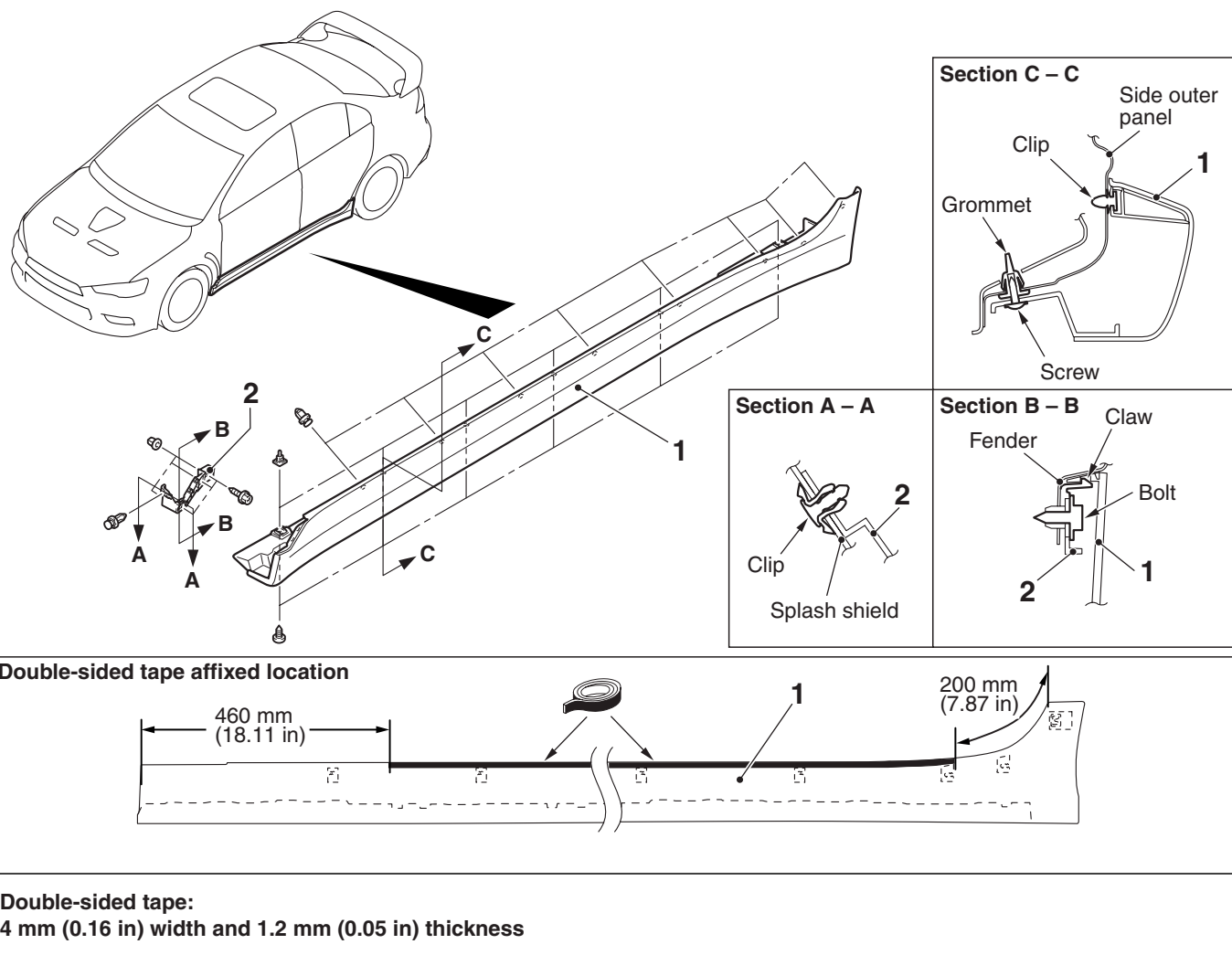
AC303415AG

- (6) Press the folded area of the tape by three stages (60°, 120°, and holding), rolling in toward the vehicle inside direction.

SIDE AIR DAM

REMOVAL AND INSTALLATION

M1511005500562



AC704709AC

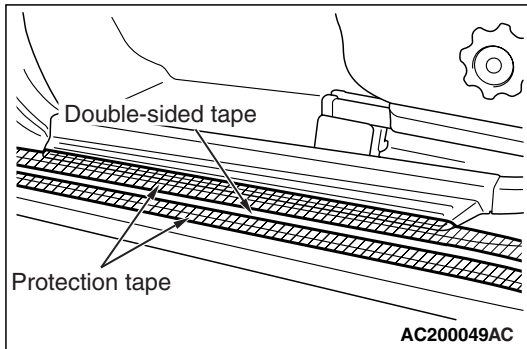
Removal steps

- <<A>> >>A<<
1. Side air dam
 2. Side air dam bracket

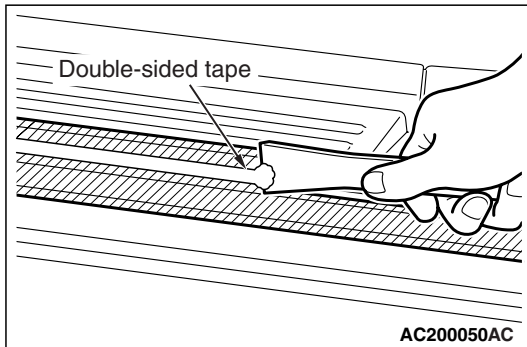
REMOVAL SERVICE POINT**<<A>> SIDE AIR DAM REMOVAL**

Gently lift and remove the side air dam. If there is any double-sided tape remaining on the side air dam, remove according to the following instructions.

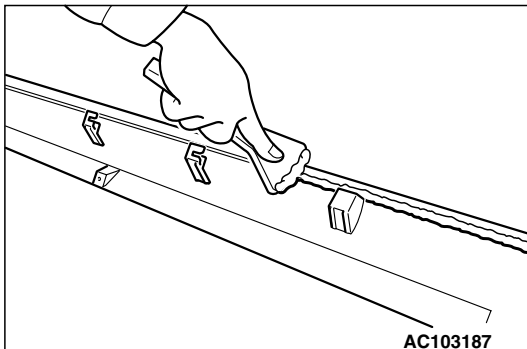
1. Remove double-sided tape remaining on the body surface (when replacing side air dam).



- (1) Attach protection tape all the way along the edges of the double-sided tape which is still adhering to the body.



- (2) Scrape off the double-sided tape with a resin spatula as possible.
- (3) Peel off the protection tape.
- (4) Use a shop towel moistened with 3M™ AAD Part number 8906 or equivalent to wipe the body.



2. Remove double-sided tape remaining on side air dam and adhere double-sided tape (when re-using side air dam).
 - (1) Scrape off the double-sided tape on the side air dam with a resin spatula as much as possible.
 - (2) Wipe the side air dam surface and clean it with a rag moistened with isopropyl alcohol.
 - (3) Remove only a small amount of the residual adhesive.
 - (4) Apply the primer as specified on the residual adhesive.
 - (5) Adhere the double-sided tape as specified on the side air dam.

INSTALLATION SERVICE POINT

>>A<< SIDE AIR DAM INSTALLATION

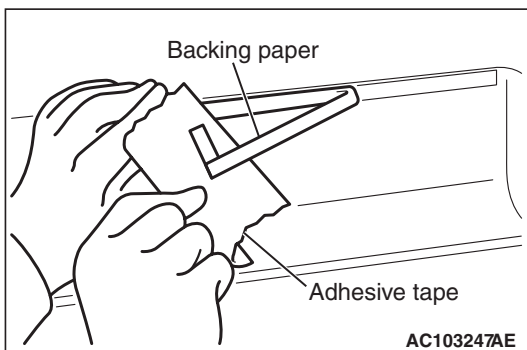
1. Tear off the double-sided tape backing paper.

NOTE: Attach the adhesive tape to the edge of the backing paper makes the backing paper tear off easier.

2. Install the side air dam.

NOTE: If the double-sided tape is difficult to affix in cold temperature, etc., warm the bonding surfaces of the body and side air dam to about 40 –60°C (104 –140°F) before affixing the tape.

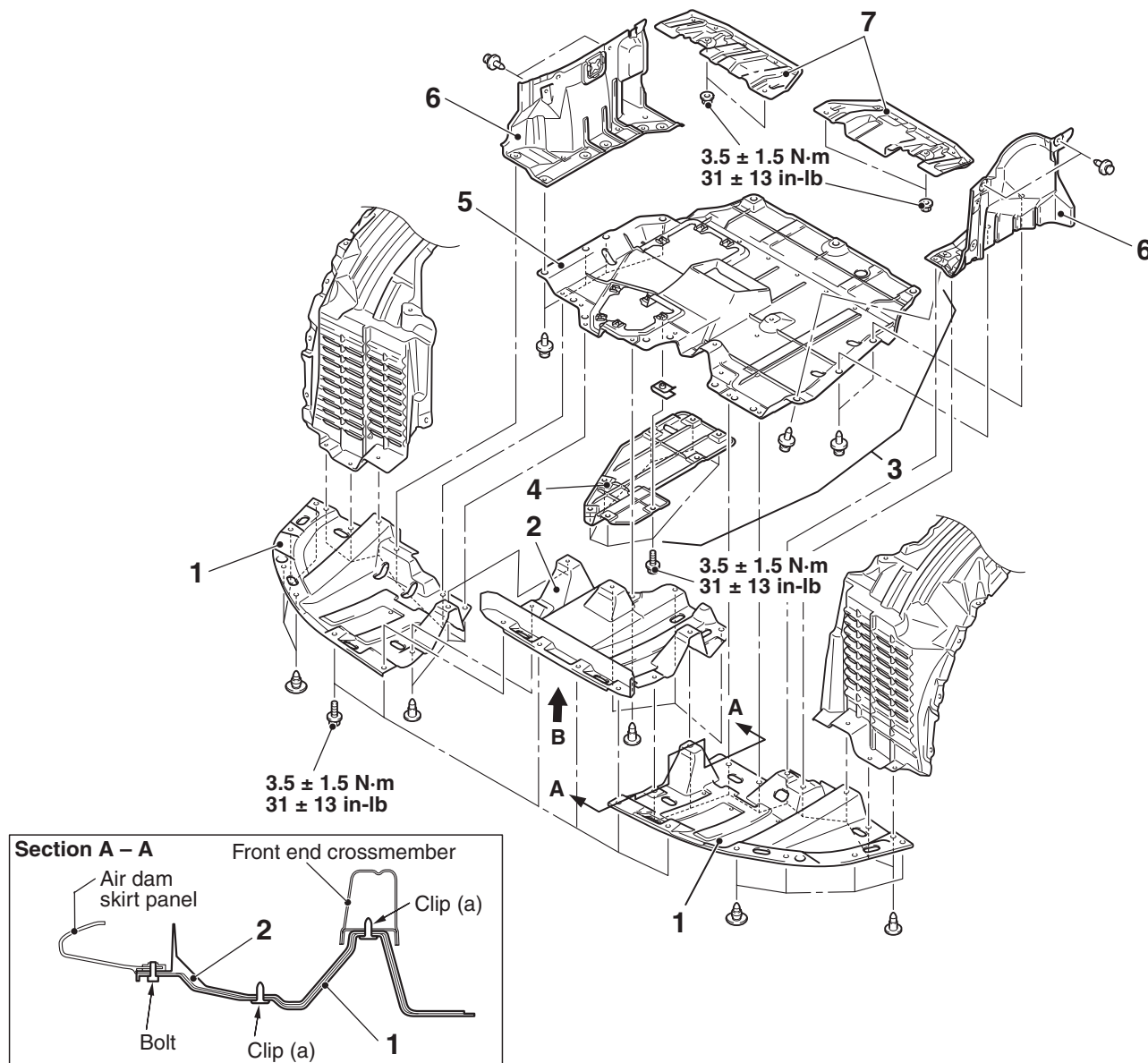
3. Firmly press in the side air dam.



UNDER COVER

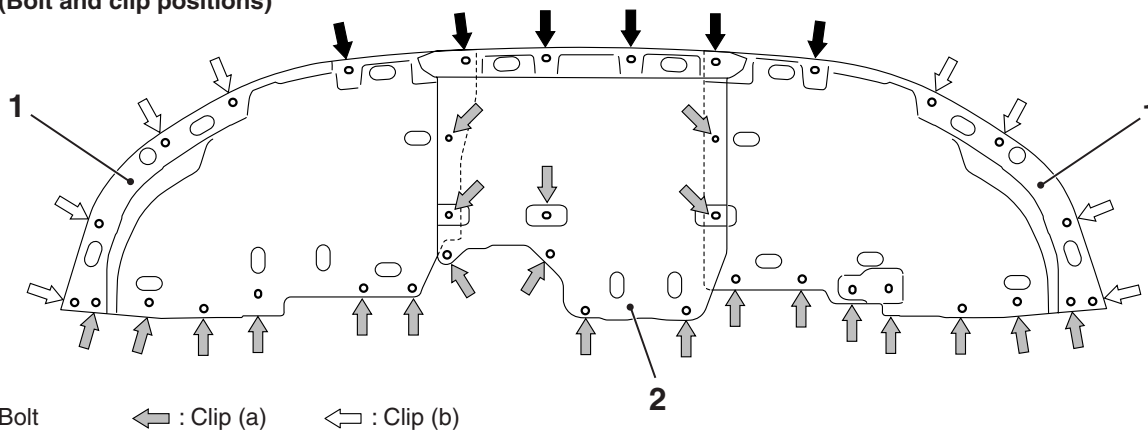
REMOVAL AND INSTALLATION

M1511019600443



AC704710AB

View B (Bolt and clip positions)



AC705190AB

Removal steps

1. Engine room under cover front A
2. Engine room under cover center
3. Engine room under cover front B assembly
4. Engine room under cover extension

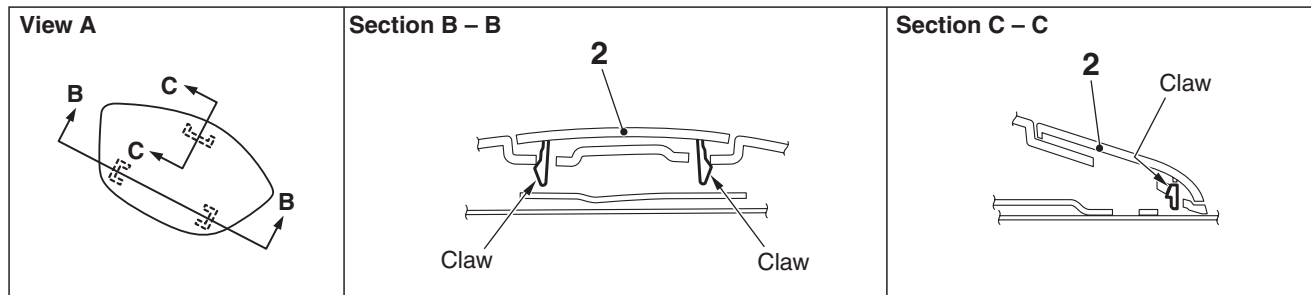
Removal steps (Continued)

5. Engine room under cover front B
6. Engine room side cover
7. Front floor under cover

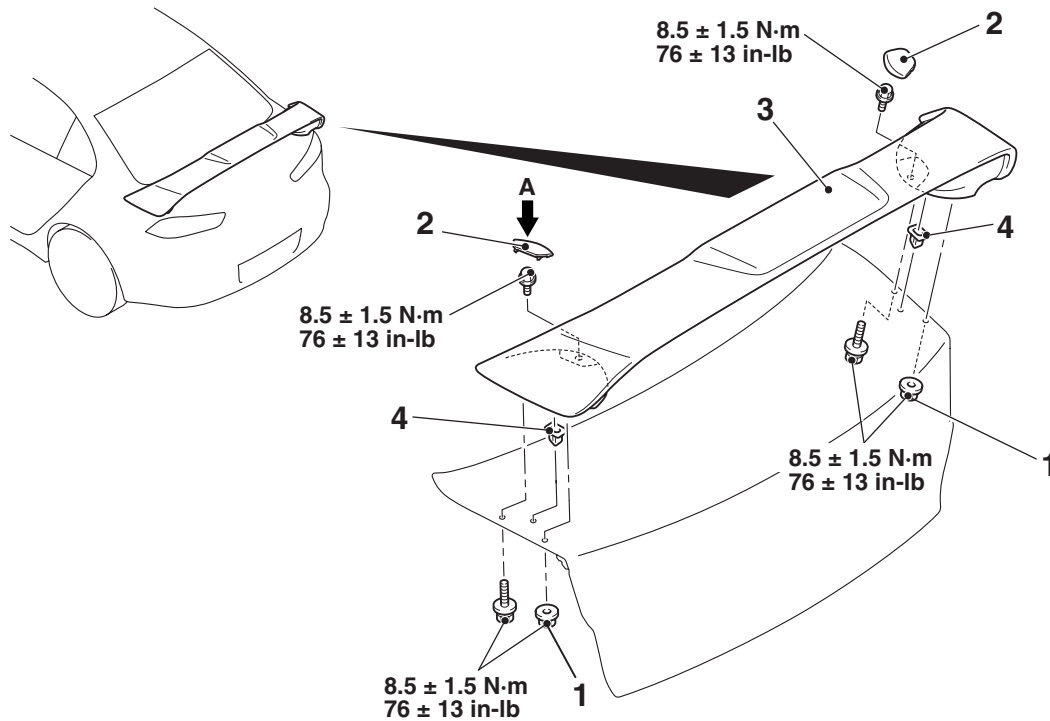
REAR SPOILER

REMOVAL AND INSTALLATION

M1511006100318



<MR>



AC708776 AD

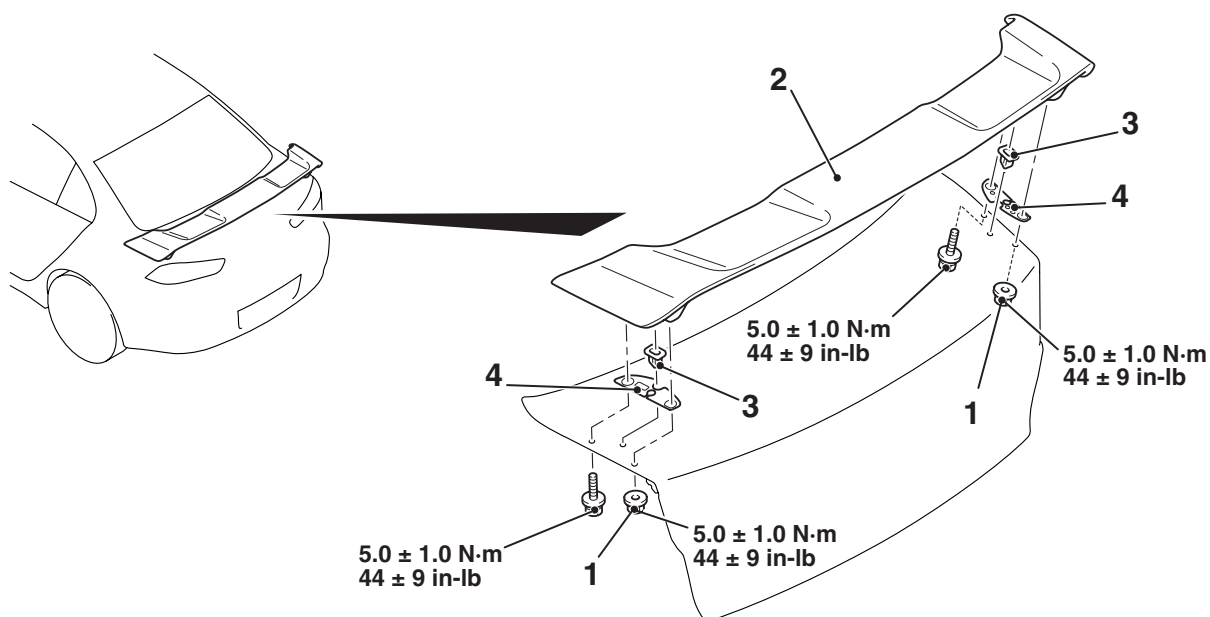
Removal steps

1. Nut
2. Cap

Removal steps (Continued)

3. Rear spoiler assembly
4. Grommet

<GSR>



AC705243 AE

Removal steps

1. Nut
2. Rear spoiler assembly

Removal steps (Continued)

3. Grommet
- >>A<< 4. Tape

INSTALLATION SERVICE POINT**>>A<< TAPE INSTALLATION**

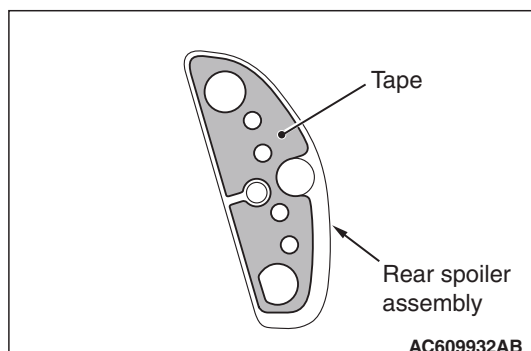
1. Installation position

⚠ CAUTION

- The ambient temperature should be 15 (59) to 40° C (104° F). Ensure that the working area is clean.
- If ambient temperature is less than 15° C (59° F), heat the tape and application surface to a temperature of 15 (59) to 40° C (104° F).
- Be careful that air bubbles are not formed under the tape.

2. Installation procedure

- (1) Wipe the tape application surface and clean it with a rag moistened with isopropyl alcohol.
- (2) Remove backing paper from the tape, and apply it to the rear spoiler assembly.



AC609932AB

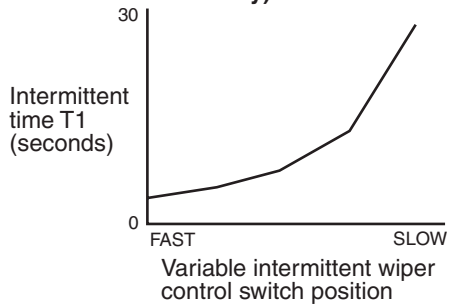
WINDSHIELD WIPER AND WASHER

GENERAL INFORMATION

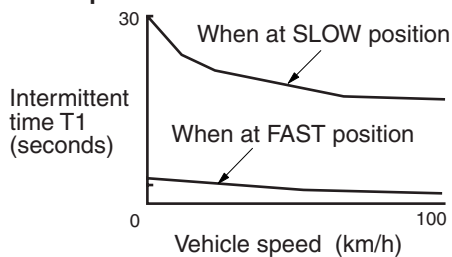
M1511000101540

WINDSHIELD WIPER AND WASHER OPERATION

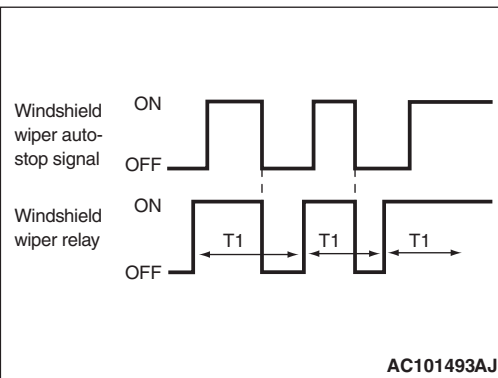
Change in intermittent time by variable intermittent wiper control switch (when vehicle is stationary)



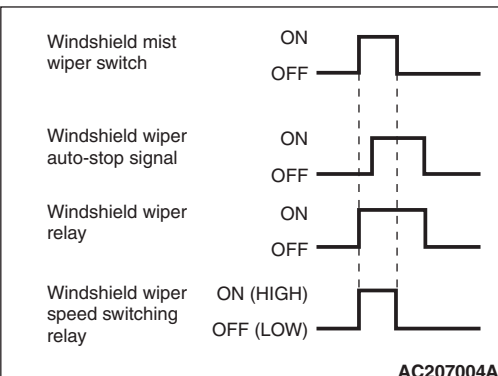
Change in intermittent time according to vehicle speed



AC101502 AC



AC101493AJ



AC207004AI

Intermittent control (Vehicle speed-dependent variable type) <Initial condition: with function>

1. ETACS calculates the windshield intermittent wiper interval T1 from the position of the windshield intermittent wiper switch on the column switch and the vehicle speed signal (sent from the combination meter to ETACS via CAN communication).

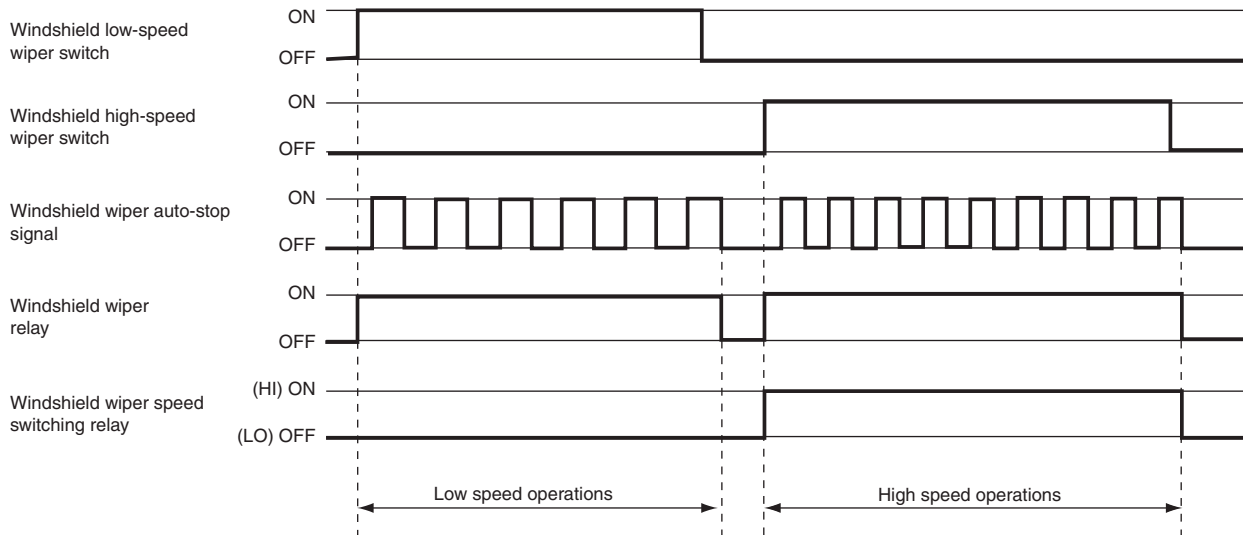
NOTE: Using the configuration function, the vehicle speed-dependent intermittent function can be invalidated (Refer to P.51-65).

2. When ETACS receives the ON signal of the windshield intermittent wiper switch, it turns the windshield wiper relay ON. When the wiper reaches the stop position, the windshield wiper auto-stop signal turns OFF, and the windshield wiper relay turns OFF. When the intermittent time T1 calculated by step 1 has elapsed after the windshield wiper relay ON, the windshield wiper relay turns ON again, and the above-mentioned operation is repeated.

Mist wiper control

When the windshield wiper mist switch on the column switch is turned ON while the ignition switch is in ACC or ON position, the column switch turns the windshield wiper relay ON. At the same time, the wiper speed switching relay turns ON (HI). When the windshield mist wiper switch is ON, the windshield wiper operates at high speed.

Low speed wiper and high speed wiper control

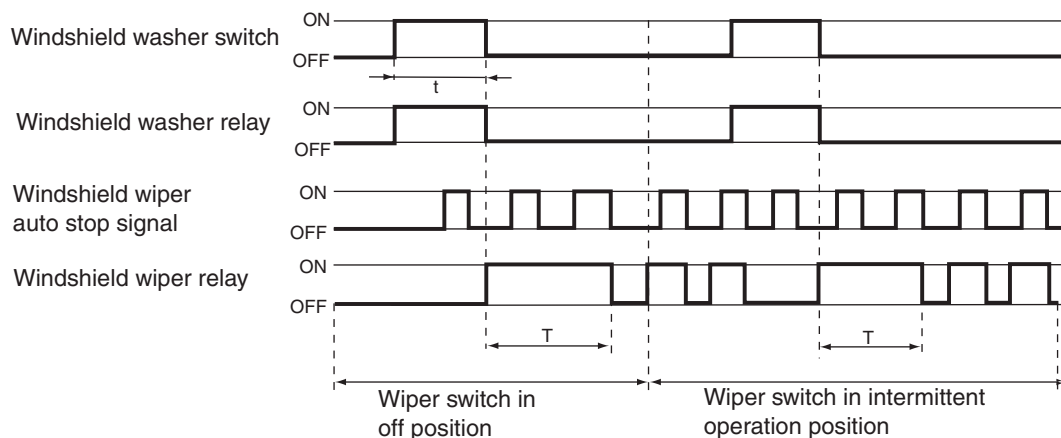


AC506610AH

When the windshield low speed wiper switch on the column switch is turned ON while the ignition switch is in ACC or ON position, the column switch turns the windshield wiper relay ON. Also, the wiper speed switching relay turns OFF (LO), and the windshield wiper operates at low speed.

When the windshield high speed wiper switch is turned ON, the windshield wiper relay turns ON. Also, the wiper speed switching relay turns ON (HI), and the windshield wiper operates at high speed.

Windshield wiper linked with washer function <Initial condition: with function>



AC707462 AB

Wiper switch	OFF position AUTO position			Low-speed or high-speed operation position
Washer switch ON time (t)	0.3 second or less	0.3 to 0.5 second	0.5 second or more	-
Intermittent wiper time (T)	0 second	2 seconds	3 seconds	3 seconds

When the windshield washer switch on the column switch is turned ON while the ignition switch is in ACC or ON position, ETACS turns the windshield washer relay ON.

When the windshield washer switch is kept ON for 0.3 seconds or longer, the windshield wiper relay (the wiper relay output time varies depending on the conditions. For details, see the table.) is turned ON, and the windshield wiper operates at high speed. The windshield wiper is turned OFF with 3 seconds delay after the windshield washer switch is turned OFF. Even when the windshield washer switch is turned ON while the windshield wiper is operating intermittently, the intermittent action starts again after the linked operation is finished.

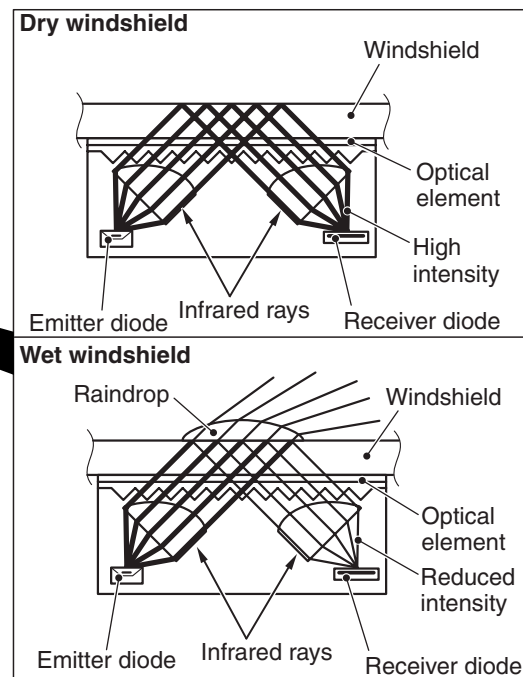
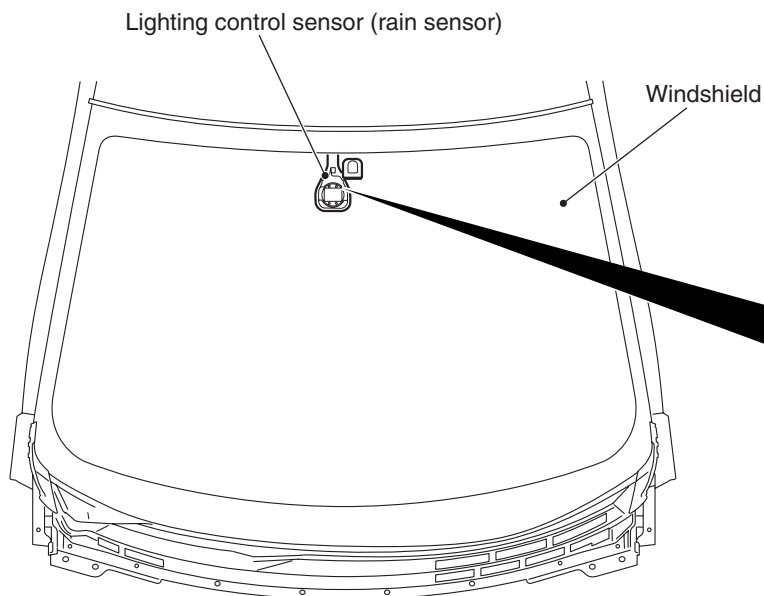
If the ignition switch is turned to ACC position while the windshield washer switch is ON, the windshield washer relay turns ON, but the windshield wiper does not perform the linked operation. When the windshield washer switch is turned OFF and then ON, the windshield wiper starts the linked operation.

NOTE:

- Using the configuration function, the washer linked windshield wiper function can be invalidated (Refer to P.51-65).
- Using the configuration function, when the washer linked windshield wiper function is invalidated, only the washer operates. It is useful to melt down the ices of the frozen windshield.

Rain sensitive wiper function

<Optional: Vehicles for USA with lighting control sensor (rain sensor)>



AC708778AB

- Lighting control sensor (rain sensor) has been installed in the upper part of the windshield to sense the raindrops and windshield wiper can be operated when the ignition switch is at ON and wiper switch is at AUTO position.
- The lighting control sensor detects the raindrops on the windshield surface using the optical element, and it automatically switches the windshield wiper operation, depending on the amount of rainfall, to the intermittent or LO/HI operation.

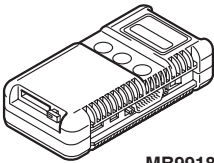
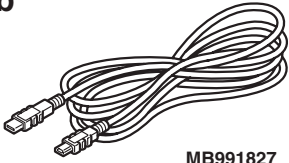
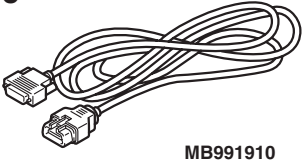
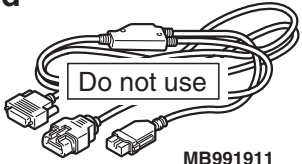
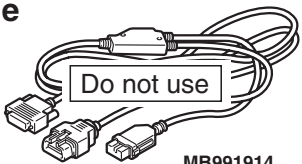
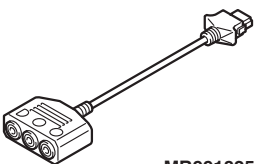
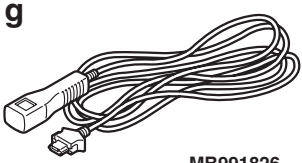
- The lighting control sensor detects the raindrops using the reflections of infrared rays, and depending on the amount of rainfall, it automatically adjusts the wiping speed.
- Dry windshield: All infrared rays emitted from the emitter diode are reflected by the windshield and directed to the receiver diode as they are.

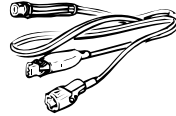
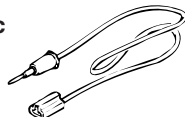
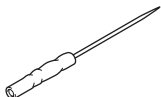
- Wet windshield: Part of the infrared rays emitted from the emitter diode are transmitted to outside of the windshield through the raindrops, and the infrared rays with reduced intensity are directed to the receiver diode.

NOTE: Using the configuration function, the rain sensitive wiper function can be invalidated (Refer to P.51-65).

SPECIAL TOOLS

M1511000602214

Tool	Tool number and name	Supersession	Application
a  MB991824 b  MB991827 c  MB991910 d  MB991911 e  MB991914 f  MB991825 g  MB991826 MB991958	MB991958 a. MB991824 b. MB991827 c. MB991910 d. MB991911 e. MB991914 f. MB991825 g. MB991826 M.U.T.-III sub assembly a. Vehicle communication interface (V.C.I.) b. M.U.T.-III USB cable c. M.U.T.-III main harness A (Vehicles with CAN communication system) d. M.U.T.-III main harness B (Vehicles without CAN communication system) e. M.U.T.-III main harness C (for Daimler Chrysler models only) f. M.U.T.-III measurement adapter g. M.U.T.-III trigger harness	MB991824-KIT <i>NOTE: G: MB991826 M.U.T.-III Trigger Harness is not necessary when pushing V.C.I. ENTER key.</i>	⚠ CAUTION M.U.T.-III main harness A (MB991910) should be used. M.U.T.-III main harness B and C should not be used for this vehicle. Windshield wiper intermittent time check

Tool	Tool number and name	Supersession	Application
a  b  c  d 	MB991223 a. MB991219 b. MB991220 c. MB991221 d. MB991222 Harness set a. Test harness b. LED harness c. LED harness adaptor d. Probe	General service tools	Continuity check and voltage measurement at harness wire or connector for loose, corroded or damaged terminals, or terminals pushed back in the connector. a. Connector pin contact pressure inspection b. Power circuit inspection c. Power circuit inspection d. Commercial tester connection
 MB992006	MB992006 Extra fine probe	—	Making voltage and resistance measurement during troubleshooting

WINDSHIELD WIPER AND WASHER DIAGNOSIS

STANDARD FLOW OF DIAGNOSTIC TROUBLESHOOTING

M1511014600318

Refer to GROUP 00 –Contents of Troubleshooting

[P.00-7](#).

TROUBLE SYMPTOM CHART

M1511015000762

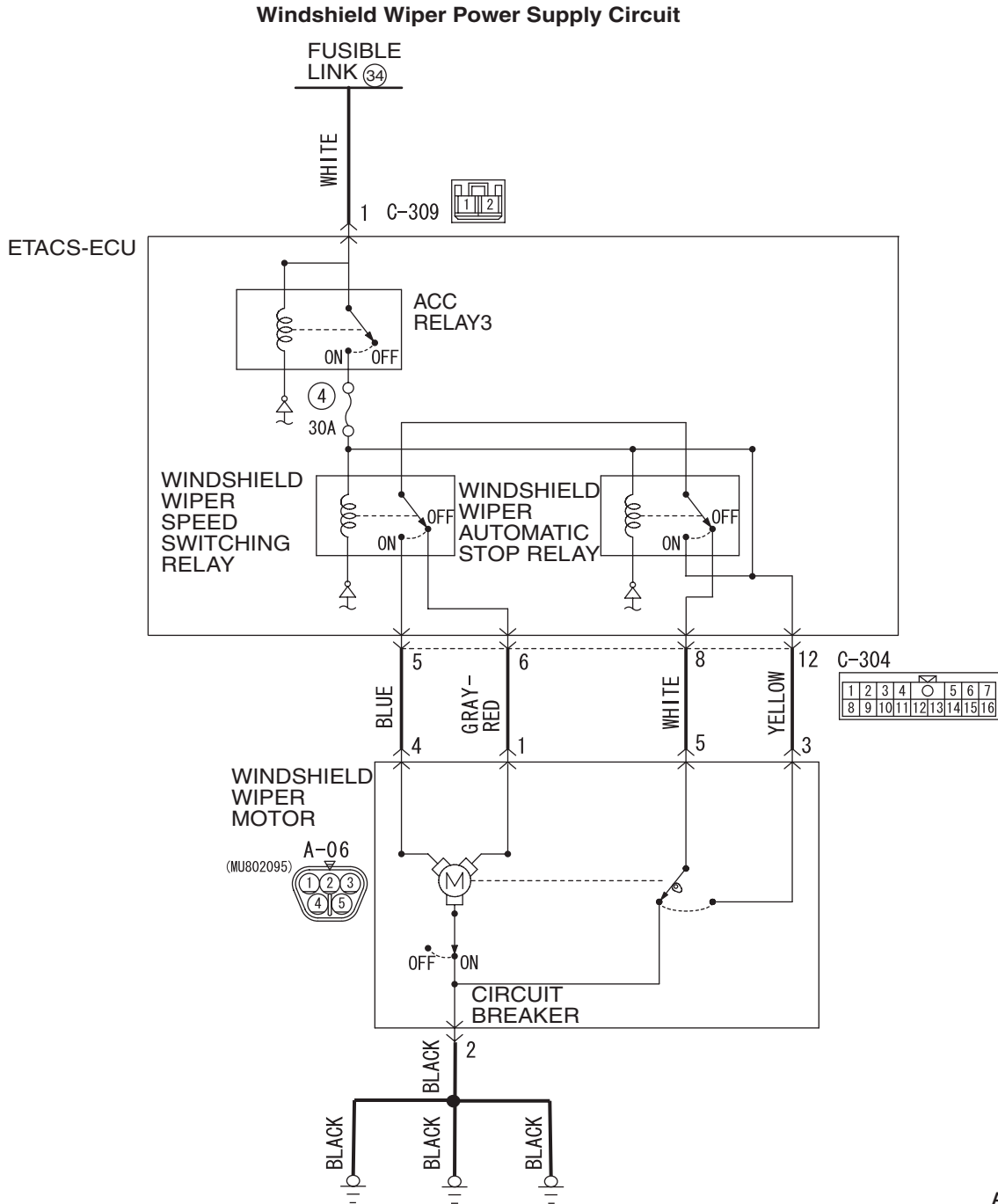
TROUBLE SYMPTOM	Inspection procedure No.	Reference page
The windshield wipers do not work at all.	1	P.51-25
The windshield wipers do not work when the wiper switch is at "INT", "Washer" or "Mist" position. However, wipers work at low speed when switch is at "Lo" or "Hi" position.	2	P.51-32
Windshield wipers do not stop at specified park position.	3	P.51-35
Windshield wipers do not work normally.	4	P.51-38
The windshield intermittent wiper interval cannot be adjusted by operating the windshield intermittent wiper volume control switch.	5	P.51-44
The windshield intermittent wiper interval is not changed according to the vehicle speed.	6	P.51-46
The rain sensitive AUTO wiper function does not work at all <Vehicles with lighting control sensor>.	7	P.51-48
The rain sensitive AUTO wiper function works even though there is no rainfall <Vehicles with lighting control sensor>.	8	P.51-53

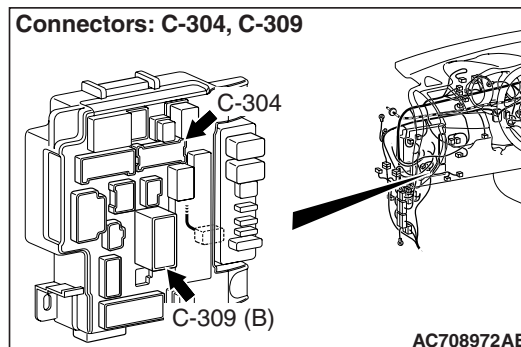
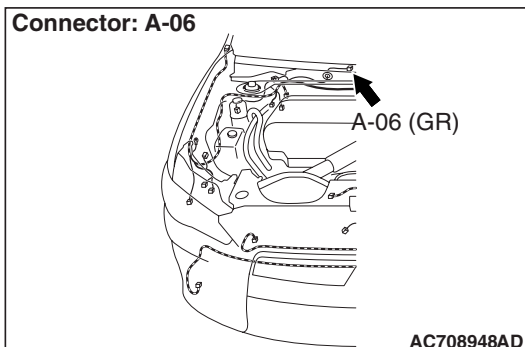
TROUBLE SYMPTOM	Inspection procedure No.	Reference page
Sometimes the rain sensitive AUTO wiper function works even though there is no rainfall <Vehicles with lighting control sensor>.	9	P.51-56
The windshield washer does not work normally.	10	P.51-59

NOTE: Even when the ETACS-ECU has failed, the windshield wipers can work at low speed as fail-safe mode.

SYMPTOM PROCEDURES

INSPECTION PROCEDURE 1: The windshield wipers do not work at all.





CIRCUIT OPERATION

- The windshield wiper and washer switch sends a signal through the column-ECU (incorporated in the column switch) to the ETACS-ECU. If the column-ECU sends a windshield wiper and washer switch "ON" signal to the ETACS-ECU, the ETACS-ECU turns on the relay (incorporated in the ETACS-ECU), thus causing the windshield wiper and washer motor to be turned on.
- If the LIN communication line is defective, the ETACS-ECU operates windshield wiper motor by using the other communication lines (wiper backup circuit) instead of that line. In this case, the windshield wiper works at low speed regardless of the windshield wiper and washer switch positions ("LO" or "HIGH").

TECHNICAL DESCRIPTION (COMMENT)

If the windshield wiper does not work at all, the windshield wiper motor, column switch (windshield wiper and washer switch) or the ETACS-ECU may be defective.

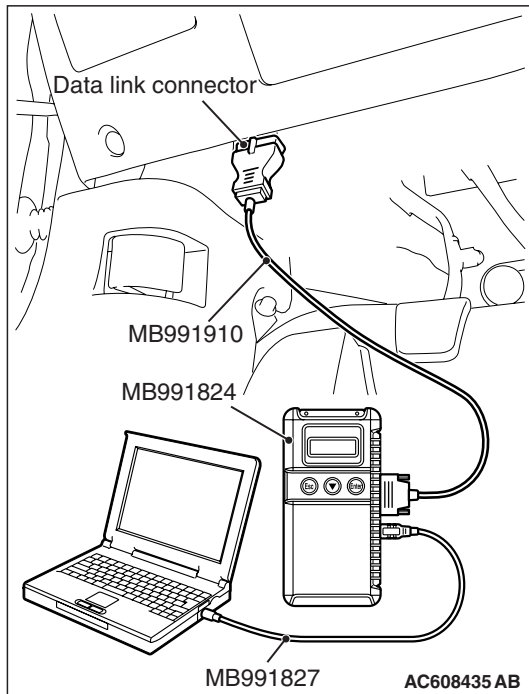
TROUBLESHOOTING HINTS

- Trouble in input signal system
- The wiring harness or connectors may have loose, corroded, or damaged terminals, or terminals pushed back in the connector
- The wiper motor may be defective
- The column switch may be defective
- The ETACS-ECU may be defective

DIAGNOSIS

Required Special Tools:

- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A



STEP 1. Use the scan tool MB991958 to check if an ETACS-related diagnostic trouble code is set.

Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the ETACS-ECU. Refer to GROUP 54A ETACS, "Diagnostic trouble code chart [P.54A-582](#)."

NO : Go to Step 2.

STEP 2. Using scan tool MB991958, check data list.

Check the input signal related to the windshield wiper. operation.

- Ignition switch: ACC
- Operate the windshield wiper switch at each switch position.

Item No.	Item name	Windshield wiper switch position	Normal condition
Item 235	Front wiper ACT	LO	ON
		HI	
		INT	ON and OFF
		MIST	ON
Item 288	ACC switch		ON

OK: Normal condition is displayed.

Q: Is the check result normal?

YES <Normal conditions are displayed for all items> :

Go to Step 3.

NO <Normal condition is not displayed for item No. 235>

: Troubleshoot the ETACS-ECU. Refer to ETACS, Diagnosis - Inspection Procedure 12 "ETACS-ECU does not receive any signal from the column switch signal." [P.54A-639](#).

NO <Normal condition is not displayed for item No. 288>

: Refer to GROUP 54A, ETACS, Diagnosis – Inspection Procedure 1 "The ignition switch (ACC) signal is not received" [P.54A-640](#).

STEP 3. Check windshield wiper motor connector A-06 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is windshield wiper motor connector A-06 in good condition?

YES : Go to Step 4.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Verify that the windshield wiper works normally.

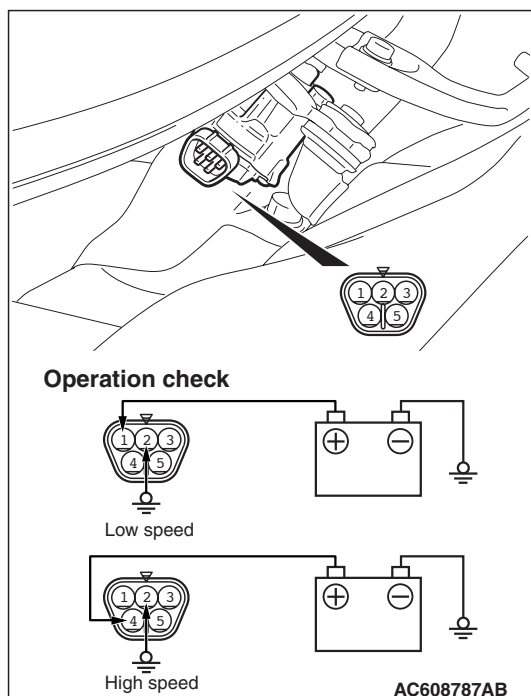
STEP 4. Check the windshield wiper motor.

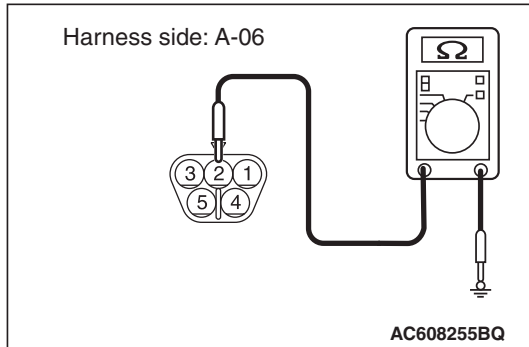
- (1) Disconnect windshield wiper motor connector A-06.
- (2) Connect a battery to the windshield wiper motor as shown. Then check that the windshield wiper motor operates normally at high and low speeds.

Q: Does the windshield wiper motor operate normally?

YES : Go to Step 5.

NO : Replace the windshield wiper motor. Verify that the windshield wiper works normally.





STEP 5. Check the ground circuit to the windshield wiper motor. Measure the resistance at the windshield wiper motor connector A-06.

- (1) Disconnect windshield wiper motor connector A-06 and measure the resistance available at the wiring harness side of the connector.
- (2) Measure the resistance value between terminal 2 and ground.
 - The resistance should be 2 ohms or less.

Q: Is the measured resistance 2 ohms or less?

YES : Go to Step 7.

NO : Go to Step 6.

STEP 6. Check the wiring harness between windshield wiper motor connector A-06 (terminal 2) and ground.

Q: Is the wiring harness between windshield wiper motor connector A-06 (terminal 2) and ground in good condition?

YES : No action is necessary and testing is complete.

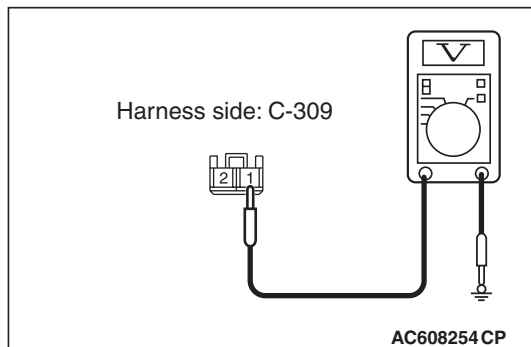
NO : The wiring harness may be damaged or the connector(s) may have loose, corroded or damaged terminals, or terminals pushed back in the connector. Repair the wiring harness as necessary. Verify the windshield wiper works normally.

STEP 7. Check ETACS-ECU connector C-309 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is ETACS-ECU connector C-309 in good condition?

YES : Go to Step 8.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Verify that the windshield wiper works normally.



STEP 8. Measure the voltage at ETACS-ECU connector C-309.

- (1) Disconnect ETACS-ECU connector C-309 and measure the resistance available at the wiring harness side of the connector.
- (2) Measure the voltage between terminal 1 and ground.
 - The voltage should measure approximately 12 volts (battery positive voltage).

Q: Is the measured voltage approximately 12 volts (battery positive voltage)?

YES : Go to Step 10.

NO : Go to Step 9.

STEP 9. Check the wiring harness between ETACS-ECU connector C-309 (terminal 1) and the fusible link (34).

- Check the power supply line for open circuits.

Q: Is the wiring harness between ETACS-ECU connector C-309 (terminal 1) and the fusible link (34) in good condition?

YES : Intermittent malfunction. Refer to GROUP 00, How to cope with intermittent malfunction [P.00-15](#).

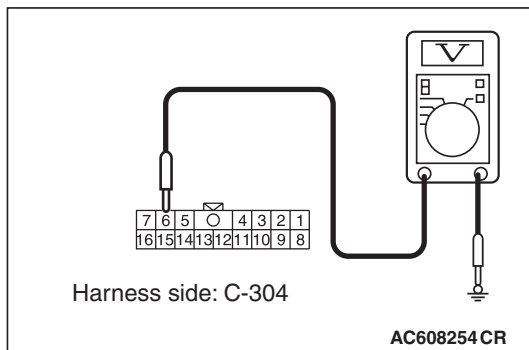
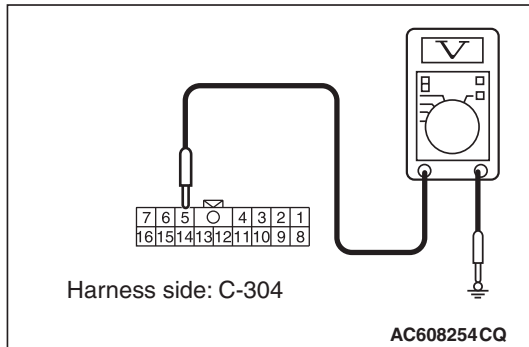
NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Cables and wire check [P.00E-12](#).

STEP 10. Check ETACS-ECU connector C-304 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is ETACS-ECU connector C-304 in good condition?

YES : Go to Step 11.

NO : Repair the damaged connector.



STEP 11. Measure the voltage at ETACS-ECU connector C-304.

- (1) Disconnect ETACS-ECU connector C-304 and measure the resistance available at the wiring harness side of the connector.
- (2) Ignition switch: ACC
- (3) windshield wiper switch: HI
- (4) Measure the voltage between C-304 ETACS-ECU connector terminal No. 5 and body ground.

OK: Battery voltage

- (5) windshield wiper switch: LO

- (6) Measure the voltage between C-304 ETACS-ECU connector terminal No. 6 and body ground.

OK: Battery voltage

Q: Is the measured voltage approximately 12 volts (battery positive voltage)?

YES : Go to Step 13.

NO : Go to Step 12.

STEP 12. Check the wiring harness between ETACS-ECU connector C-304 (terminals 5, 6) and windshield wiper motor connector A-06 (terminals 4, 1).

- Check the input and output lines for open or short circuit.

Q: Is the wiring harness between ETACS-ECU connector C-304 (terminals 5, 6) and windshield wiper motor connector A-06 (terminals 4, 1) in good condition?

YES : Go to Step 13.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Cables and wire check [P.00E-12](#).

STEP 13. Retest the system.

Check that the windshield wipers work normally.

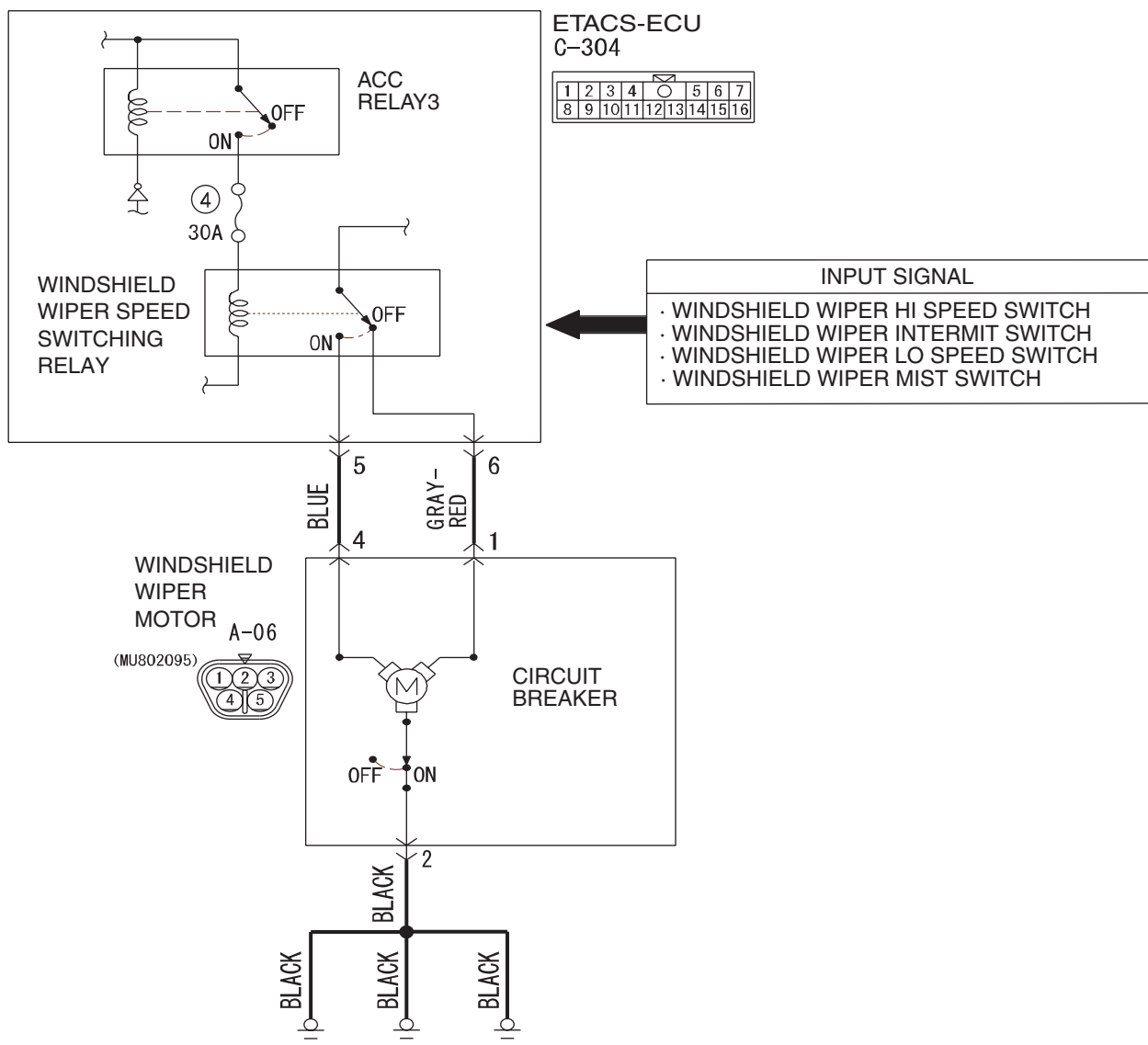
Q: Is the check result normal?

YES : Intermittent malfunction. Refer to GROUP 00, How to cope with intermittent malfunction [P.00-15](#).

NO : Replace the ETACS-ECU.

INSPECTION PROCEDURE 2: The windshield wipers do not work when the wiper switch is at the "INT", "Washer" or "Mist" position. However, the wipers work at low speed when the switch is at the "Lo" or "Hi" position.

Windshield Wiper Motor Circuit



AC708841AB

TECHNICAL DESCRIPTION (COMMENT)

This system may be at fail-safe mode if the LIN communication line is defective.

If the system cannot receive any signal from the column switch (windshield wiper and washer switch) due to an open circuit in the LIN communication line or other reasons, the system will enter the fail-safe mode when the ignition switch is at the "ACC" position.

TROUBLESHOOTING HINTS

- The wiring harness or connectors may have loose, corroded, or damaged terminals, or terminals pushed back in the connector
- The column switch may be defective
- The ETACS-ECU may be defective
- The LIN bus line may be defective

DIAGNOSIS

Required Special Tools:

- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

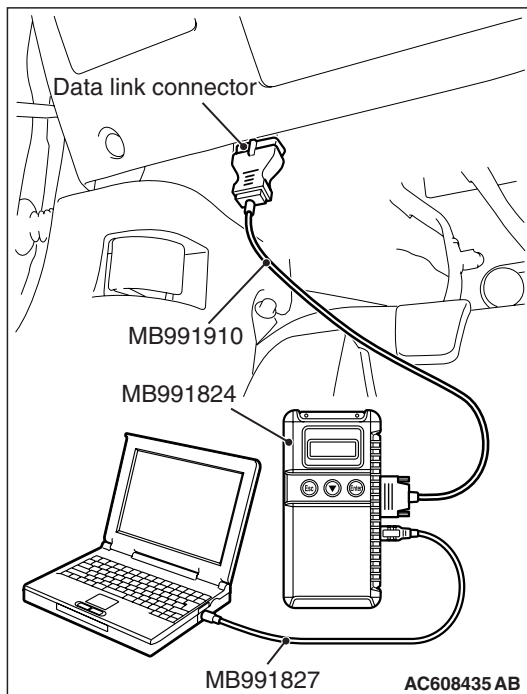
STEP 1. Use the scan tool MB991958 to check if an ETACS-related diagnostic trouble code is set.

Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the ETACS-ECU. Refer to GROUP 54A ETACS, "Diagnostic trouble code chart [P.54A-582](#)."

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, check data list.

Check the input signal related to the windshield wiper operation.

- Ignition switch: ACC
- Operate the windshield wiper switch at each switch position.

Item No.	Item name	Windshield wiper switch position	Normal condition
Item 235	Front wiper ACT	LO	ON
		HI	
		INT	ON and OFF
		MIST	ON
Item 236	Front wiper Lo/Hi	LO	OFF
		HI	ON
		INT	OFF
		MIST	ON

OK: Normal condition is displayed.

Q: Is the check result normal?

YES : Go to Step 3.

NO : Troubleshoot the ETACS-ECU. Refer to ETACS, Diagnosis - Inspection Procedure 12 "ETACS-ECU does not receive any signal from the column switch signal." [P.54A-639](#).

STEP 3. Retest the system.

Check that the windshield wipers work normally.

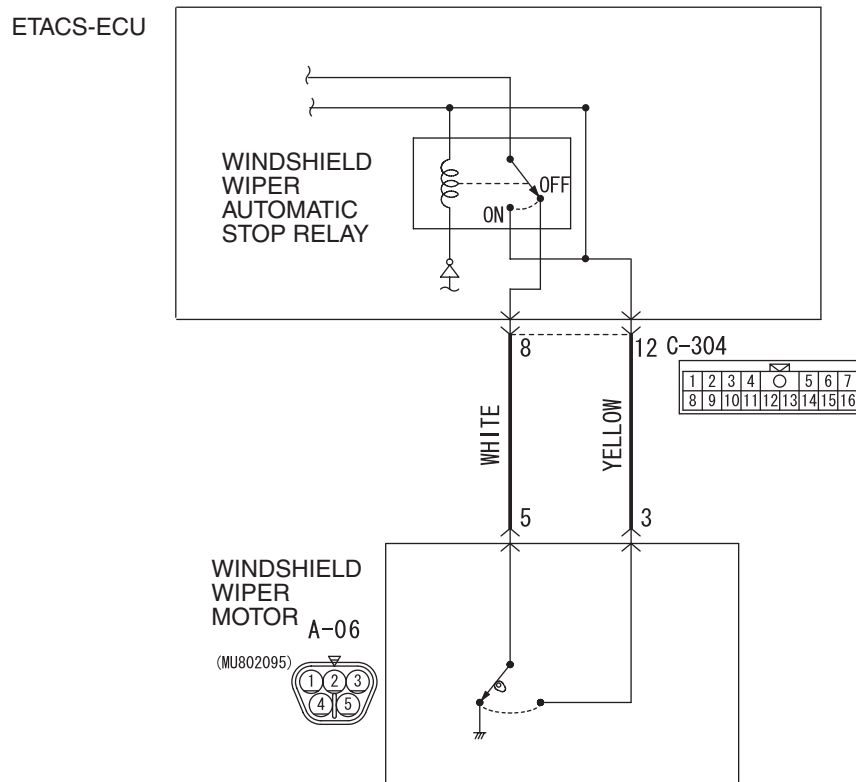
Q: Is the check result normal?

YES : Intermittent malfunction. Refer to GROUP 00, How to cope with intermittent malfunction [P.00-15](#).

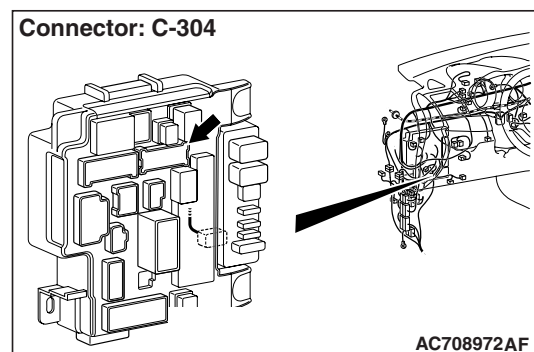
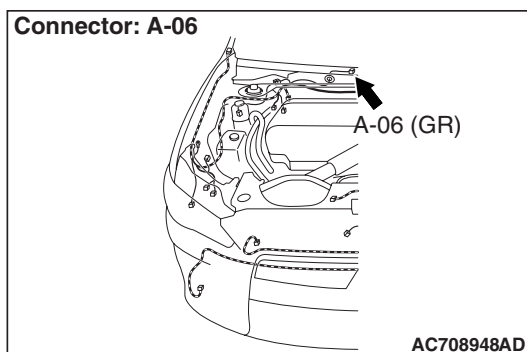
NO : Replace the ETACS-ECU.

INSPECTION PROCEDURE 3: Windshield wipers do not stop at the specified park position.

Windshield Wiper Automatic Stop Relay Circuit



AC708843AB



TECHNICAL DESCRIPTION (COMMENT)

If the windshield wipers do not stop at predetermined park position, the windshield wiper motor or the ETACS-ECU may be defective.

TROUBLESHOOTING HINTS

- The wiring harness or connectors may have loose, corroded, or damaged terminals, or terminals pushed back in the connector
- The windshield wiper motor may be defective
- The ETACS-ECU may be defective

DIAGNOSIS**Required Special Tools:**

- MB991223: Harness Set
- MB992006: Extra Fine Probe

STEP1. Check windshield wiper motor connector A-06 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is windshield wiper motor connector A-06 in good condition?

YES : Go to Step 2.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Verify that the windshield wiper works normally.

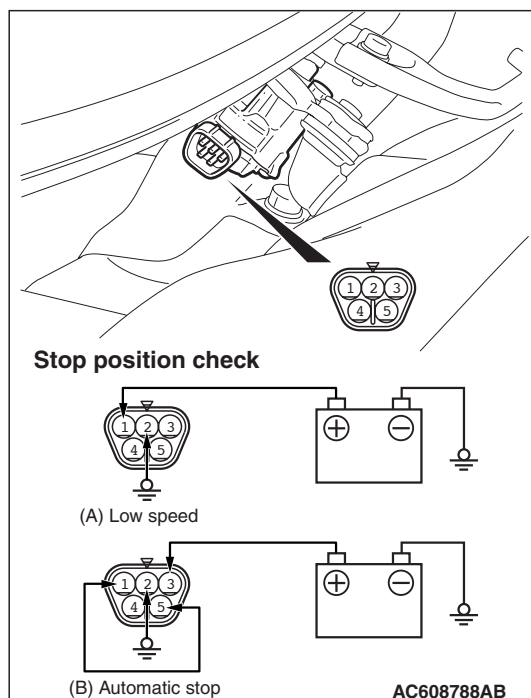
STEP 2. Check the windshield wiper motor.

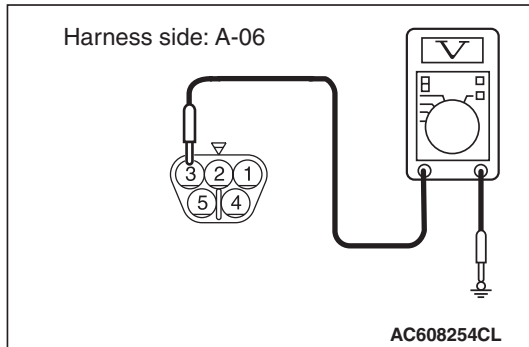
- (1) Disconnect windshield wiper motor connector A-06.
- (2) Connect the vehicle battery to the windshield wiper motor connector as shown, and operate the windshield wiper at low speed. While the windshield wiper is working, disconnect the battery at positions other than the specified park position to stop the windshield wiper motor.
- (3) When the battery is connected as shown, the motor should run at low speed, and then stop at the specified park position.

Q: Does the windshield wiper motor operate normally?

YES : Go to Step 3.

NO : Replace the windshield wiper motor. The windshield wiper should now stop at the specified park position.





STEP 3. Check the ground circuit to the windshield wiper motor. Measure the resistance at the windshield wiper motor connector A-06.

- (1) Disconnect windshield wiper motor connector A-06 and measure the voltage available at the wiring harness side of the connector.
- (2) Turn the ignition switch to the "ACC" position.
- (3) Measure the voltage between terminal 3 and ground.
 - The voltage should measure approximately 12 volts (battery positive voltage).

Q: Is the measured voltage approximately 12 volts (battery positive voltage)?

YES : Go to Step 6.

NO : Go to Step 4.

STEP 4. Check ETACS-ECU connector C-304 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is ETACS-ECU connector C-304 in good condition?

YES : Go to Step 5.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#).

STEP 5. Check the wiring harness between ETACS-ECU connector C-304 (terminals 8, 12) and windshield wiper motor connector A-06 (terminals 3, 5).

- Check the output lines for open or short circuits.

Q: Is the wiring harness between ETACS-ECU connector C-304 (terminals 8, 12) and windshield wiper motor connector A-06 (terminals 3, 5) in good condition?

YES : Go to Step 6.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Cables and wire check [P.00E-12](#).

STEP 6. Retest the system.

Check that the windshield wipers stops at the specified park position.

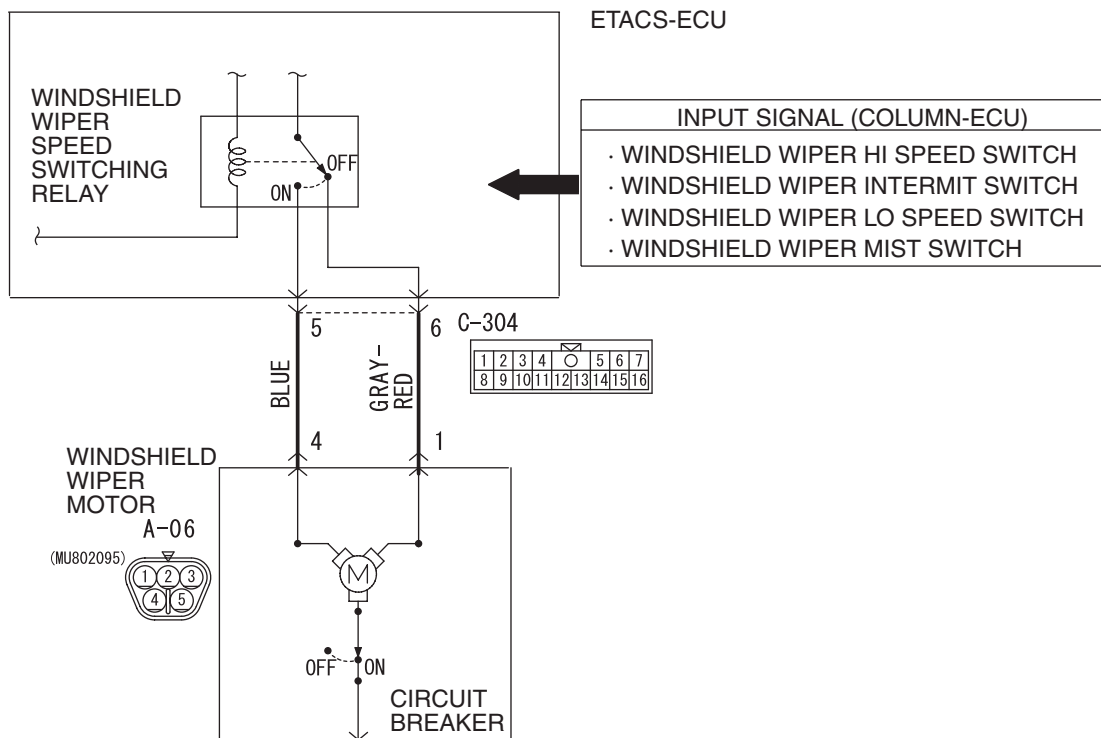
Q: Is the check result normal?

YES : Intermittent malfunction. Refer to GROUP 00, How to cope with intermittent malfunction [P.00-15](#).

NO : Replace the ETACS-ECU.

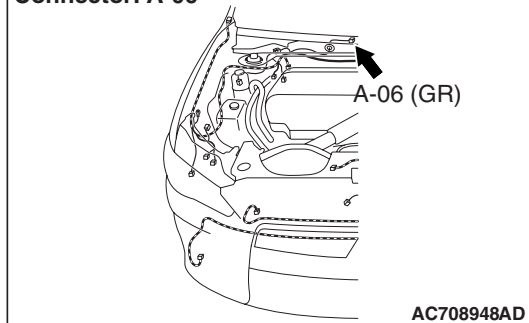
INSPECTION PROCEDURE 4: Windshield Wipers do not work normally.

Windshield Wiper Motor Drive Circuit

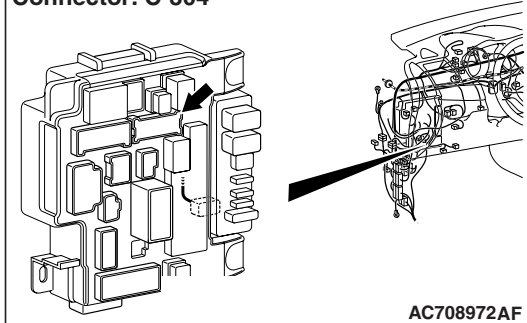


AC708844AB

Connector: A-06



Connector: C-304



TECHNICAL DESCRIPTION (COMMENT)

If either of the windshield wiper switch positions is defective, the windshield wiper motor, column switch (windshield wiper and washer switch) or the ETACS-ECU may be defective.

TROUBLESHOOTING HINTS

- Trouble in input signal system
- The wiring harness or connectors may have loose, corroded, or damaged terminals, or terminals pushed back in the connector
- The wiper motor may be defective
- The column switch may be defective
- The ETACS-ECU may be defective

DIAGNOSIS

Required Special Tools:

- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

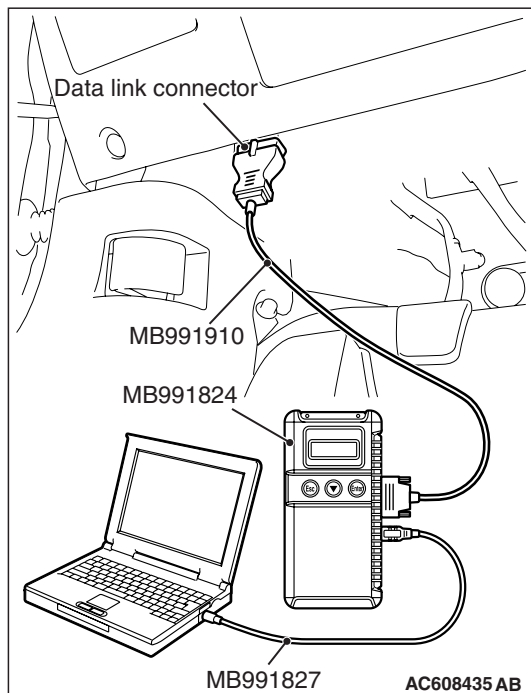
STEP 1. Use the scan tool MB991958 to check if an ETACS-related diagnostic trouble code is set.

Connect the scan tool. Refer to "How to connect the scan tool (M.U.T.-III) [P.54B-4](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the ETACS-ECU. Refer to Diagnostic trouble code chart [P.54A-582](#)."

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, check data list.

Check the input signal related to the windshield wiper operation.

- Ignition switch: ACC
- Operate the windshield wiper switch at each switch position.

Item No.	Item name	Windshield wiper switch position	Normal condition
Item 235	Front wiper ACT	LO	ON
		HI	
		INT	ON and OFF
		MIST	ON
Item 236	Front wiper Lo/Hi	LO	OFF
		HI	ON
		INT	OFF
		MIST	ON

OK: Normal condition is displayed.

Q: Is the check result normal?

YES : Go to Step 3.

NO : Troubleshoot the ETACS-ECU. Refer to ETACS, Diagnosis - Inspection Procedure 12 "ETACS-ECU does not receive any signal from the column switch signal." [P.54A-639](#).

STEP 3. Check that the windshield wipers work.

Check that the windshield wipers work at high speed and the mist mode.

Q: Is the check result normal?

YES : Go to Step 4.

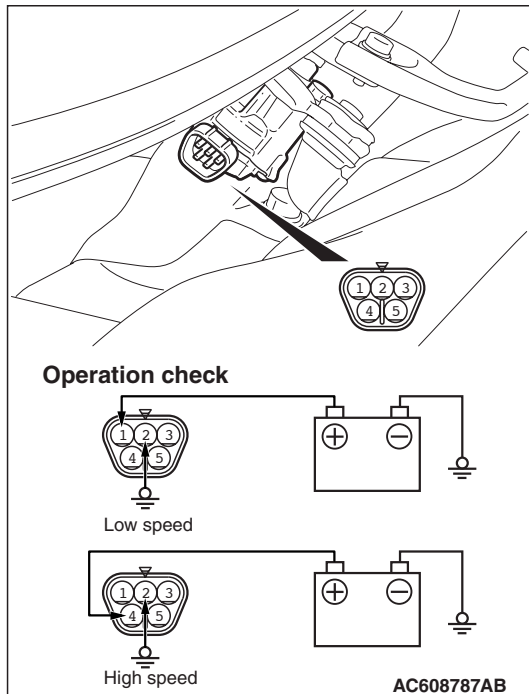
NO : Go to Step 10.

STEP 4. Check windshield wiper motor connector A-06 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is windshield wiper motor connector A-06 in good condition?

YES : Go to Step 5.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Verify that the windshield wiper operates normally when the windshield wiper switch is moved to each position.



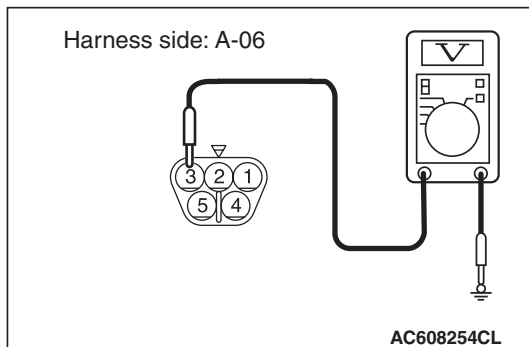
STEP 5. Check the windshield wiper motor.

- (1) Disconnect windshield wiper motor connector A-06.
- (2) Connect a battery to the windshield wiper motor as shown. Then check if the windshield wiper motor operates normally at high and low speeds.

Q: Does the windshield wiper motor operate normally?

YES : Go to Step 6.

NO : Replace the windshield wiper motor. Verify that the windshield wiper operates normally when the windshield wiper switch is moved to each position.



STEP 6. Measure the voltage at windshield wiper motor connector A-06

- (1) Disconnect the connector, and measure the voltage at the wiring harness side.
- (2) Ignition switch: ACC
- (3) windshield wiper switch: LO
- (4) Measure the voltage between A-06 windshield wiper motor connector terminal No. 3 and body ground.

OK: Battery voltage

Q: Is the measured voltage approximately 12 volts (battery positive voltage)?

YES : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15](#)).

NO : Go to Step 7.

STEP 7. Check ETACS-ECU connector C-304 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is ETACS-ECU connector C-304 in good condition?

YES : Go to Step 8.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Verify that the windshield wiper operates normally when the windshield wiper switch is moved to each position.

STEP 8. Check the wiring harness between windshield wiper motor connector A-06 (terminals 1) and ETACS-ECU connector C-304 (terminals 6).

Q: Is the wiring harness between windshield wiper motor connector A-06 (terminals 1) and ETACS-ECU connector C-304 (terminals 6) in good condition?

YES : Go to Step 9.

NO : The wiring harness may be damaged or the connector(s) may have loose, corroded or damaged terminals, or terminals pushed back in the connector. Repair the wiring harness as necessary. Verify that the windshield wiper operates normally when the windshield wiper switch is moved to each position.

STEP 9. Retest the system.

Check that the windshield wipers work normally by moving the switch to each position.

Q: Is the check result normal?

YES : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15](#)).

NO : Replace the ETACS-ECU.

STEP 10. Check the A-06 windshield wiper motor connector

Q: Is the check result normal?

YES : Go to Step 11.

NO : Repair the connector concerned.

STEP 11. Check the windshield wiper motor.

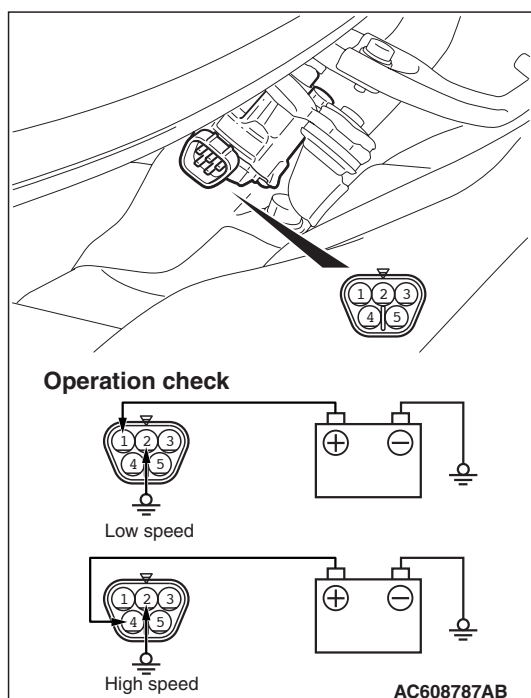
(1) Disconnect windshield wiper motor connector A-06.

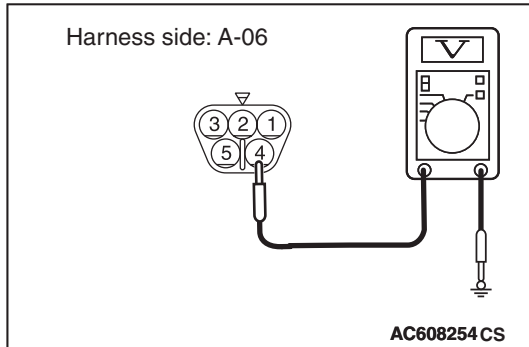
(2) Connect a battery to the windshield wiper motor as shown. Then check if the windshield wiper motor operates normally at high and low speeds.

Q: Does the windshield wiper motor operate normally?

YES : Go to Step 12.

NO : Replace the windshield wiper motor. Verify that the windshield wiper operates normally when the windshield wiper switch is moved to each position.





STEP 12. Measure the voltage at the A-06 windshield wiper motor connector.

- (1) Disconnect the connector, and measure the voltage at the wiring harness side.
- (2) Ignition switch: ACC
- (3) Windshield wiper switch: HI, MIST
- (4) Measure the voltage between A-06 windshield wiper motor connector terminal No.4 and body ground.

OK: Battery voltage

Q: Is the check result normal?

YES : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15](#)).

NO : Go to Step 13.

STEP 13. Check ETACS-ECU connector C-304 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is ETACS-ECU connector C-304 in good condition?

YES : Go to Step 14.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Verify that the windshield wiper operates normally when the windshield wiper switch is moved to each position.

STEP 14. Check the wiring harness wires between C-304 ETACS-ECU connector terminal No. 5 and A-05 windshield wiper motor connector terminal No.4.

- Check the output lines for open or short circuit.

Q: Is the check result normal?

YES : Go to Step 15.

NO : Repair the wiring harness.

STEP 15. Retest the system.

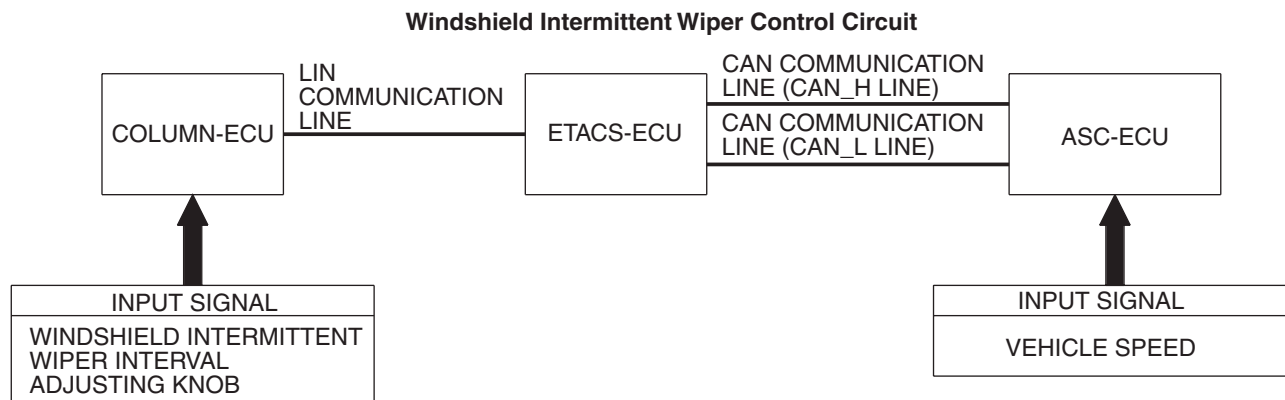
Check that the windshield wipers work normally by moving the switch to each position.

Q: Is the check result normal?

YES : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15](#)).

NO : Replace the ETACS-ECU.

INSPECTION PROCEDURE 5: The windshield intermittent wiper interval cannot be adjusted by operating the windshield intermittent wiper interval control switch.



AC609166 AD

TECHNICAL DESCRIPTION (COMMENT)

If the windshield intermittent wiper interval is not changed by operating the windshield intermittent wiper interval adjusting knob or according to the vehicle speed, the column switch or the ETACS-ECU may be defective.

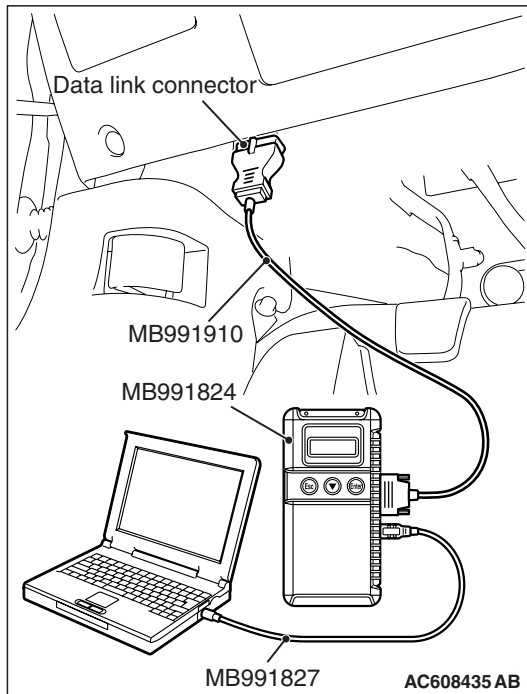
TROUBLESHOOTING HINTS

- Trouble in input signal system
- The wiring harness or connectors may have loose, corroded, or damaged terminals, or terminals pushed back in the connector
- The column switch may be defective
- The ETACS-ECU may be defective

DIAGNOSIS

Required Special Tools:

- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A



STEP 1. Use the scan tool MB991958 to check if an ETACS-related diagnostic trouble code is set.

Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the ETACS-ECU. Refer to GROUP 54A ETACS, "Diagnostic trouble code chart [P.54A-582](#)."

NO : Go to Step 2.

STEP 2. Check the ETACS configuration function.

Use the ETACS configuration function to check that "Front wiper operation" is set to "Variable INT" or "Speed sensitive".

Q: Is the check result normal?

YES : Go to Step 3.

NO : Use the ETACS configuration function to set "Front wiper operation" to "variable INT" or "Speed sensitive". (Refer to [P.51-65](#).)

STEP 3. Using scan tool MB991958, check data list.

Check the input signal related to the windshield wiper operation.

- Ignition switch: ACC
- Rotate the windshield wiper interval control from (+) to (-) side.

Item No.	Display on scan tool	Check conditions	Normal condition
Item 359	Front wiper (interval volume)	Rotate the windshield wiper interval control from (+) to (-) side.	Value changes from (+) to 254 (-)

OK: Normal condition is displayed.

Q: Is the check result normal?

NO : Go to Step 4.

YES : Replace the column switch.

STEP 4. Retest the system.

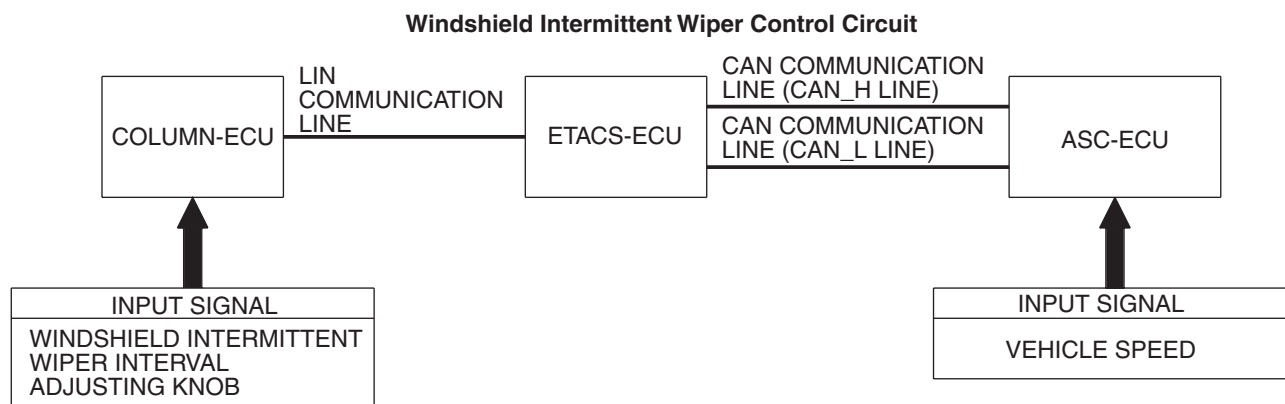
Check that the windshield wiper interval changes when the windshield wiper interval control is rotated.

Q: Is the check result normal?

YES : Intermittent malfunction. Refer to GROUP 00, How to cope with intermittent malfunction [P.00-15](#).

NO : Replace the ETACS-ECU.

INSPECTION PROCEDURE 6: The windshield intermittent wiper interval is not changed according to the vehicle speed.



AC609166 AD

TECHNICAL DESCRIPTION (COMMENT)

If the intermittent wiper interval does not depend on the vehicle speed, the input circuit of the vehicle speed signal and the ETACS-ECU may be defective. Alternatively, the vehicle speed-dependent wiper may be set to "disabled" by using the configuration function.

TROUBLESHOOTING HINTS

- The wiring harness or connectors may have loose, corroded, or damaged terminals, or terminals pushed back in the connector
- The combination meter may be defective
- The ETACS-ECU may be defective

DIAGNOSIS

Required Special Tools:

- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

STEP 1. Retest the system.

Check that the windshield intermittent wiper interval can be adjusted by operating the windshield intermittent wiper interval control switch.

Q: Is the check result normal?

YES : Go to Step 2.

NO : Refer to Inspection Procedure 5 "The windshield intermittent wiper interval cannot be adjusted by operating the windshield intermittent wiper interval control switch" [P.51-44](#).

STEP 2. Check the ETACS configuration function.

Use the ETACS configuration function to check that "Front wiper operation" is set to "Variable INT" or "Speed sensitive".

Q: Is the check result normal?

YES : Go to Step 3.

NO : Use the ETACS configuration function to set "Front wiper operation" to "variable INT" or "Speed sensitive". (Refer to [P.51-65](#).)

STEP 3. Using scan tool MB991958, check data list.

Check the input signal related to the combination meter.

- Drive the vehicle and change vehicle speed.

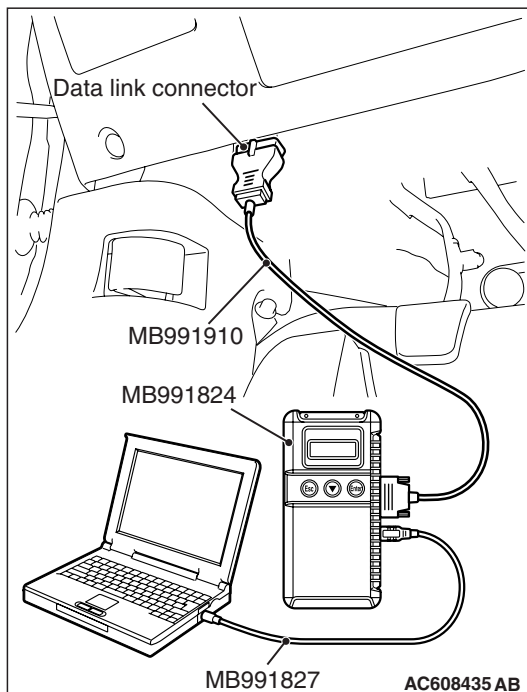
Item No.	Display on scan tool	Check condition	Normal condition
Item 80	speedometer	Speedometer and scan tool displays meet	

Q: Is the check result normal?

YES : Go to Step 4.

NO <Normal condition is not displayed for item No. 80.>

- : Troubleshoot the speedometer (Refer to GROUP 54A -The combination meter, Diagnosis - Inspection Procedure 2: "The speedometer does not work (the other meters work)." [P.54A-65](#))



STEP 4. Retest the system.

Check that the intermittent wiper interval depends on the vehicle speed.

Q: Is the check result normal?

YES : Intermittent malfunction. Refer to GROUP 00, How to cope with intermittent malfunction [P.00-15](#).

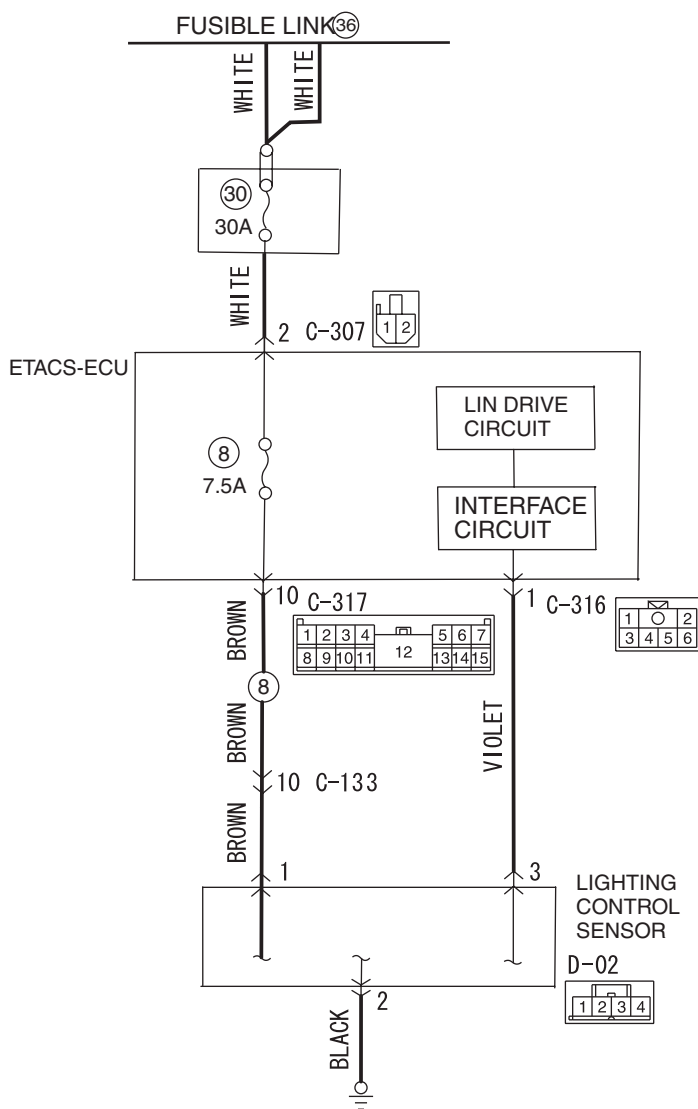
NO : Replace the ETACS-ECU.

INSPECTION PROCEDURE 7: The rain sensitive AUTO wiper function does not work at all <Vehicles with lighting control sensor>.

CAUTION

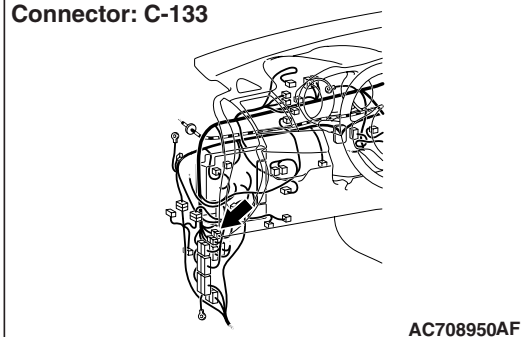
Whenever ECU is replaced, ensure that the input and output signal circuits are normal.

Raindrops Sensing Wiper Function Control Circuit



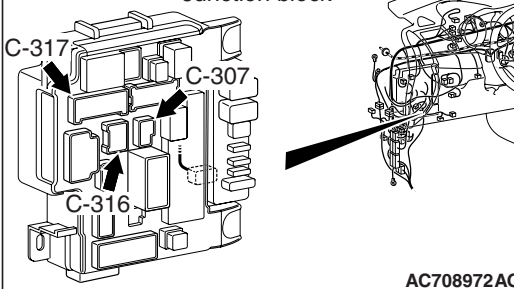
AC708845AB

Connector: C-133

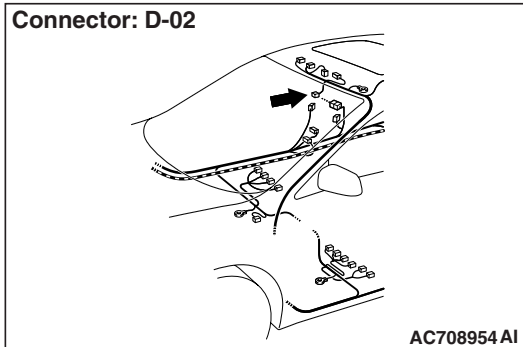


AC708950AF

Connectors: C-307, C-316, C-317
Junction block



AC708972AG



CIRCUIT OPERATION

When the column switch is in the AUTO position, this function automatically adjusts the wiping speed of windshield wiper by detecting the rain fall through lighting control sensor.

TECHNICAL DESCRIPTION (COMMENT)

The windshield wiper motor, the column switch, the lighting control sensor, the harness connector, or the ETACS-ECU may be defective.

TROUBLESHOOTING HINTS

- Defective column switch (column-ECU)
- Malfunction of the lighting control sensor
- Defective windshield wiper motor
- Malfunction of ETACS-ECU
- Damaged wiring harness and connectors

DIAGNOSIS

Required Special Tools:

- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

STEP 1. Lighting control sensor installation surface check

Visually check the presence of scratches or air bubbles (diameter of 5 mm or more) on the windshield to which the lighting control sensor is installed.

Q: Is the check result normal?

YES : Go to Step 2.

NO : Replace the windshield (Refer to GROUP 42A – Windshield Removal and Installation [P.42A-19](#)).

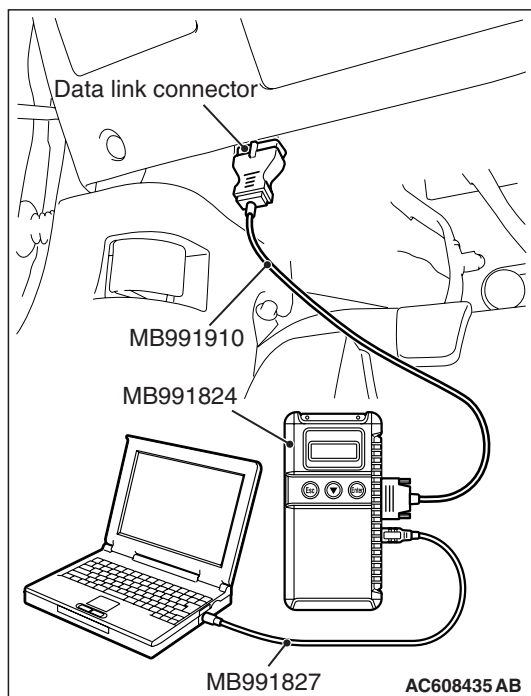
STEP 2. Windshield wiper operation check

Check that the windshield wipers work normally.

Q: Is the check result normal?

YES : Go to Step 3.

NO : Refer to trouble symptom chart [P.51-23](#).



STEP 3. Use the scan tool MB991958 to check if an ETACS-related diagnostic trouble code is set.

Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the ETACS-ECU. Refer to GROUP 54A ETACS, "Diagnostic trouble code chart [P.54A-582](#)."

NO : Go to Step 4.

STEP 4. Using scan tool MB991958, read the lighting control sensor diagnostic trouble code.

Check if a lighting control sensor diagnostic trouble code is set. Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

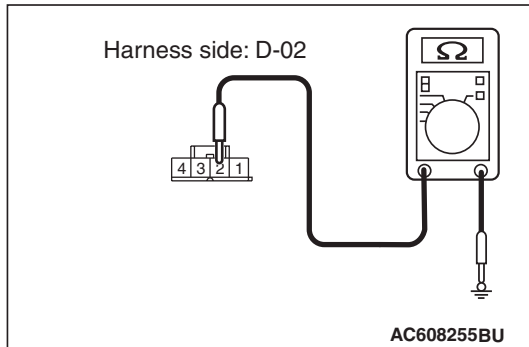
YES : Diagnose the lighting control sensor. [Diagnostic Trouble Code: (Refer to GROUP 54A –Diagnosis Code Chart [P.54A-107](#)) or Service Data ([P.51-64](#)).]

NO : Go to Step 5.

STEP 5. Connector check: D-02 Lighting control sensor connector**Q: Is the check result normal?**

YES : Go to Step 6.

NO : Repair the damaged connector.



STEP 6. Resistance measurement at the D-02 lighting control sensor connector

- (1) Disconnect the connector, and measure the resistance at the wiring harness.
- (2) Measure the resistance between D-02 lighting control sensor connector terminal No.2 and body ground.

OK: Continuity exists (2 Ω or less)

Q: Is the check result normal?

YES : Go to Step 8.

NO : Go to Step 7.

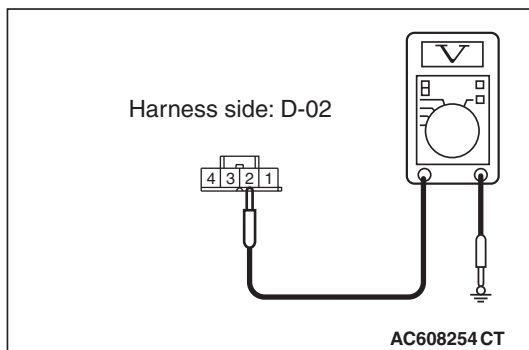
STEP 7. Check the wiring harness between D-02 lighting control sensor connector terminal No.2 and body ground.

- Check the ground wires for open circuit.

Q: Is the check result normal?

YES : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15.](#))

NO : Repair the wiring harness.



STEP 8. Voltage measurement at the D-02 lighting control sensor connector

- (1) Disconnect the connector, and measure the voltage at the wiring harness side.
- (2) Measure the voltage between D-02 lighting control sensor connector terminal No.1 and body ground.

OK: Battery voltage

Q: Is the check result normal?

YES : Go to Step 10.

NO : Go to Step 9.

STEP 9. Check the wiring harness between D-02 lighting control sensor connector terminal No.1 and fusible link (36).

NOTE: Prior to the wiring harness inspection, check intermediate connector C-133, ETACS-ECU connectors C-307 and C-317, and repair if necessary.

- Check the power supply line for open circuit.

Q: Is the check result normal?

YES : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15.](#))

NO : Repair the wiring harness.

STEP 10. Connector check: C-316 ETACS-ECU connector

Q: Is the check result normal?

YES : Go to Step 11.

NO : Repair the damaged connector.

STEP 11. Measure the voltage at the C-316 ETACS-ECU connector.

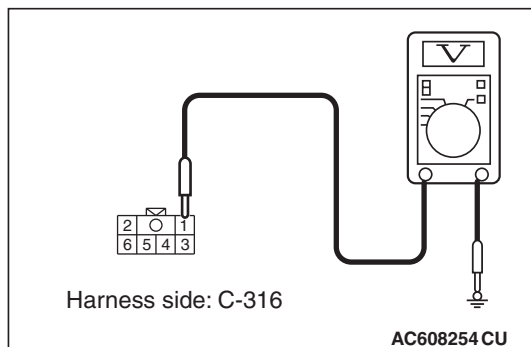
- (1) Disconnect the connector, and measure the voltage at the wiring harness side.
- (2) Measure the voltage between C-316 ETACS-ECU connector terminal No. 1 and body ground.

OK: Battery voltage

Q: Is the check result normal?

YES : Replace the Lighting control sensor.

NO : Go to Step 12.



STEP 12. Check the wiring harness between C-316 ETACS-ECU connector terminal No.1 and D-02 lighting control sensor connector terminal No.3.

- Check the input and output lines for open or short circuit.

Q: Is the check result normal?

YES : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15.](#))

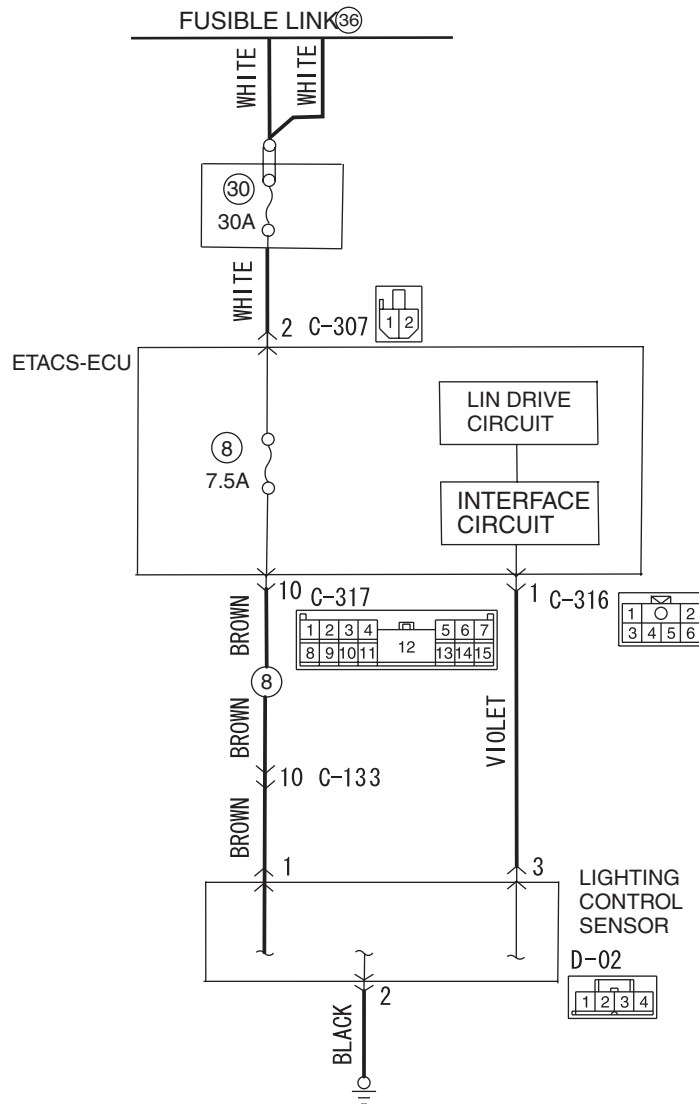
NO : Repair the wiring harness.

INSPECTION PROCEDURE 8: The rain sensitive AUTO wiper function works even though there is no rainfall <Vehicles with lighting control sensor>.

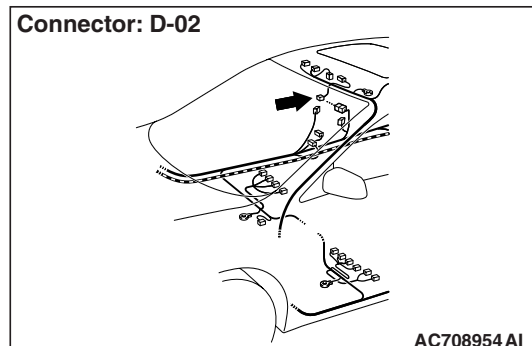
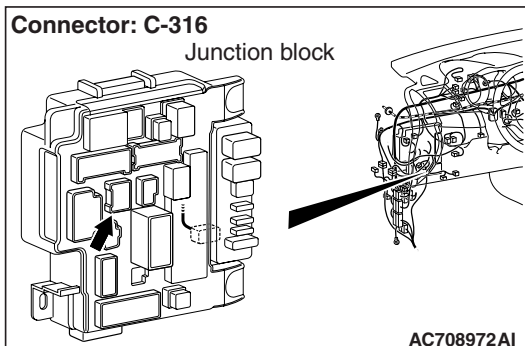
CAUTION

Whenever ECU is replaced, ensure that the input and output signal circuits are normal.

Raindrops Sensing Wiper Function Control Circuit



AC708845AB



TECHNICAL DESCRIPTION (COMMENT)

The lighting control sensor, the harness connector, or the ETACS-ECU may be defective.

TROUBLESHOOTING HINTS

- Malfunction of the lighting control sensor
- Malfunction of ETACS-ECU
- Damaged wiring harness and connectors

DIAGNOSIS**Required Special Tools:**

- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

STEP 1. Lighting control sensor installation surface check

Visually check the presence of scratches or air bubbles (diameter of 5 mm or more) on the windshield to which the lighting control sensor is installed.

Q: Is the check result normal?

YES : Go to Step 2.

NO : Replace the windshield (Refer to GROUP 42A – Windshield Removal and Installation [P.42A-19](#)).

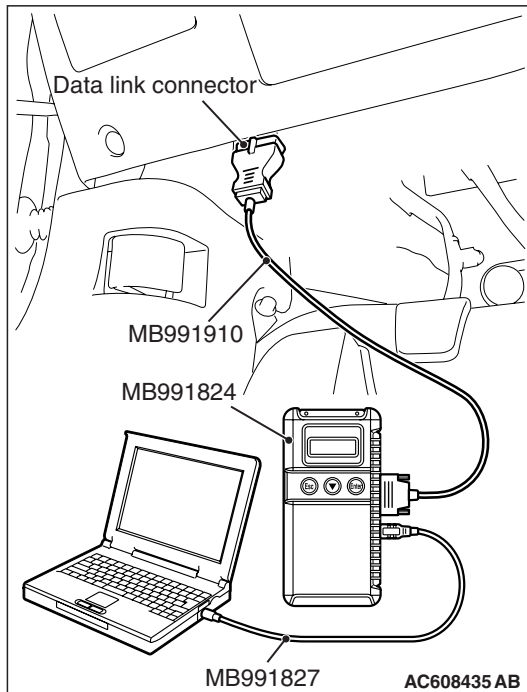
STEP 2. Windshield wiper operation check

Check that the windshield wipers work normally.

Q: Is the check result normal?

YES : Go to Step 3.

NO : Refer to Inspection procedure 3 [P.51-35](#).



STEP 3. Use the scan tool MB991958 to check if an ETACS-related diagnostic trouble code is set.

Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the ETACS-ECU. Refer to GROUP 54A ETACS, "Diagnostic trouble code chart [P.54A-582](#)."

NO : Go to Step 4.

STEP 4. Using scan tool MB991958, read the lighting control sensor diagnostic trouble code.

Check if a lighting control sensor diagnostic trouble code is set. Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the lighting control sensor. [Diagnosis Code: (Refer to GROUP 54A –Diagnosis Code Chart [P.54A-107](#)) or service data ([P.51-64](#)).]

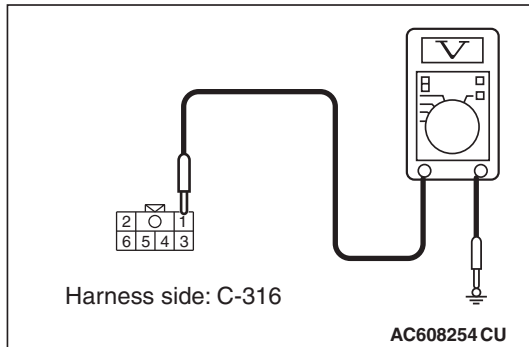
NO : Go to Step 5.

STEP 5. Connector check: C-316 ETACS-ECU connector

Q: Is the check result normal?

YES : Go to Step 6.

NO : Repair the damaged connector.

**STEP 6. Measure the voltage at the C-316 ETACS-ECU connector.**

- (1) Disconnect the connector, and measure the voltage at the wiring harness side.
- (2) Measure the voltage between C-316 ETACS-ECU connector terminal No. 1 and body ground.

OK: Battery voltage**Q: Is the check result normal?****YES** : Replace the Lighting control sensor.**NO** : Go to Step 7.**STEP 7. Check the wiring harness between C-316 ETACS-ECU connector terminal No.1 and D-02 lighting control sensor connector terminal No.3.**

- Check the input and output lines for open or short circuit.

Q: Is the check result normal?**YES** : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15.](#))**NO** : Repair the wiring harness.

INSPECTION PROCEDURE 9: Sometimes the rain sensitive AUTO wiper function works even though there is no rainfall < Vehicles with lighting control sensor>.

CAUTION

Whenever ECU is replaced, ensure that the input and output signal circuits are normal.

TECHNICAL DESCRIPTION (COMMENT)

The lighting control sensor may be defective or a failure in the lighting control sensor (rain sensor) adaptation is suspected.

TROUBLESHOOTING HINTS

- Malfunction of the lighting control sensor
- Lighting control sensor (rain sensor) adaptation failure

DIAGNOSIS**Required Special Tools:**

- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

STEP 1. Lighting control sensor (rain sensor) installation surface check

Visually check the presence of scratches or air bubbles (diameter of 5 mm or more) on the windshield to which the lighting control sensor is installed.

Q: Is the check result normal?

YES : Go to Step 2.

NO : Replace the windshield (Refer to GROUP 42A – Windshield Removal and Installation [P.42A-19](#)).

STEP 2. Windshield wiper operation check

Check that the windshield wipers work normally.

Q: Is the check result normal?

YES : Go to Step 3.

NO : Refer to Inspection procedure 3 [P.51-35](#).

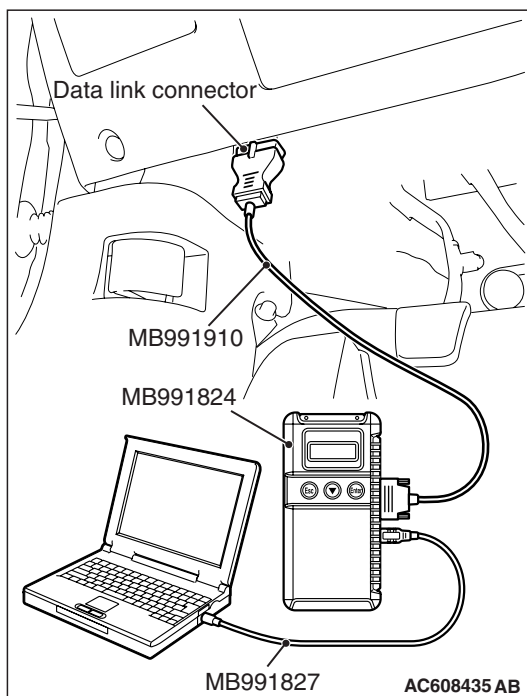
STEP 3. Use the scan tool MB991958 to check if an ETACS-related diagnostic trouble code is set.

Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the ETACS-ECU. Refer to GROUP 54A ETACS, "Diagnostic trouble code chart [P.54A-582](#)."

NO : Go to Step 4.



STEP 4. Using scan tool MB991958, read the lighting control sensor diagnostic trouble code.

Check if a lighting control sensor diagnostic trouble code is set. Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

YES : Diagnose the lighting control sensor. [Diagnosis Code: (Refer to GROUP 54A –Diagnosis Code Chart [P.54A-107](#)) or service data ([P.51-64](#)).]

NO : Go to Step 5.

STEP 5. Lighting control sensor (rain sensor) installation surface check

Check that the lighting control sensor (rain sensor) is installed to the windshield glass firmly.

Q: Is the check result normal?

YES : Go to Step 6 after completion of the lighting control sensor (rain sensor) adaptation ([P.51-76](#)).

NO : Install the lighting control sensor (rain sensor) to the windshield glass correctly (Refer to GROUP 54A – Lighting control sensor removal and Installation [P.54A-187](#)).

STEP 6. Retest the system.

Check the lighting control sensor (rain sensor) after completion of the lighting control sensor (rain sensor) adaptation

- (1) Check that the windshield glass surface is dry.
- (2) Pour the windshield glass surface where the lighting control sensor is installed.

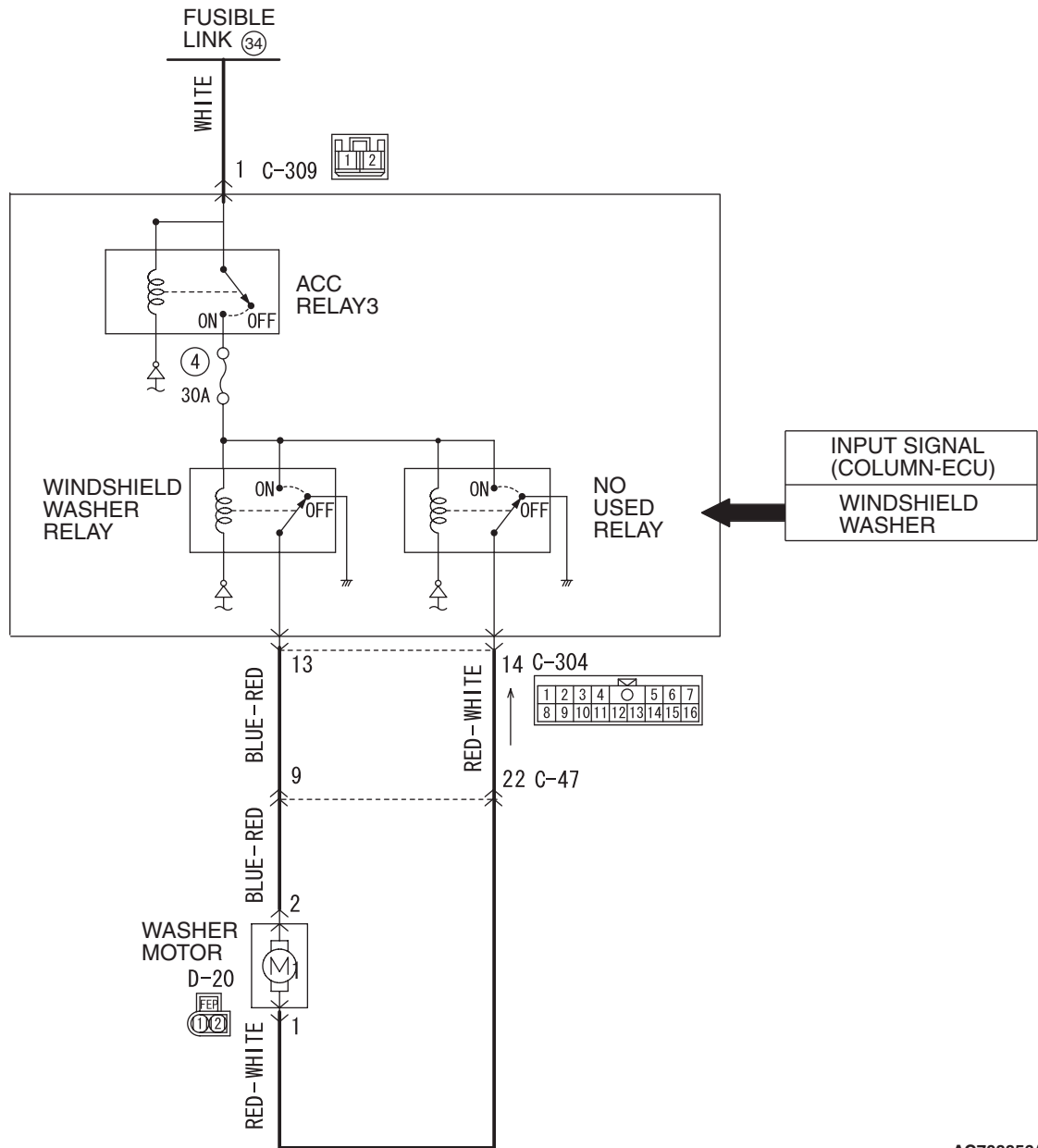
Q: Does the windshield wiper operate?

Operate one or more : Intermittent malfunction (Refer to GROUP 00 –How to Use Troubleshooting/Inspection Service Points - How to Cope with Intermittent Malfunction [P.00-15](#)).

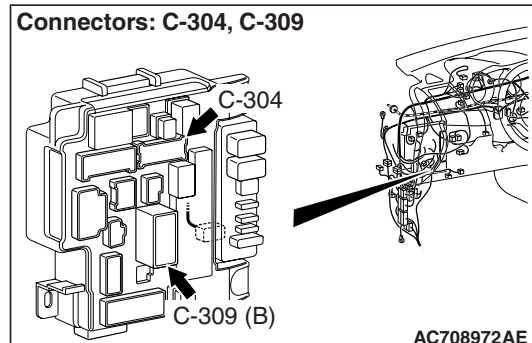
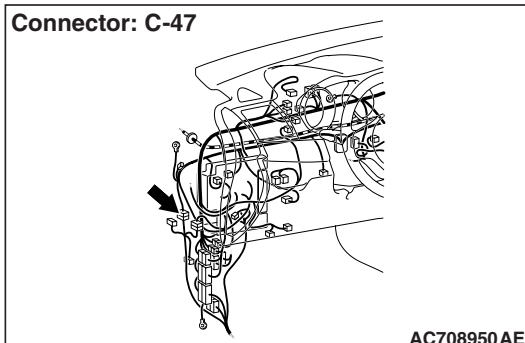
Does not operate : Troubleshoot the rain sensitive wiper function (Refer to Inspection procedure 7 [P.51-48](#)).

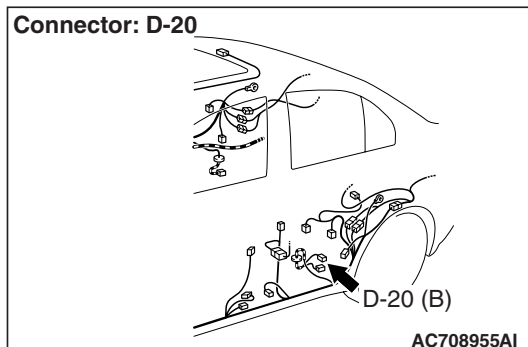
INSPECTION PROCEDURE 10: The windshield washer does not work normally.

Windshield Washer Motor Circuit



AC708856AB





CIRCUIT OPERATION

The windshield washer switch sends a signal through the column-ECU (incorporated in the column switch) to the ETACS-ECU. If the column-ECU sends a windshield washer switch "ON" signal to the ETACS-ECU, the ETACS-ECU turns on the relay (incorporated in the ETACS-ECU), thus causing the washer motor to be turned on.

TECHNICAL DESCRIPTION (COMMENT)

If the windshield washer does not work normally, the washer motor, the column switch (windshield wiper and washer switch) or the ETACS-ECU may be defective.

TROUBLESHOOTING HINTS

- The wiring harness or connectors may have loose, corroded, or damaged terminals, or terminals pushed back in the connector
- The washer motor may be defective
- The column switch may be defective
- The ETACS-ECU may be defective

DIAGNOSIS

Required Special Tools:

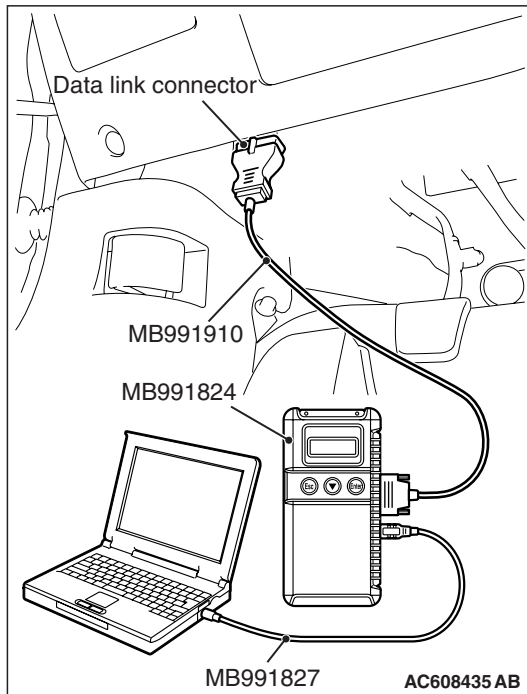
- MB991223: Harness Set
- MB992006: Extra Fine Probe
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

STEP 1. Verify the windshield wiper operation.

Q: Does the windshield wiper operate normally?

YES : Go to Step 2.

NO : Refer to Inspection Procedure 1 "The windshield wipers do not work at all [P.51-25](#)."



STEP 2. Use the scan tool MB991958 to check if an ETACS-related diagnostic trouble code is set.

Connect the scan tool. Refer to GROUP 54A ETACS, "How to connect the scan tool (M.U.T.-III) [P.54A-579](#)."

Q: Is the diagnostic trouble code set?

- YES :** Diagnose the ETACS-ECU. Refer to GROUP 54A ETACS, "Diagnostic trouble code chart [P.54A-582](#)."
NO : Go to Step 3.

STEP 3. Check the input signal related to the windshield washer operation.

- Ignition switch: ACC
- Windshield washer switch: ON

Item No.	Display on scan tool	Normal condition
Item 237	Front washer	ON

OK: Normal condition is displayed.

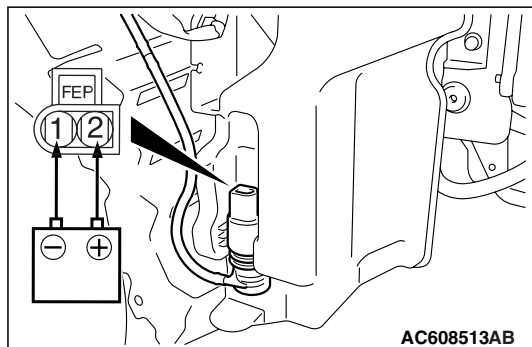
Q: Is the check result normal?

- NO :** Go to Step 4.
YES : Refer to GROUP 54A –ETACS, Input signal chart [P.54A-639](#).

STEP 4. Check windshield washer motor connector D-20 for loose, corroded or damaged terminals, or terminals pushed back in the connector.

Q: Is windshield washer motor connector D-20 in good condition?

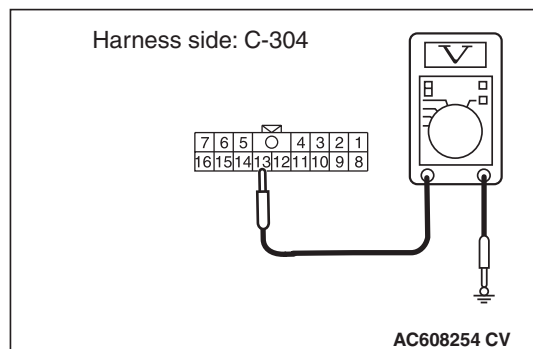
- YES :** Go to Step 5.
NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Verify that the windshield wiper works normally.

**STEP 5. Check the operation of windshield washer motor.**

- (1) Disconnect windshield washer motor connector D-20 and check at windshield washer motor connector side.
- (2) Fill the windshield washer tank with washer fluid.
- (3) When battery voltage is applied between terminals 1 and 2, washer fluid should spray out.

Q: Does the washer motor operate normally?**YES :** Go to Step 6.**NO :** Replace the washer motor. Verify that the windshield washer works normally.**STEP 6. Check ETACS-ECU connector C-304 for loose, corroded or damaged terminals, or terminals pushed back in the connector.****Q: Is ETACS-ECU connector C-304 in good condition?****YES :** Go to Step 7.**NO :** Repair or replace the damaged component(s). Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Verify that the windshield washer works normally.**STEP 7. Measure the voltage at ETACS-ECU connector C-304.**

- (1) Disconnect ETACS-ECU connector C-304 and measure the resistance available at the wiring harness side of the connector.
- (2) Ignition switch: ACC
- (3) windshield washer switch: ON
- (4) Measure the voltage between C-304 ETACS-ECU connector terminal No. 13 and body ground.

OK: Battery voltage**Q: Is the measured voltage approximately 12 volts (battery positive voltage)?****YES :** Go to Step 10.**NO :** Go to Step 8.

STEP 8. Check the wiring harness between windshield washer motor connector D-20 (terminal 2) and ETACS-ECU connector C-304 (terminal 13).

- Check the input lines for open or short circuit.

Q: Is the wiring harness between windshield washer motor connector D-20 (terminal 2) and ETACS-ECU connector C-304 (terminal 13) in good condition?

YES : Go to Step 9.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Cables and wire check [P.00E-12](#).

STEP 9. Check the wiring harness between windshield washer motor connector D-20 (terminal 1) and ETACS-ECU connector C-304 (terminal 14).

NOTE: Prior to the wiring harness inspection, check intermediate connector C-47 and repair if necessary.

- Check the input lines for open or short circuit.

Q: Is the wiring harness between windshield washer motor connector D-20 (terminal 1) and ETACS-ECU connector C-304 (terminal 14) in good condition?

YES : Go to Step 10.

NO : Repair or replace the damaged component(s). Refer to GROUP 00E, Cables and wire check [P.00E-12](#).

STEP 10. Retest the system.

Check that the windshield washers work normally.

Q: Is the check result normal?

YES : Intermittent malfunction. Refer to GROUP 00, How to cope with intermittent malfunction [P.00-15](#).

NO : Replace the ETACS-ECU.

DATA LIST REFERENCE TABLE

M1511015100048

LIN (LIGHTING CONTROL SENSOR)

Item No.	Scan tool display	Check condition	Normal condition
7005	RLS Rain sensor ON/OFF	When the wiper switch is AUTO position	ON
		When a switch (except the wiper switch) is AUTO position	OFF
7006	RLS Rain sensor sensitivity	Changes from 1 to 5 according to the wiper volume.	1, 2, 3, 4, 5
7008	RLS Wiper auto stop SW	When the windshield wiper is in operation	Park
		Other than above	Outside park
7013	RLS Wiper control output	When the operation is not requested from lighting control sensor (rain sensor).	OFF
		When the LO operation is requested from lighting control sensor (rain sensor)	LO
		When the HI operation is requested from lighting control sensor (rain sensor)	HI
7020	RLS RS measurement value(RS1)	When the lighting control sensor (rain sensor) detects raindrops	The sensor 1 detects raindrops and the value changes.
7021	RLS RS measurement value(RS2)	When the lighting control sensor (rain sensor) detects raindrops	The sensor 2 detects raindrops and the value changes.
7022	RLS RS adaptation value(RS1)	When initializing after the adaptation (sensor 1)	The amount of output when initializing (sensor 1)
7023	RLS RS adaptation value(RS2)	When initializing after the adaptation (sensor 2)	The amount of output when initializing (sensor 2)
7024	RLS RS adaptation gain level	When initializing after the adaptation	The calibrated value of initializing

ON-VEHICLE SERVICE

WINDSHIELD INTERMITTENT WIPER INTERVAL CHECK

M1511023600112

1. Check that the intermittent wiper interval is changed as the windshield intermittent wiper interval control is operated.
2. Turn the windshield intermittent wiper switch to the intermittent operation position. Use scan tool MB991958 to set a simulated vehicle speed with the wiper volume held. The intermittent wiper interval should be changed as the simulated vehicle speed is changed.
3. If either of above is defective, carry out the troubleshooting. (Refer to [P.51-44](#))

LIGHTING CONTROL SENSOR (RAIN SENSOR) INSPECTION

M1511028400049

Under clear weather (windshield glass is dry), turn the ignition switch to "ON" position and wiper switch to "AUTO" position. Check that the wiper works when applying water to the upper part of the windshield glass where the lighting control sensor is installed. If there is a malfunction, perform the troubleshooting (Refer to [P.51-23](#)).

CONFIGURATION FUNCTION

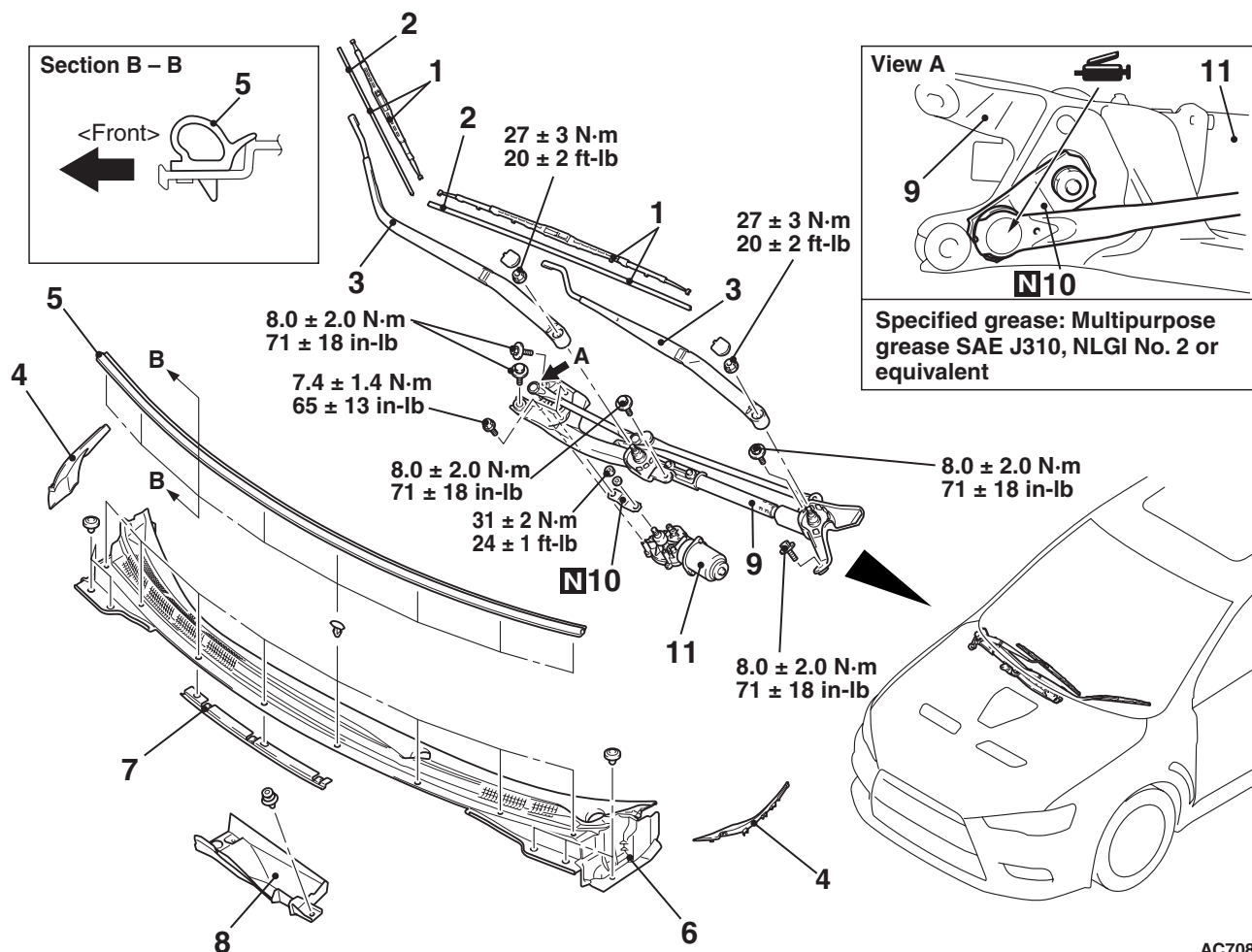
M1511027400251

Using the ETACS system of scan tool MB991958, the following functions can be programmed. The programmed information is held even when the battery is disconnected.

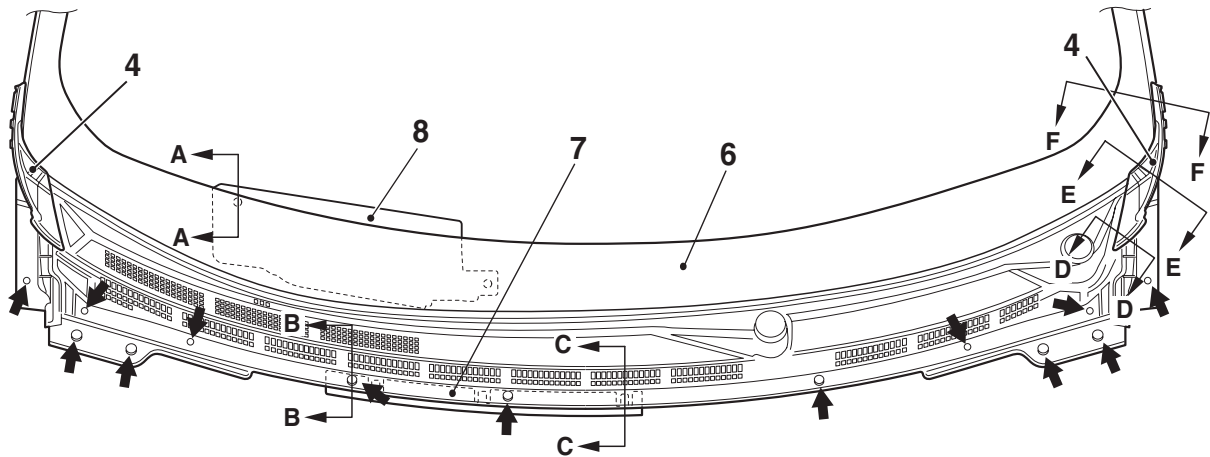
Adjustment item (scan tool MB991958 display)	Adjustment item	Adjusting contents (scan tool MB991958 display)	Adjusting contents
Front wiper operation	Adjustment of the intermittent windshield wiper operation <vehicles without auto light>	Normal INT	Intermittent wiper interval is fixed to 4 seconds.
		Variable INT	Intermittent wiper interval is calculated only by the wiper volume control.
		Speed Sensitive	Intermittent wiper interval is calculated according to the intermittent wiper volume control and vehicle speed (default).
	Adjustment of the intermittent windshield wiper operation <vehicles with auto light>	Normal INT	Intermittent wiper interval is fixed to 4 seconds.
		Variable INT	Intermittent wiper interval is calculated only by the wiper volume control.
		Speed Sensitive	Intermittent wiper interval is calculated according to the intermittent wiper volume control and vehicle speed.
		Rain Sensitive	Intermittent wiper interval is calculated according to the intermittent wiper volume control and lighting control sensor (default).
	Disabling or enabling washer-linked wiper function	Only washer	No function
Front wiper washer		Washer&wiper	With function (default)

WINDSHIELD WIPER
REMOVAL AND INSTALLATION

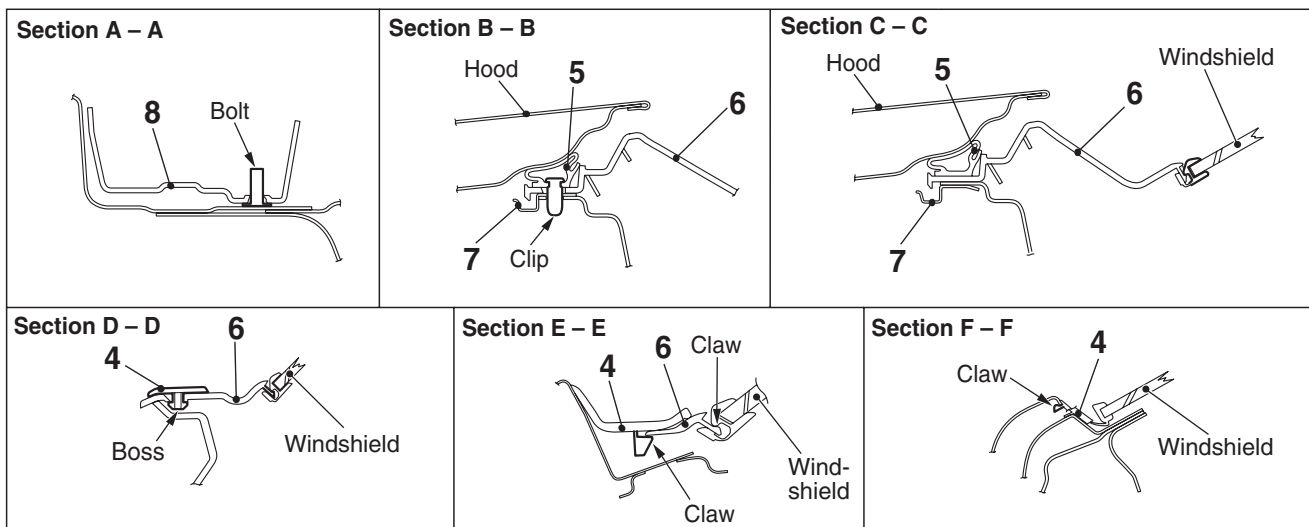
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← : Clip positions



AC709005AB

Wiper blade removal steps

- >>C<< 1. Wiper blade assembly
>>A<< 2. Wiper blade

Windshield wiper motor and wiper link assembly removal steps

- >>C<< 3. Wiper arm
4. Front deck garnish cover A
5. Hood weatherstrip
6. Front deck garnish
7. Front deck garnish cover B
8. Front deck cover
<<A>> >>B<< 9. Wiper link assembly

Windshield wiper motor and wiper link assembly removal steps (Continued)

- <> 10. Wiper motor link plate
<> 11. Windshield wiper motor

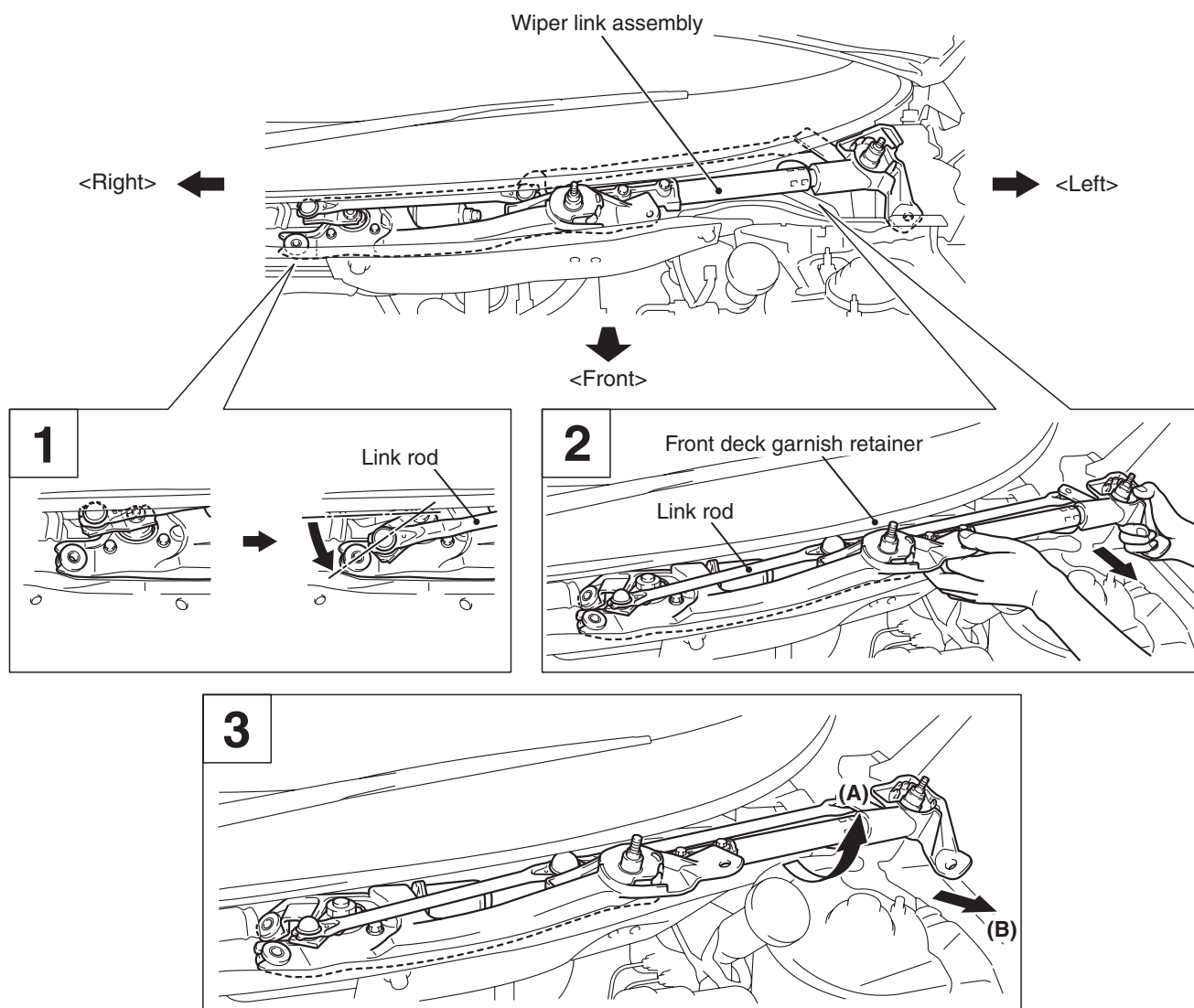
Required Special Tool:

- MB991958: M.U.T.-III sub-assembly
- MB991223: Harness set
- MB992006: Extra fine probe

NOTE: For removal and installation of the wiper and washer switch, refer to GROUP 54A, Column switch P.54A-278.

REMOVAL SERVICE POINTS

<<A>> WIPER LINK ASSEMBLY REMOVAL



AC609130AE

1. Manually turn the link rod of the wiper link assembly forward to the position indicated in the figure.

⚠ CAUTION

Another mechanic fully pushes up the hood so that the left side of link assembly is pulled out easily.

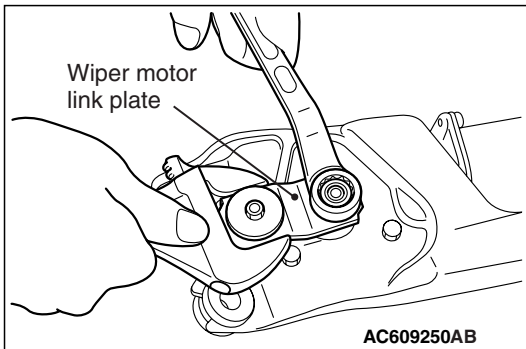
2. Pull out the attachment section of the left side of wiper link assembly forward until the link rod bumps against the front deck garnish retainer.

⚠ CAUTION

When pulling out the link assembly forward, be careful not to make a contact with the windshield glass.

3. While turning the link assembly upward (A), pull out the entire link assembly diagonally forward left (B).

<> WIPER MOTOR LINK PLATE/WINDSHIELD WIPER MOTOR REMOVAL



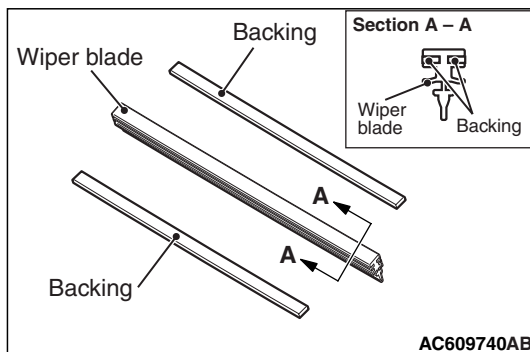
1. To disconnect the wiper motor link plate from the wiper motor shaft, use an appropriate tool to counter-hold the plate to prevent it from turning.
2. Remove the wiper motor from the wiper link assembly.

INSTALLATION SERVICE POINTS

>>A<< WIPER BLADE INSTALLATION

CAUTION

For good windshield wiper wiping performance, use windshield wipers without a curve in the backing of the wiper blade.



>>B<< WIPER LINK ASSEMBLY/WIPER MOTOR LINK PLATE/WINDSHIELD WIPER MOTOR INSTALLATION

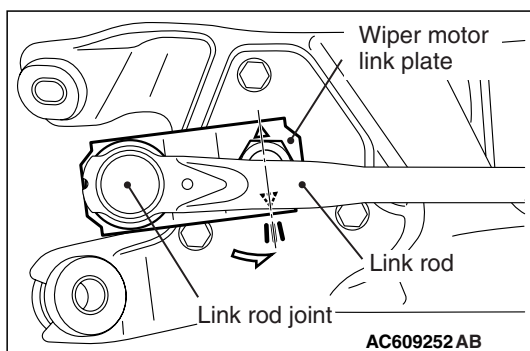
1. Confirm that the wiper motor has set to automatic stop position (Refer to [P.51-70](#)).

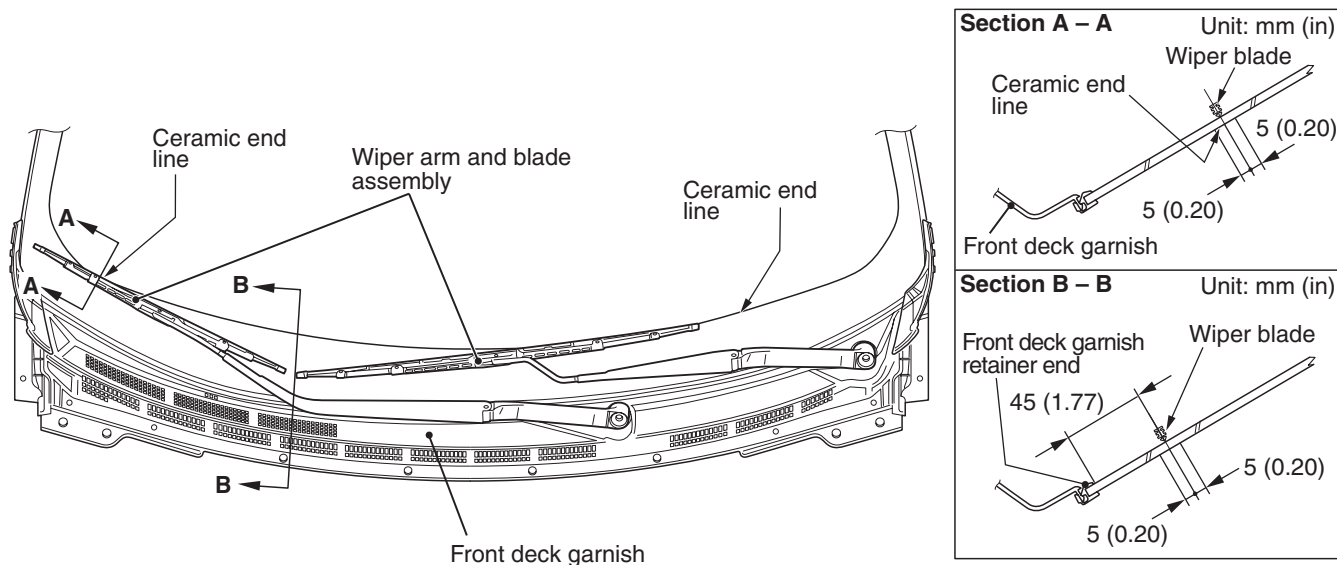
CAUTION

Always replace the wiper motor link plate with new one.

2. Set the wiper motor link plate onto the wiper motor shaft; aligning the triangle mark on the plate to the center line (of 3 lines) on the wiper link assembly as shown in the illustration.
3. Tighten the attaching nut of wiper motor link plate and wiper motor shaft. Use an appropriate tool to counter-hold the plate to prevent it from turning.
4. Apply grease to the inside of link rod joint (as required) and connect the link rod to wiper motor link plate.

Specified grease: Multipurpose grease SAE J310, NLGI No.2 or equivalent



>>C<< WIPER ARM AND BLADE ASSEMBLY
INSTALLATION

AC612983AD

Set the wiper arm and blade assembly at the specified positions.

A(Passenger's side): Ceramic end line
 $\pm 5 \text{ mm}(0.20 \text{ in})$

B(Driver's side): Front deck garnish end
 $45 \pm 5 \text{ mm}(1.77 \pm 0.20 \text{ in})$

INSPECTION

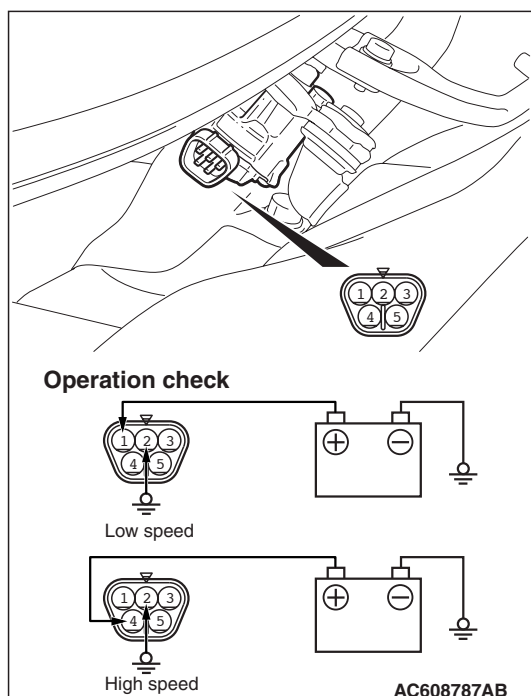
M1511008000317

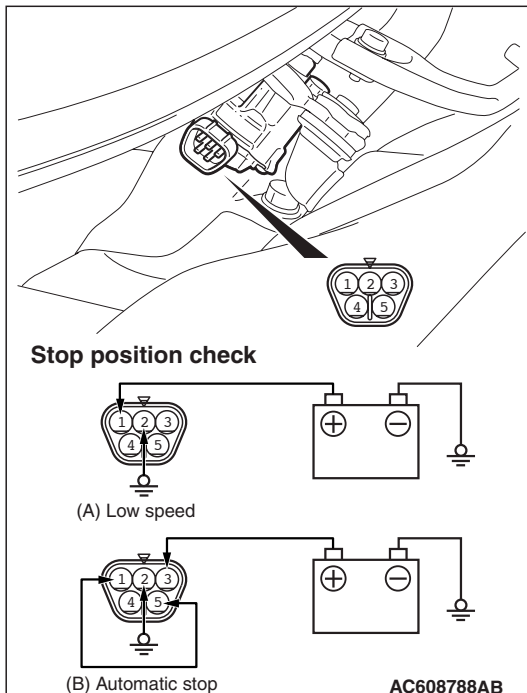
WINDSHIELD WIPER MOTOR CHECK

The windshield wiper motor assembly should be installed to the vehicle body and the harness connector should be disconnected when checking the wiper motor.

WINDSHIELD WIPER MOTOR AT LOW OR HIGH SPEED OPERATION

Connect the battery to the windshield wiper motor to inspect the operation of motor rotation at low or high speed.



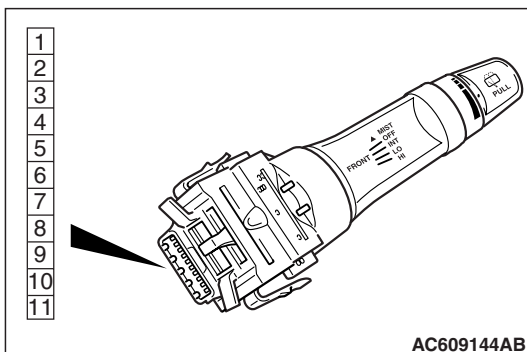


WINDSHIELD WIPER MOTOR AT STOP POSITION OPERATION

1. Connect the battery to the windshield wiper motor as shown in the illustration (A).
2. Run the windshield wiper motor at low speed, then disconnect the battery in the middle of the motor rotation and check to see that the motor stops.
3. Connect the battery to the windshield wiper motor as shown in the illustration (B).
4. Connect the terminals of the windshield wiper motor connector as shown in the illustration (B).
5. Check to see that the windshield wiper motor runs at low speed and then stops at the automatic stop position.

WINDSHIELD WIPER SWITCH CHECK

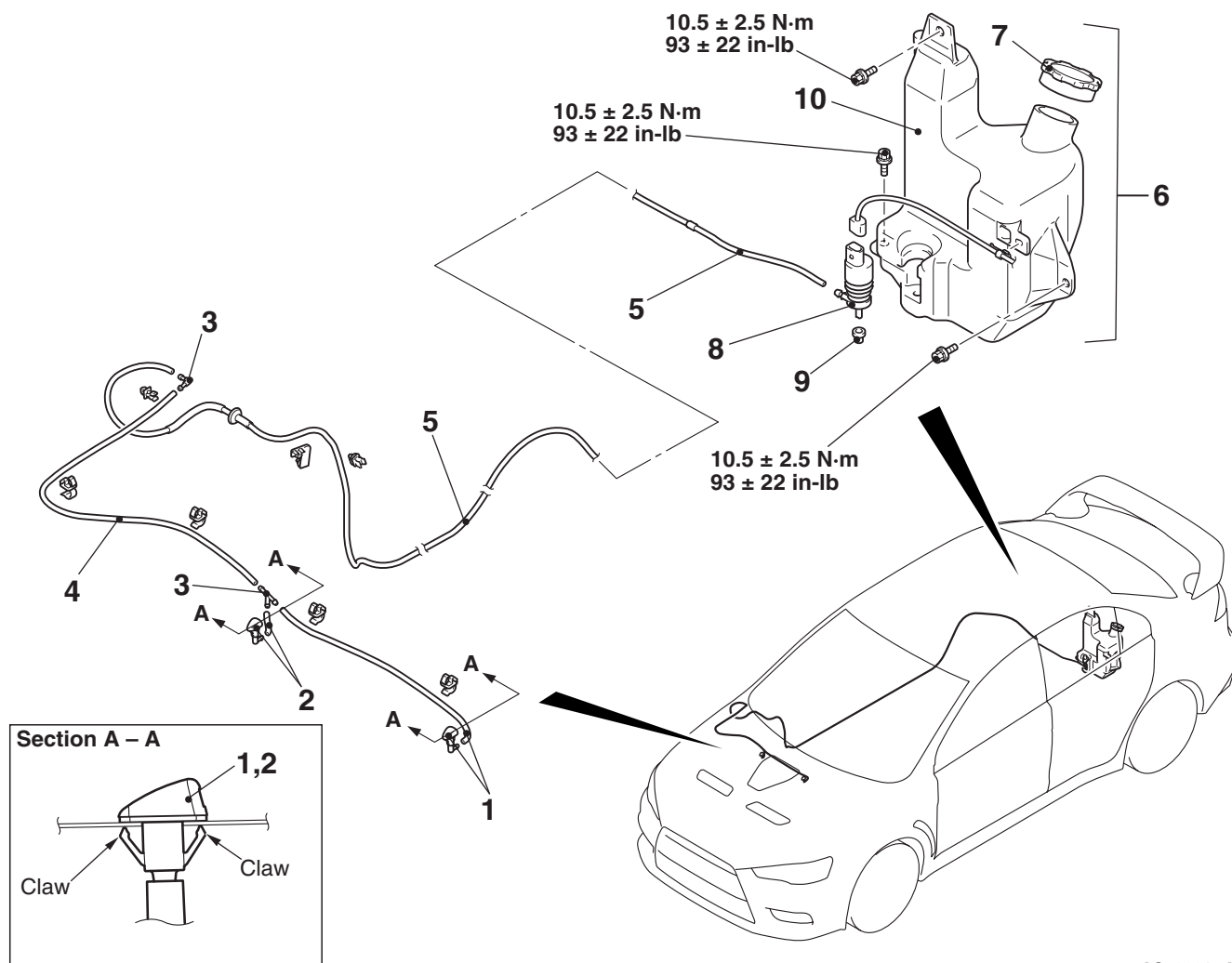
Check continuity between the switch terminals.



Switch position	Tester connection	Specified condition
OFF	6 – 11, 6 – 10, 6 – 9, 6 – 8	Open circuit
Windshield mist wiper switch	6 – 11	Continuity (Less than 2 ohms)
Windshield intermittent wiper switch	6 – 10	
Windshield low-speed wiper switch	6 – 9	
Windshield high-speed wiper switch	6 – 8	

WINDSHIELD WASHER
REMOVAL AND INSTALLATION

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Windshield washer nozzle removal steps

- Connection of windshield washer hose A
 - Hood insulator A and B (Refer to GROUP 42A, Hood [P.42A-7](#))
1. Windshield washer nozzle assembly (LH)
 2. Windshield washer nozzle assembly (RH)
 3. Joint

Windshield washer hose removal steps

- Front scuff plate, rear scuff plate, center pillar trim lower, trunk room trim front (Refer to GROUP 52A, Interior trim [P.52A-11](#))
 - Connection of washer nozzle/washer motor
3. Joint

Windshield washer hose removal steps (Continued)

4. Windshield washer hose A
5. Windshield washer hose B

Windshield washer tank removal steps

- Trunk room trim front (Refer to GROUP 52A, Interior trim [P.52A-11](#))
 - Connection of windshield washer hose
 - Windshield washer motor connector connection
6. Windshield washer tank assembly
 7. Washer tank cap
 8. Windshield washer motor
 9. Grommet
 10. Windshield washer tank

<<A>>

Windshield washer motor removal steps

- Connection of windshield washer hose B
- Windshield washer motor connector connection

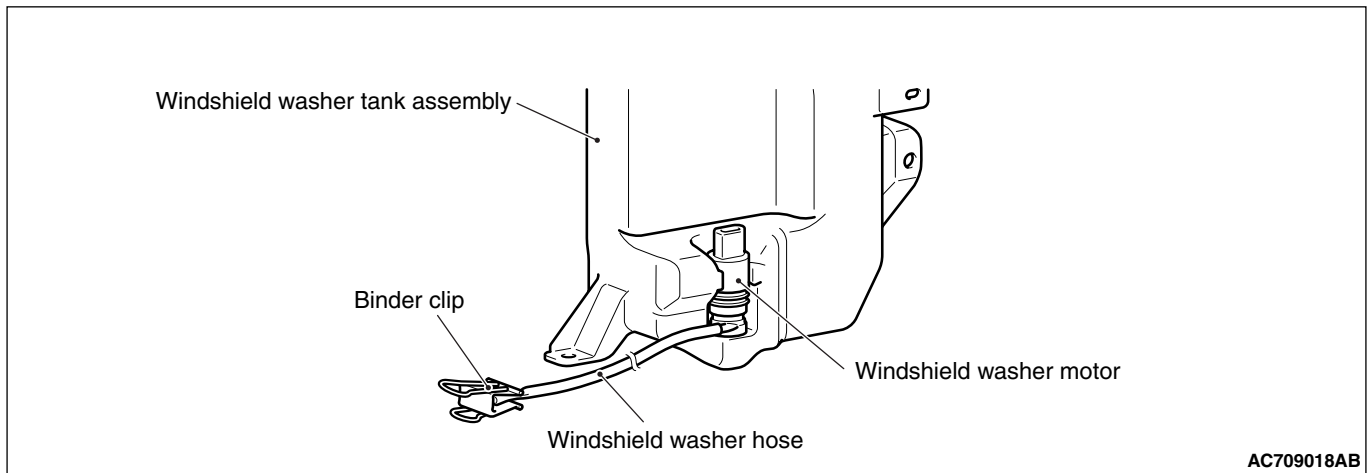
Windshield washer motor removal steps (Continued)

8. Windshield washer motor
9. Grommet

NOTE: For removal and installation of the wiper and washer switch, refer to GROUP 54A, Column switch P.54A-278.

REMOVAL SERVICE POINT

<<A>> WINDSHIELD WASHER TANK ASSEMBLY REMOVAL



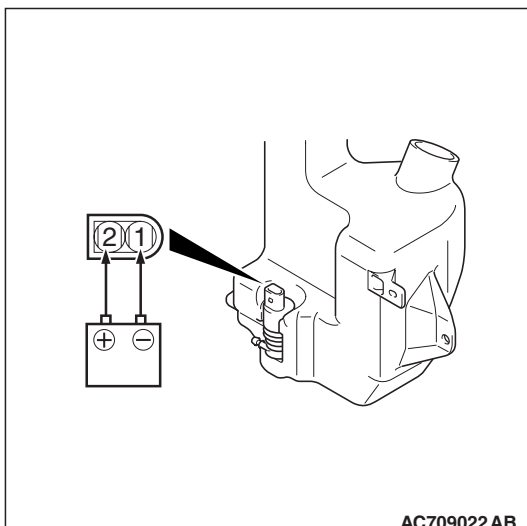
When removing the windshield washer tank assembly, close the windshield washer hose by a binder clip as shown, to prevent the windshield washer from flowing out.

INSPECTION

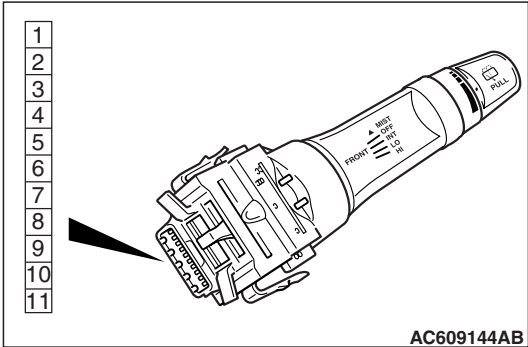
M1511008300385

WINDSHIELD WASHER MOTOR CHECK

1. Remove the washer tank assembly with the washer hose attached. Then fill the washer tank with water.
2. Check to see that the water is vigorously sprayed when connecting the positive battery terminal to terminal number 2 and terminal number 1 to ground.



WINDSHIELD WASHER SWITCH CHECK

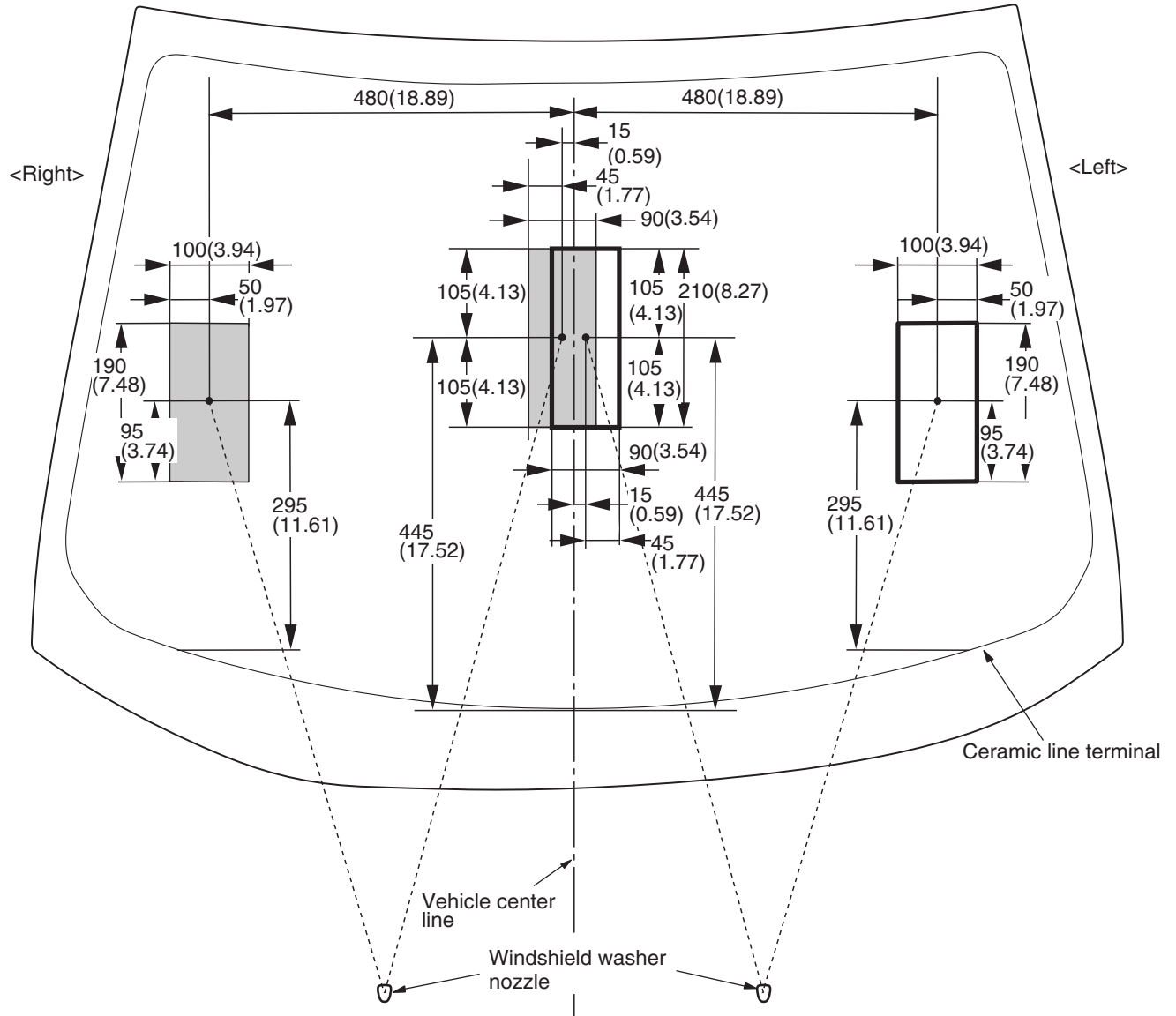


Switch position	Tester connection	Specified condition
OFF	6 –7	Open circuit
Windshield washer switch ON	6 –7	Continuity (Less than 2 ohms)

WINDSHIELD WASHER FLUID EJECTION CHECK

NOTE: The splashing point cannot be adjusted by the nozzle. If the splashing point of the washer fluid is different from the figure below, replace the nozzle.

Unit: mm(in)



■ : Right side washer

□ : Left side washer

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LIGHTING CONTROL SENSOR

REMOVAL AND INSTALLATION

Refer to GROUP 54A-Lighting control sensor (P.54A-187).

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ADAPTATION

M1511028600010

 CAUTION

- Before performing the adaptation, check if the lighting control sensor (rain sensor)-related diagnostic trouble code is set. (if set, refer to GROUP 54A, Headlight Diagnostic Trouble Code Chart P.54A-107).
 - Turn the wiper switch to the OFF position.
1. Clean the windshield in fine weather.
 2. Wipe the surface of the windshield thoroughly, and check that the surface is dry.
 3. Turn the ignition switch to the ON position.
 4. Turn the ignition switch to the LOCK (OFF) position.

 CAUTION

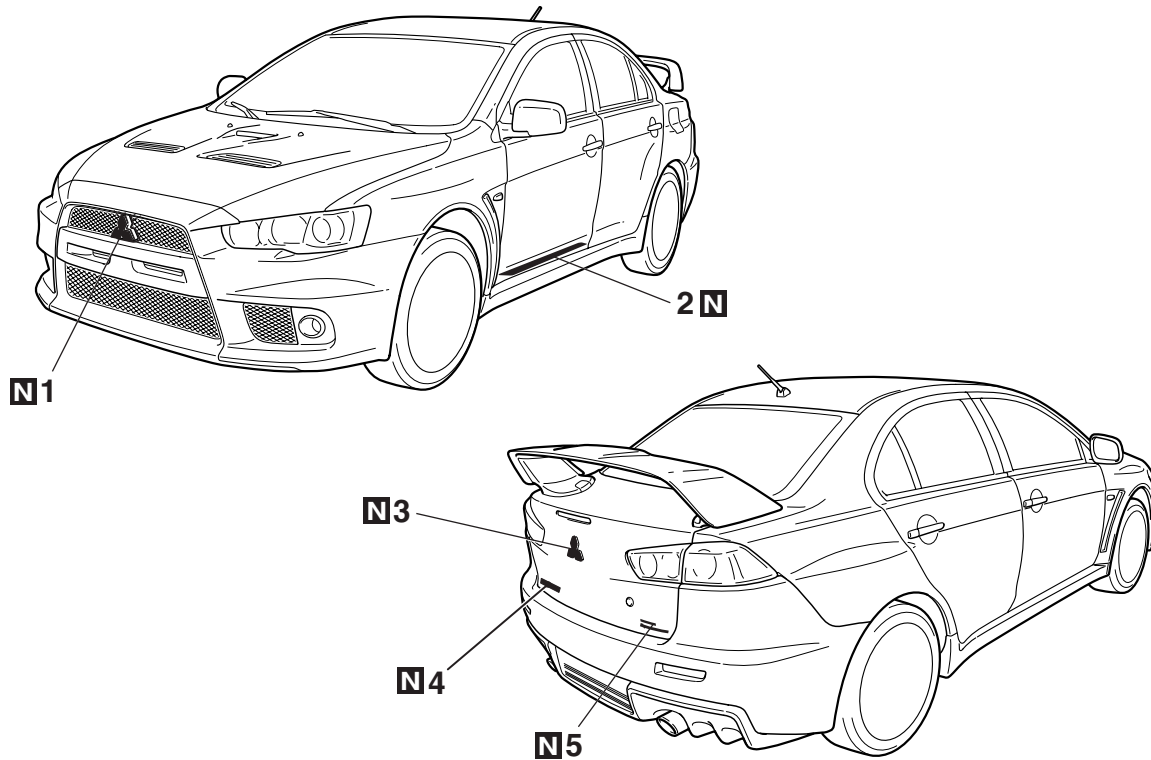
Before connecting or disconnecting scan tool MB991958, always turn the ignition switch to the LOCK (OFF) position.

5. Connect the scan tool MB991958 to the data link connector.
6. Turn the ignition switch to the ON position.
7. Wipe the windshield surface of the lighting control sensor section thoroughly, and check that the surface is dry.
8. Select "LIN" on the "System Select" screen, and press the "OK" button.
9. Select "Rain light sensor" on the "System Select" screen, and press the "OK" button.
10. Select "Special Function" on the "Rain light sensor" screen.
11. Select "Rain Sensor Adaptation" on the "Special Function" screen.
12. Press the "OK" button, and execute the "Rain Sensor Adaptation."
13. Press the "OK" button after the execution screen is displayed.
14. Press the "OK" button after "Completed" is displayed.

MARK

REMOVAL AND INSTALLATION

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AC708418 AD

Removal

1. Front three-diamond mark (Refer to P.51-4.)
- >>A<< 2. Front step plate

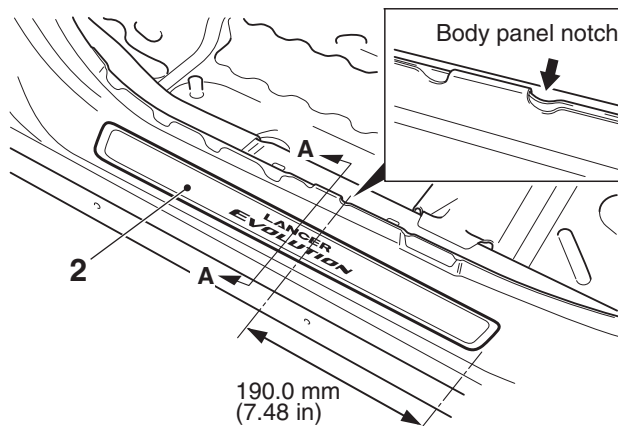
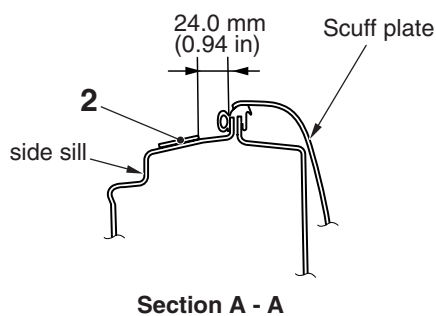
Removal (Continued)

- >>A<< 3. Rear three-diamond mark
- >>A<< 4. MITSUBISHI mark
- >>A<< 5. LANCER EVOLUTION mark

INSTALLATION SERVICE POINT

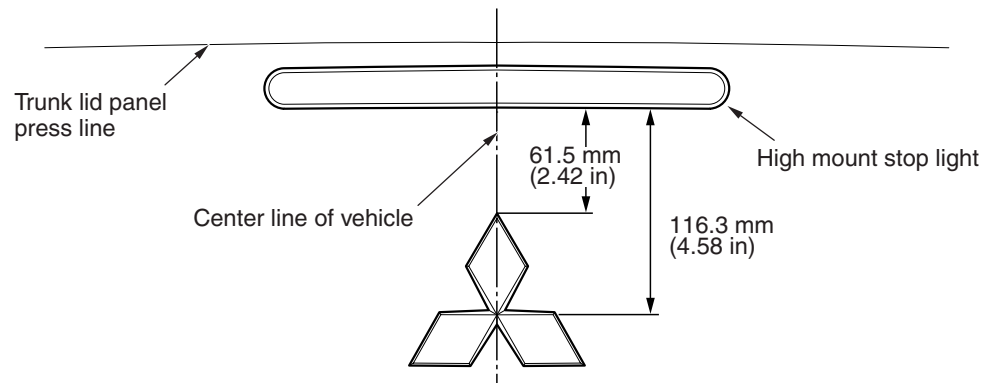
>>A<< MARK INSTALLATION

2. Front step plate



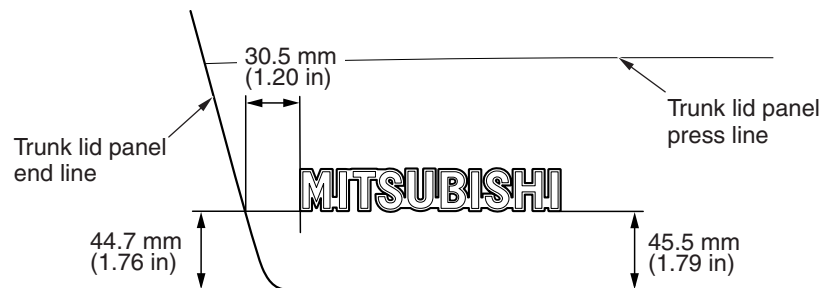
AC704761AB

3. Rear three-diamond mark



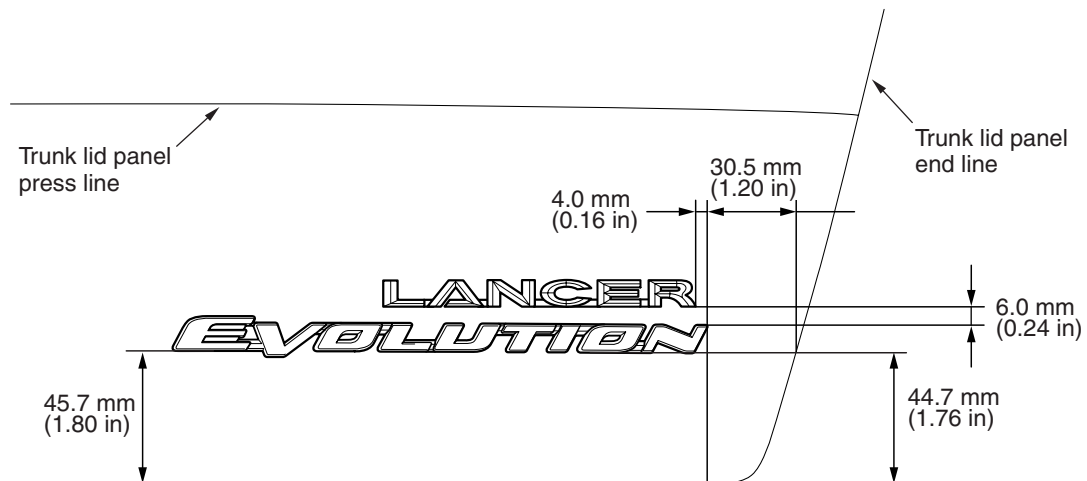
AC609448 AE

4. MITSUBISHI mark



AC609134 AE

5. LANCER EVOLUTION mark



AC708779 AB

1. Installation position

Attach each mark to the position shown in the illustration.

2. Installation procedure

- (1) Use 3M™ AAD Part number 8906 or equivalent to clean the mark installation surfaces on the body.

CAUTION

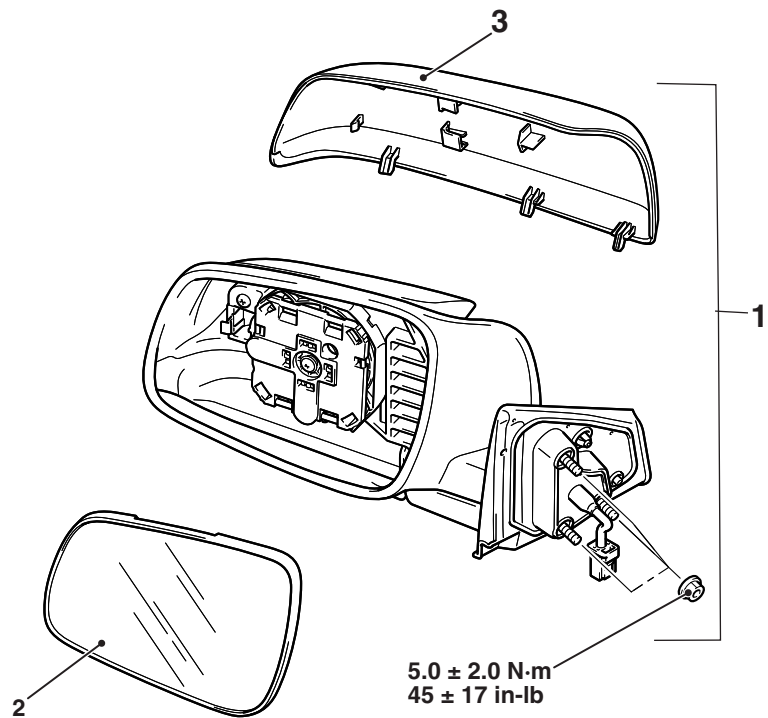
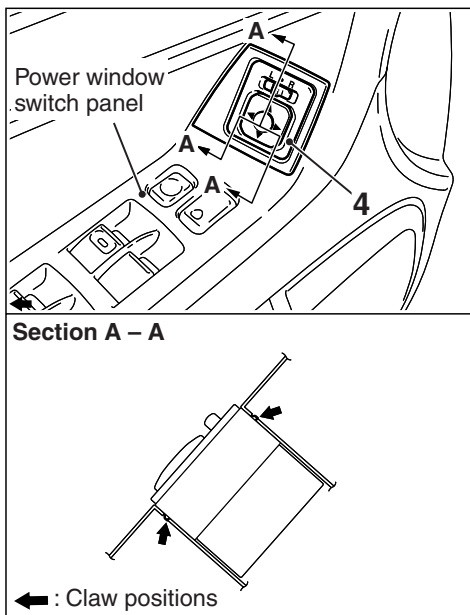
When attaching the marks, the ambient temperature should be 20 – 38° C (68 – 100° F) and the air should be completely free of dust. If the ambient temperature is lower than 20° C (60° F), the marks and the places on the vehicle body where the marks are to be attached should be heated to 20 – 30° C (68 – 86° F).

- (2) Peel off the protection sheet on the back of the marks to affix it in position.

DOOR MIRROR

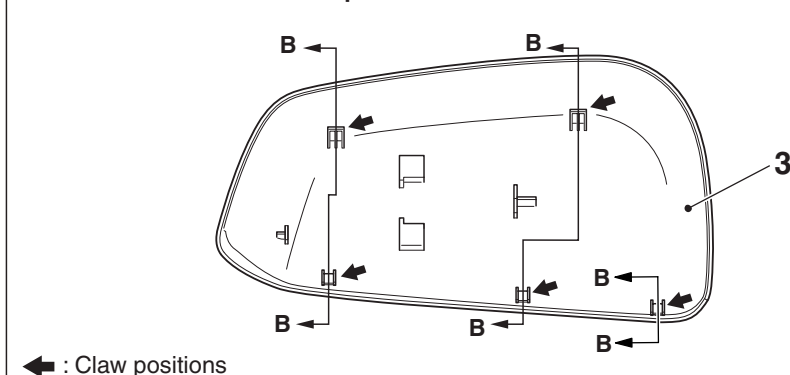
REMOVAL AND INSTALLATION

M1511006400933

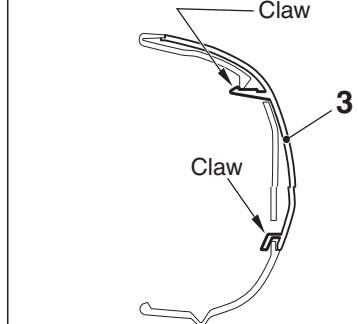


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Door mirror outer cover claw positions



Section B – B



AC705416AD

Door mirror assembly removal steps

- Tweeter (Refer to GROUP 54A, Speaker [P.54A-575](#))
 - Door mirror connector connection
1. Door mirror assembly

<<A>> >>A<<

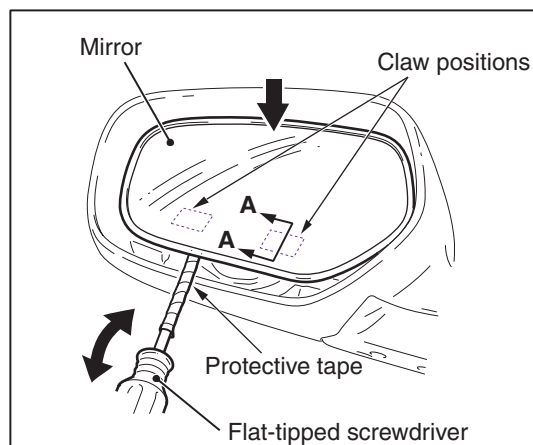
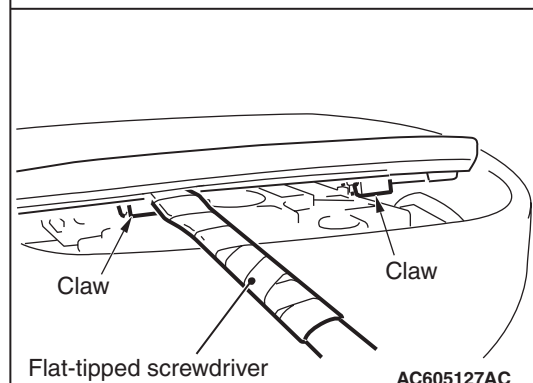
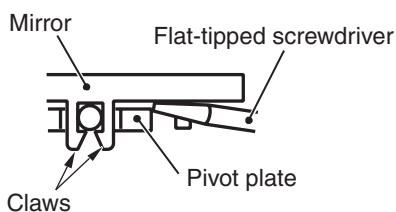
Door mirror outer cover removal steps

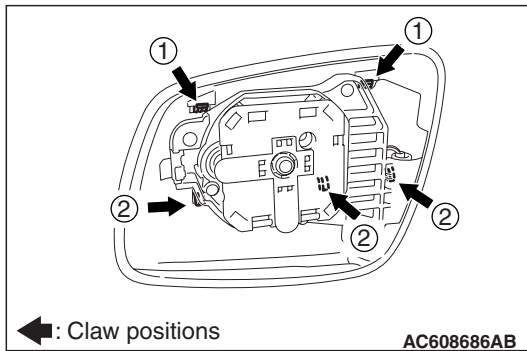
2. Mirror
 3. Door mirror outer cover
- Remote controlled mirror switch removal steps**
- Front door trim (Refer to GROUP 52A, Door Trim [P.52A-15](#))
4. Remote controlled mirror switch

REMOVAL SERVICE POINT**<<A>> DOOR MIRROR OUTER COVER REMOVAL****⚠ CAUTION**

The claws of the mirror are prone to break when working in cold temperatures. Always use a hair drier or the like to warm up the mirror claws and its periphery to 20° C or higher prior to works. When the mirror is heated too quickly from its cold state, it may be broken.

1. Slant the mirror upward with your hands. Then insert a flat-tipped screwdriver wrapped with protective tape between the pivot plate and mirror through the cut-out from behind the mirror. Now pry off the mirror claws and release the lower side of the mirror as shown in the illustration.
2. Release the upper side of the mirror from the tabs while pulling out the mirror downwards.

**Section A - A**



3. Remove the door mirror outer cover by disengaging the claws in the numerical order of the illustration from the mirror body side.

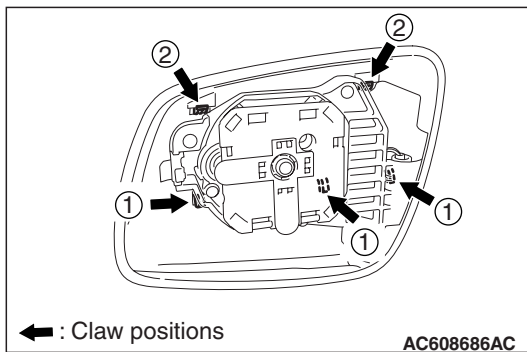
INSTALLATION SERVICE POINT

>>A<< DOOR MIRROR OUTER COVER INSTALLATION

⚠ CAUTION

Tap the claw positions securely to confirm that they are engaged securely.

Install the door mirror outer cover by engaging the claws in the numerical order of the illustration.

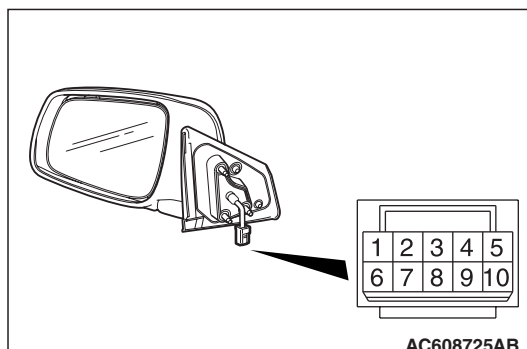


INSPECTION

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REMOTE CONTROLLED MIRROR OPERATION CHECK

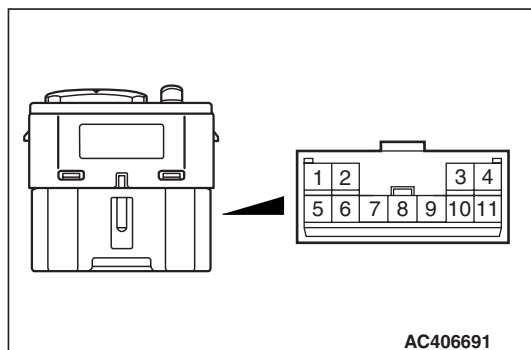
Check that the mirror moves as described in the table when each terminal is connected to the battery.



Battery connection	Direction operation
- Connect terminal 8 to the negative battery terminal. - Connect terminal 6 to the positive battery terminal.	Up
- Connect terminal 8 to the positive battery terminal. - Connect terminal 6 to the negative battery terminal.	Down
- Connect terminal 8 to the negative battery terminal. - Connect terminal 7 to the positive battery terminal.	Right

Battery connection	Direction operation
- Connect terminal 8 to the positive battery terminal. - Connect terminal 7 to the negative battery terminal.	Left

DOOR MIRROR CONTROL SWITCH CONTINUITY CHECK



Switch position		Tester connection	Specified condition
OFF		9 -2, 9 -3, 9 -6, 9 -10, 9 -11, 1 -2, 1 -3, 1 -6, 1 -10, 1 -11	Open circuit
Left side	OFF	9 -6, 9 -10, 9 -11, 1 -6, 1 -10, 1 -11	Open circuit
	Up	1 -6, 9 -11	Continuity exists (2 Ω or less)
	Down	1 -11, 6 -9	
	Right	1 -6, 9 -10	
	Left	1 -10, 6 -9	
Right side	OFF	9 -2, 9 -3, 9 -6, 1 -2, 1 -3, 1 -6	Open circuit
	Up	1 -6, 3 -9	Continuity exists (2 Ω or less)
	Down	1 -3, 6 -9	
	Right	1 -6, 2 -9	
	Left	1 -2, 6 -9	